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Supplementary information

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Direct photo-oxidation of methane to methanol over a mono-iron-hydroxyl site

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Supplementary Methods

Materials

2,2'-bipyridine-5,5'-dicarboxylic acid (H_2bpydc, 98%) was purchased from Tokyo Chemical Industry (TCI) Co., Ltd. Zirconium(IV) chloride ($ZrCl_4$, 99.9%), 2,2′-bipyridyl (bpy, >99%), ruthenium(III) chloride trihydrate ($RuCl_3{\cdot}3H_2O$), sodium tungstate dihydrate ($Na_2WO_4{\cdot}2H_2O$, >99%), phosphoric acid (H_3PO_4, 85%), lithium chloride (LiCl, >99%), tetrabutylammonium bromide (TBABr), sodium acetate (NaOAc, >99%), glacial acetic acid (HOAc, >99.7%), sodium metavanadate ($NaVO_3$, >99.9%), $FeCl_3{\cdot}6H_2O$ (99%), hydrochloric acid (HCl, 37%), potassium phosphate tribasic (K_3PO_4, >98%), ammonium molybdate ($(NH_4)_6Mo_7O_{24}$), *p*-benzoquinone, 2-aminoterephthalic acid, terephthalic acid, 1,4-dioxane (>99%), 5,5-dimethyl-1-pyrroline N-oxide (DMPO for EPR experiments), ethanol (EtOH), methanol (CH_3OH), N,N'-dimehylformaminde (DMF), tetrahydrofuran (THF), acetone, benzene (anhydrous and ACS reagent), hydroxyl peroxide (30% H_2O_2), deuterated reagents, $H_2{}^{18}O$ and $^{13}CH_4$ (99.995%) were purchased from Sigma-Aldrich Co., UK. Ultra-pure research grade (99.999%) argon, nitrogen, oxygen, compressed air and methane were purchased from BOC or Air Liquide. All the reagents, solvents, and gases are commercially available and used as received.

General Experimental Methods

^{1}H NMR spectra were recorded on a B400 Bruker Avance III and B500 Bruker Avance II+ spectrometers and referenced to the proton resonance resulting from incomplete deuteration of D_2O-d_2 (δ = 4.79), $CDCl_3$ (δ = 7.26) and DMSO-d_6 (δ = 2.50). Fourier transform infrared spectra (FTIR) were measured with a Thermo Scientific Nicolet iS5-IR spectrometer in attenuated total-reflection (ATR) sampling mode across the range 4000-400 cm^{-1}. Elemental analyses for Zr, Ru, P, W, V, and Fe were performed on a Thermo iCap 6300 inductively-coupled plasma optical emission spectroscopy (ICP-OES) and analyses for C, H and N were carried out on a CE-440 elemental analyzer (EAI Company). Mass spectrometry was performed on a Bruker Esquire electrospray ion-trap (ESI) spectrometer with samples dissolved in the appropriate solvent, and data are reported as m/z. Thermal gravimetric analyses (TGA) were performed under a flow of air (100 mL/min) with a heating rate of 2 °C/min using a TA SDT-600 thermogravimetric analyzer (TA Company). Powder X-ray diffraction (PXRD) patterns were recorded on a Philips X'pert X-ray diffractometer (40 kV and 30 mA) using Cu Kα radiation (λ = 1.5406 Å). UV-Vis spectra were measured with Hitachi U-3010 spectrophotometer.

Adsorption isotherms for N_2 were recorded at 77 K on a 3Flex or Tristar gas sorption analyzer (Micromeritics Instruments Corporation) to analyze the porosity of the catalysts. The samples were prepared by solvent exchange and outgassed at 80 ^oC under vacuum for 12 h. The specific surface area was calculated using the Brunauer-Emmett-Teller (BET) method in the range of P/P_0 = 0.05-0.3. Micro- and meso-porous surface areas, pore volumes, and mean pore diameters were evaluated by the Horvath-Kawazoe (HK) method and the Barrett-Joyner-Halenda (BJH) method from the adsorption branches of the isotherms.

The morphology and size of MOF crystals were measured by scanning electron microscopy (SEM) on a Quanta FEG 650 microscope. The ratios of Zr/Ru/Fe/W/V in the sample were quantified by energy-dispersive X-ray spectroscopy (EDX) using multiple regions over a sample on a Bruker XTrace instrument. Transmission electron microscopy (TEM) were performed on Phillips Analytical FEI Tecnai F30 electron microscope operating at an electron acceleration voltage of 300 kV. X-ray photoelectron spectroscopy (XPS) measurements were carried out with a Qtac-100 LEISS-XPS spectrometer with a hemispherical electron energy analyzer and a home-made reaction chamber. Monochromatic Al Kα X-ray source (1486.6 eV, anode operating at 300 W) was used as the excitation source. The error in the measured binding energies was ±0.2 eV. The binding energies in all spectra were calibrated according to C 1s peak at 284.8 eV, and XPS binding energies were taken from the

NIST XPS database (https://srdata.nist.gov/xps/main_search_menu.aspx). Quantum efficiency measurements were performed using the standard ferrioxalate actinometric method.[1]

In situ synchrotron FTIR microspectroscopy experiments were carried out using the B22: Multimode InfraRed Imaging and Microspectroscopy (MIRIAM) beamline at the Diamond Light Source. Spectra were collected (512 scans) in the range 4000–650 cm^{-1} with 4 cm^{-1} resolution and infrared spot size at the sample of approximately 15 × 15 μm.

Cyclic voltammetry (CV) and photocurrent measurements were conducted in a standard three-electrode glass cell connected with a CHI760e electrochemical workstation. Electrochemical impedance spectroscopy (EIS) of samples were measured on an electrochemical workstation (Zahner Zennium). Emission spectra were measured on an Edinburgh Instrument FP920 Phosphorescence. Time-domain lifetimes were measured using the Time-Correlated Single Photon Counting (TCSPC) method on a Lifetime Spectrometer equipped with 450 W steady state xenon lamp, a 405 nm ps pulsed diode laser and a Hamamatsu R928P red-sensitive photomultiplier in peltier, air cooled housing. Kinetic data were acquired following 405 nm pulsed excitation using time correlated single photon counting and lifetimes determined by applying a tail fit exponential function and the goodness of fit judged by minimising Chi2 and residuals squared.

Synthetic Procedure for UiO-MOF Catalysts

$ZrCl_4$ (10.0 mg, 43 μmol), H_2bpydc (9.0 mg, 37 μmol), [Ru(bpy)$_2$(H_2bpydc)]Cl_2 (2.0 mg, 4 μmol), and glacial acetic acid (82 μL, 1.44 mmol) were dissolved in 1.5 mL DMF and sealed in a 5 mL vial. The vial was heated at 100 °C for 72 h, and, after cooling to room temperature, the solid product, MOF-Ru, was collected and sequentially washed several times with DMF and THF and dried at 50 °C under vacuum overnight (yield: 51%). ICP-OES analysis confirmed the Zr/Ru ratio of 20 indicating 0.30 per {Zr_6} cluster.

Post-synthetic metalation of mono-iron-hydroxyl species in MOF-Ru is similar with PMOF-RuFe(OH) to obtain MOF-RuFe(OH). The Fe-loading of MOF-RuFe(OH) was also determined by ICP-OES analysis to be 4.0 wt%, corresponding to a Fe/Zr ratio of 0.51, and 3.0 Fe per {Zr_6} cluster.

Analysis of Products

The reactor was connected directly to a gas chromatographic (GC) analyser equipped with both thermal conductivity detector (TCD) and flame ionisation detector (FID) for analysis of the gaseous products. Before analysis, the gas in the reactor was used to sweep the GC lines 3 times. The amount of CH_4 was quantified by GC using an external standard with a Bruker Matrix MG5 FTIR spectrometer, calibrated using known concentrations of target gases. The conversion of CH_4 was calculated according to the following equation:

$$CH_4\ conversion\% = \frac{mol_{initial\ CH_4} - mol_{CH_4 after\ reaction}}{mol_{initial\ CH_4}} \times 100\%$$

The other gases (e.g., CO_2 on TCD, C_xH_y on FID) were quantified by the following equation by using quantified CH_4 as the standard.

$$mol_{gas} = f(correction\ factor) \times Area_{gas} \times \frac{mol_{CH_4}}{Area_{CH_4}}$$

After analysing the gaseous phase, the solid catalyst was separated from the liquor and the liquor analysed by GC-FID with 1,4-dioxane as an internal standard. The selectivity and productivity of CH_3OH were calculated by the following equations:

$$Methanol\ Selectivity\ \% = \frac{mol_{produced\ methanol}}{mol_{initial\ CH_4} - mol_{CH_4 after\ reaction}} \times 100\%$$

$$Methanol\ productivity\ (\mu mol \cdot g_{cat}^{-1} \cdot h^{-1}) = \frac{produced\ methanol\ (\mu mol)}{m_{cat}(g) \times reaction\ time\ (h)}$$

The products in the liquid phase were also quantified by ^{1}H NMR spectroscopy. Typically, 0.7 mL of liquor after reaction (with 1,4-dioxane as internal standard) was mixed with 0.2 mL of D_2O to prepare a solution for NMR measurement.

Isotopic Experiments

The isotopic experiments were performed with PMOF-RuFe(OH) using the general procedure for photocatalytic methane oxidation in batch mode except with isotopic sources. After the reaction, the products were collected and analyzed by GC-MS.

Leaching Test

For the leaching experiment, 10.0 mg of PMOF-RuFe(OH) catalyst in 3 mL of H_2O was used. After reaction with irradiation for 20 h under 1 atm of CH_4 and O_2, the solid catalyst was removed *via* centrifugation, and the supernatant filtered and split into 2 samples - one for GC analyses and one for a second reaction. The reactor was re-filled with CH_4 and O_2 gas for further reaction tests under irradiation. No additional product was observed in the second reaction.

Recycling Experiments

The best-performing catalyst, PMOF-RuFe(OH), was chosen for the recycling test for photocatalytic oxidation of CH_4. After the general catalytic procedure, the headspace gas was analyzed by GC using a TCD, and the solid PMOF-RuFe(OH) was collected from the reaction mixture *via* centrifugation, and the supernatant analyzed by GC to quantify the yield of CH_3OH. The recovered solid catalyst was washed for 3 times with H_2O and added again to the reactor together with 3 mL of fresh H_2O. The reactor was then re-filled with CH_4 and O_2 gases for a second reaction under irradiation. The catalyst was reused without any significant drop in activity for 10 consecutive runs.

X-Ray Absorption Spectroscopic (XAS) Analysis

X-ray absorption data for iron and vanadium were collected at 10-BM beamline at the Advanced Photon Source (APS) at the Argonne National Laboratory and at the BL11S beamline at the Aichi Synchrotron Radiation Centre (AichiSR), respectively. Spectra were collected at the Fe K-edge (7112 eV) in transmission mode and the V K-edge (5465 eV) in fluorescence mode. The X-ray beam was monochromatized by a Si(111) monochromator and detuned by 50% to reduce the contribution of higher-order harmonics below the level of noise. Multiple X-ray absorption spectra were collected at room temperature for each sample. Samples were ground and mixed with polyethylene glycol (PEG) and packed in a 6-shooter sample holder to achieve adequate absorption length.

Data were processed using the Athena and Artemis programs of the IFEFFIT package based on FEFF 6.[2,3] Fitting of the EXAFS data was performed using the Artemis program of the IFEFFIT package and was performed with a k-weight of 2 in R-space. Refinement was performed by optimizing an amplitude factor S_0^2 and energy shift ΔE_0, which are common to all paths, in addition to parameters for bond length (ΔR) and Debye-Waller factor (σ^2).

Adsorption Isotherms for CH_4

Gravimetric sorption isotherms for CH_4 were recorded on a Hiden Isochema Intelligent Gravimetric Analyzer (IGA) system under ultra-high vacuum using a turbo pumping system. The temperature was maintained between 298-338 K using a temperature-programmed water bath. In a typical gas adsorption experiment, 50.0 mg of acetone-exchanged MOF was loaded into the IGA system and activated at 353 K under dynamic high vacuum (10^{-10} bar) for 1 day to give a fully desolvated sample.

^{57}Fe Mössbauer spectroscopy

The ^{57}Fe Mössbauer spectrum of PMOF-RuFe(OH) was recorded at room temperature using a ^{57}Co source in Pd. The spectrometer was operated with a triangular velocity waveform and a NaI scintillation detector was used for γ-ray detection. Velocity calibration was performed with an α-Fe foil.

Computational Setup

Density Functional Theory (DFT) calculations were carried out using the Gaussian 09 Program.[4] The structures of chemical species in solution-phase and gas-phase were fully optimized by using the B3LYP functional.[5,6] The 6-31+G (d) basis set was used for C, H, N, O and P, and the LanL2DZ (Lanl-2-double-zeta) was used for Fe, Ru, Zr, W and V. Solvation effects were accounted for by using the implicit solvation model, *i.e.*, SMD.[7] The model contains a solvent molecule in a cavity surrounded by a continuous polarizable dielectric medium, which responds to the distributed solute charges according to its dielectric constant. The solvent was set as water. The structures were optimised using the method described above. For the calculation of solvation free energies, both electrostatic and nonelectrostatic (dispersion, cavitation) energy contributions were considered. No explicit water was included in the catalyst models. The corresponding single point energy calculation was based on def2TZVP basis set for accuracy. The catalyst structure was modeled using $[(bpy)Fe(OH)(H_2O)_3]^{2+}$, the $\{Zr_6\}$ cluster with formate as a capping groups, $[Ru(bpy)_3]^{2+}$ and PW_9V_3. As $bpydc^{2-}$ ligands are coordinated to Zr_6 clusters, the coordinated carboxyl groups can be regarded as neutral groups (similar to ester), and thus the $[(bpy)Fe^{III}(OH)(H_2O)_3]^{2+}$ fragment was used instead of $[(bpydc)Fe(OH)(H_2O)_3]$. The bpy was fixed and optimized, and CH_4 together with associated oxygen coordinated to the iron center were free to move.

Vibrational frequencies and polarization vectors were calculated using CP2K (http://www.cp2k.org)[8], based on the mixed Gaussian and plane-wave scheme[9] and the Quickstep module[10]. The calculation used molecularly optimized Double-Zeta-Valence plus Polarization (DZVP) basis set[11], Goedecker-Teter-Hutter pseudopotentials[12], and the Perdew-Burke-Ernzerhof (PBE) exchange correlation functional[13]. The plane-wave energy cutoff was 400 Ry. The DFT-D3 level correction for dispersion interactions, as implemented by Grimme et al[14], was applied, with a cutoff distance of 15 Å. The calculation was performed on Gamma point only, with no symmetry constraint. A cubic unit cell with a lattice constant of ~26.9 Å was used for the calculation for each system. Structural disorder and various possible local coordinates are represented within the unit cell, but their long-range effects are suppressed due to the periodic boundary conditions. Structural optimization was performed using the Broyden-Fletcher-Goldfarb-Shannon (BFGS) optimizer, until the maximum force is below 0.00045 Ry/Bohr (0.011 eV/Å). The finite displacement method was used for the phonon calculation, with incremental displacement of 0.01 Bohr (0.0053 Å). The INS spectrum was simulated using the OClimax software.[15]

Supplementary Text

Catalyst Model and Calculation of Values of pK_a

To determine the protonation states of $[(bpy)Fe(OH)_x(H_2O)_{4-x}]^{n+}$ under the neutral pH conditions of the reaction, the pK_a of the possible acidic sites, including coordinated water and hydroxyl groups, were calculated. The acidity constant (pK_a) of a species X–H is defined as:

$$\boldsymbol{pK_a(XH) = -\log K_a = \frac{\Delta G_{ps}}{2.303RT}} \qquad (\boldsymbol{eq.S1})$$

where ΔG_{ps} is the Gibb's free energy of protonated X–H in the solution phase, and K_a is the equilibrium constant[16]. Here, the ΔG_{ps} is calculated from the thermodynamic cycle (Scheme S1).[17]

$$\begin{tikzcd}
\mathrm{XH(g)} \arrow[r, "\Delta G_{pg}"] \arrow[d, "\Delta G_s(\mathrm{XH})"'] & \mathrm{X^-(g)} \arrow[d, "\Delta G_s(\mathrm{X^-})"'] & + & \mathrm{H^+(g)} \arrow[d, "\Delta G_s(\mathrm{H^+})"'] \\
\mathrm{XH(aq)} \arrow[r, "\Delta G_{ps}"] & \mathrm{X^-(l)} & + & \mathrm{H^+(l)}
\end{tikzcd}$$

Scheme S1. Thermodynamic cycle used for calculating pK_a of X-H.

ΔG_{pg} and ΔG_{ps} are the gas- and solution-phase free energy of deprotonation, respectively, and $\Delta G_s(XH)$, $\Delta G_s(X^-)$ and $\Delta G_s(H^+)$ are the free energies of solvation for XH, X^- and H^+, respectively. From this cycle, ΔG_{ps} can be derived as:

$$\Delta G_{ps} = \Delta G_s(X^-) + \Delta G_s(H^+) + \Delta G_{pg} - \Delta G_s(XH) \qquad (eq.S2)$$

Thus, substituting *eq.S2* in *eq.S1*, the pK_a of X-H can be obtained from *eq.S3*:

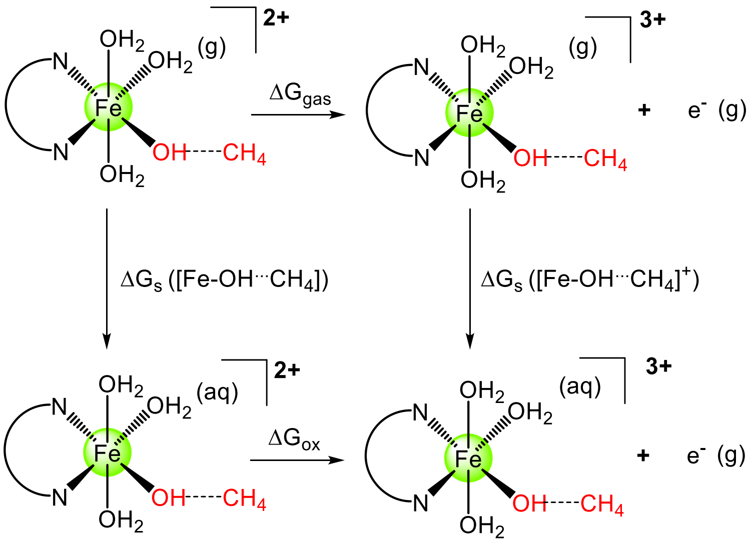
$$pK_a(XH) = \frac{\Delta G_s(X^-) + \Delta G_s(H^+) + \Delta G_{pg} - \Delta G_s(XH)}{2.303RT} \qquad (eq.S3)$$

The experimental value (-264.0 kcal mol^{-1}) was used for the free energy of solvation of a proton $\Delta G_s(H^+)$.[18] All values for G_{pg} (gas-phase free energy) were calculated by the CBS-QB3 method.[19,20] All values of ΔG_s (free energy of dissolution) were calculated based on the M05-2X method combined with the SMD solvent model.[21,22] Different methods were used here to ensure the accuracy and efficiency of calculation, in line with literature reports.[23,24]

The pK_a of all possible acidic sites of the model were calculated, and the results are summarised in Figure S11. The first deprotonation of $[[(bpy)Fe(H_2O)_4]^{3+}$ gives the complex with the hydroxyl group in the equatorial plane (pK_a = 6.74) as opposed to the axial position (pK_a = 7.74). As would be expected protonation of $[[(bpy)Fe(H_2O)_4]^{3+}$ gives 4+ cations with significant acidity, pK_a = -17.97 and -18.97, while deprotonation of $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ has values of pK_a of 10.72 and 21.12. Under the reaction conditions of this study (pH = 7.0) the equilibrium structure $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ was obtained as the catalyst model with one coordinated hydroxyl group and three water molecules on the iron site.

Calculation of the Oxidation Potential

To verify the proposed redox cycle, the oxidation potential of adsorbed CH_4 on $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ within the species $[(bpy)Fe(OH)(H_2O)_3{\cdots}CH_4]^{2+}$ {abbreviated to [Fe-OH···CH_4]} was calculated based upon transfer of one electron to form $\cdot CH_3$ radical.[25] A free energy cycle based upon $[(bpy)Fe(OH)(H_2O)_3{\cdots}CH_4]^{2+}$ as the reactant was developed (Scheme S2) to assist the calculation. The free energy change associated with the top side of the cycle is the gas-phase oxidation potential, which is closely related to the adiabatic ionization potential. Free energy changes associated with the left-hand and right-hand sides of the cycle are free energies of solvation, and thus the bottom side of the cycle refers to the oxidation potential in solution.



Scheme S2. Thermodynamic cycle used for calculating oxidation potential. The initial model represents $[(bpy)Fe(OH)(H_2O)_3{\cdots}CH_4]^{2+}$.

Oxidation potentials are usually reported relative to a reference potential, typically the normal hydrogen electrode (NHE); this convention, then, refers to the free energy change for the net reaction resulting from addition of the NHE half-reaction to the half-reaction at the bottom of the free energy cycle. The half reaction of normal hydrogen electrode is shown below.

$$H^+(aq) + e^-(g) \xrightarrow{DG = -4.44 eV} 1/2\ H_2(g)$$

Thus, the aqueous oxidation potential of a closed-shell neutral solute X is determined from Scheme S2 as:

$$\Delta G_{ox}([Fe\text{-}OH\cdots CH_4]v\ s\ NHE) =$$
$$\Delta G_{gas} + \Delta G_s([Fe\text{-}OH\cdots CH_4]^+) - \Delta G_s([Fe\text{-}OH\cdots CH_4]\,) - 4.44\ eV \quad (eq.S4)$$

All values of G_{gas} were calculated based on the CBS-QB3 method. All values of ΔG_s were calculated based on the M05-2X method combined with the SMD solvent model to obtain accurate free energy of dissolution. The calculated oxidation potential of [Fe-OH···CH_4] to [Fe-OH···CH_4]$^+$ is 0.84V (*vs* NHE), while the $E_{PS^+/PS} = 0.91\ V\ (vs\ NHE)$, which is sufficient to oxidize the absorbed CH_4 and to drive the reaction.

Calculation of the Activation of CH_4

To further support the proposed mechanism of CH_4 activation, the profile of energy change from CH_4 to $\cdot CH_3$ radical was calculated. The Gibbs energy was calculated using B3LYP/6-31G* to optimise the structure and obtain thermal correction for the Gibbs free energy. The thermal correction contains the correction factor of entropy caused by the changes of vibrational/translational/rotational degrees of freedom. Figure 4D gives the corresponding energy changes and the intermediates during the activation process of CH_4. Adsorption of CH_4 onto $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ gives a stabilization energy of -10.6 kJ mol^{-1}, which thus has a pre-activation effect on CH_4. On oxidation of the Fe center, along with the calculated potentials, the barrier is 81.0 kJ mol^{-1} (0.84 V potential vs NHE) and the potential of photosensitizer of 87.8 kJ mol^{-1} (0.91 V potential vs NHE) was coupled into this step to give an overall energy change of -7.0 kJ mol^{-1}, which is thermodynamically favorable. The oxidative intermediate $[Fe\text{-}OH\cdots CH_4]^+$ will then go through a transition state leading to the formation of the $\bullet CH_3$ radical with a barrier of only 17.4 kJ mol^{-1}, the energy for which is delivered by the photosensitizer. After crossing the transition state, the energy drops by 10.6 kJ mol^{-1} to the final state. The overall energy change profile for CH_4 activation shows that the rate determine step is the electron injection from PS* to the iron centre.

Justification of choice of methods in DFT calculations

The DFT calculation was employed (i) to optimise the intermediate structure and calculate the Gibbs free energies associated with the activation of methane and (ii) to calculate the pK_a value in various structural models to identify the local structure of Fe site. Each of the calculations is different in nature for a system of such complexity, and the most appropriate method (based upon literature survey) was therefore chosen for each calculation rather than using one method throughout the study. Such methodology, using different DFT methods for calculations, has been widely reported previously in the literature.[26,27]

Inelastic Neutron Scattering (INS)

Direct visualization of the interaction between adsorbed CH_4 and the active sites is crucial to understanding the molecular details of adsorption and activation of CH_4 into CH_3OH. INS is a powerful neutron spectroscopy technique to investigate the dynamics (particularly for H-containing compounds, such as hydrocarbons) of host-guest interactions because it has several unique advantages:

- INS spectroscopy is ultra-sensitive to the vibrations of hydrogen atoms, and hydrogen is ten times more visible than other elements due to its high neutron cross-section.
- The technique is not subject to any optical selection rules. All vibrations are active and, in principle, measurable.
- INS observations are not restricted to the centre of the Brillouin zone (gamma point) as is the case for optical techniques.
- INS spectra can be readily and accurately modelled: the intensities are proportional to the concentration of elements in the sample and their cross-sections, and the measured INS intensities relate straightforwardly to the associated displacements of the scattering atom. Treatment of background correction is also straightforward.
- Neutrons penetrate deeply into materials and pass readily through the walls of metal containers making neutrons ideal to measure bulk properties of the material.
- INS spectrometers cover the whole range of the molecular vibrational spectrum, 0-500 meV (0-4000 cm^{-1}).
- INS data can be collected at low temperature (at 5 K in this case), where the thermal motion of the MOF material and the adsorbed CH_4 molecules is significantly reduced.
- Calculation of the INS spectra by DFT vibrational analysis can be readily achieved, and DFT calculations relate directly to the INS spectra, and, in the case of solid-state calculations, there are no approximations other than the use of DFT eigenvectors and eigenvalues to determine the spectral intensities.

Supplementary Figures

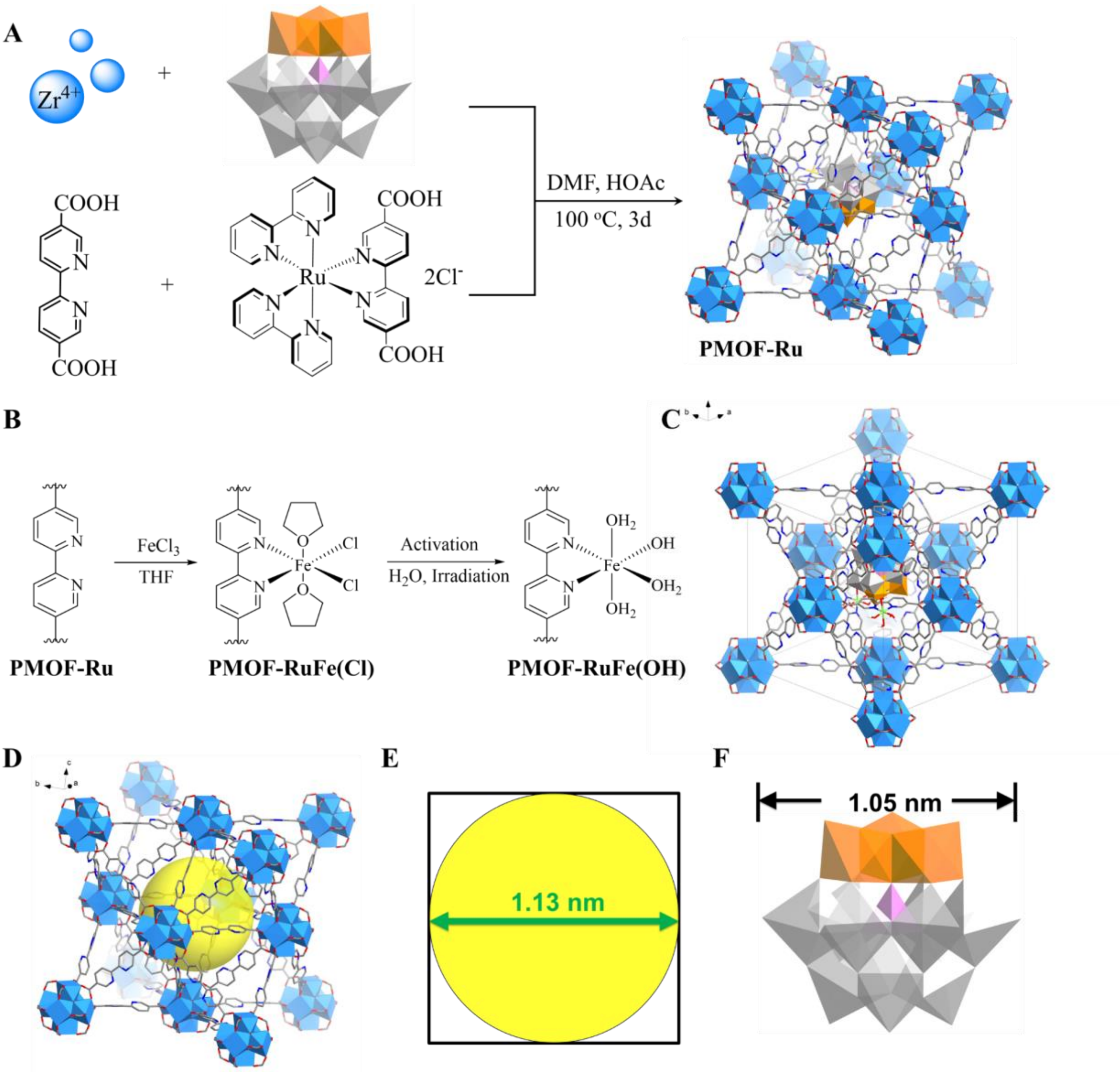


Figure S1. Preparation of MOF catalysts. (A) Synthesis of PMOF-Ru via charge-assisted self-assembly. (B) Post-modification binding of $Fe(OH)(OH_2)_3$ species to form PMOF-RuFe(OH). (C) An illustration of the structure of PMOF-RuFe(OH) within the unit cell (black line). It is worth noting that statistically the various components will be distributed homogeneously throughout the structure. The occupancies of atoms are given in the simulated CIF files. (D) Structure of octahedral cage in UiO-67 without guest species. (E) The diameter of the circle representing the cavity within the octahedral cage. (F) View of the *Keggin*-type polyanion in $[PW_9V_3O_{40}]^{6-}$. Due to the nature of the one-pot synthesis, most likely the $[Ru(bpy)_3]^{2+}$ moieties are pushed out of the cage, and/or some of the ligands are missing to accommodate the $[PW_9V_3O_{40}]^{6-}$ cluster.

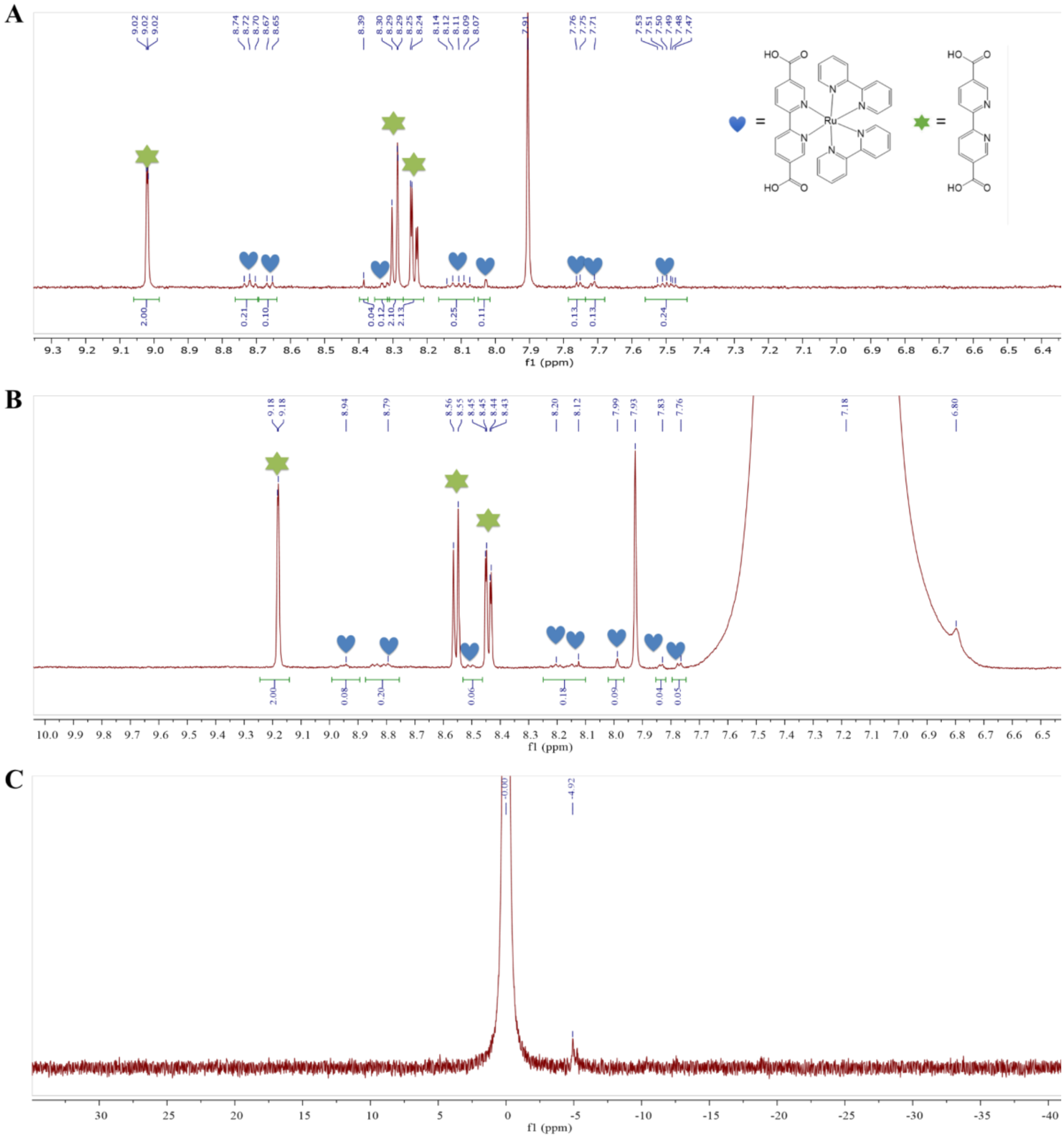


Figure S2. NMR spectra. (A) ^1H NMR spectrum of MOF-Ru digested in $K_3PO_4/D_2O/DMSO\text{-}d_6$. (B) ^1H NMR spectra of PMOF-Ru digested in $H_3PO_4/DMSO\text{-}d_6$. Peaks at δ = ~7.9 and 8.38 ppm are assigned to DMF and formate (from DMF decomposition) guest molecules, respectively. (C) ^{31}P NMR spectrum of digested PMOF-Ru referenced to H_3PO_4 (δ =0 ppm). Note that the chemical shifts of polyvanadotungstates lie in the range of + 10 to -20ppm with respect to H_3PO_4.

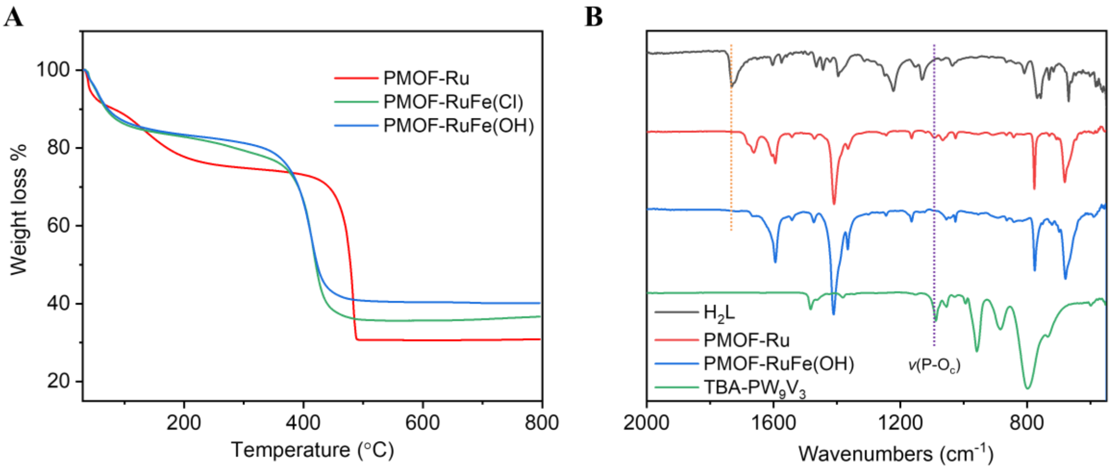


Figure S3. Characterization of PMOF-Ru, PMOF-RuFe(Cl) and PMOF-RuFe(OH). (A) TGA curves: initial weight loss (25-300 °C) corresponds to the removal of adsorbed solvents and the subsequent weight loss (300-500 °C) corresponds to ligands [43% for PMOF-Ru, 40.2% for PMOF-RuFe(Cl) and 39.5% for PMOF-RuFe(OH), respectively]. The increased amount of residual material (30.3%, 35.7% and 40.7%) indicates successful post-synthetic metalation. (B) IR transmission spectra of PMOF-Ru and PMOF-RuFe(OH). No peaks for free carboxylate ligands were observed indicating that all the ligands are coordinated to metal centers. The successful incorporation of PW_9V_3 in PMOF-Ru was evidenced by ^{31}P NMR spectroscopy ($\delta = -12.71$ ppm) and by infrared spectroscopy ($\nu_{P\text{–}Oc}$: 1085 cm^{-1}). Additionally, upon introduction of PW_9V_3 as guest molecules, the strong Bragg peaks at $2\theta = 5.7$ and 11.4^o of PMOF-Ru become weaker and the peaks at $2\theta = 6.6$ and 7.7^o grow in intensity. This is in good agreement with the reported structures of polyoxometallate@UiO.[28,29]

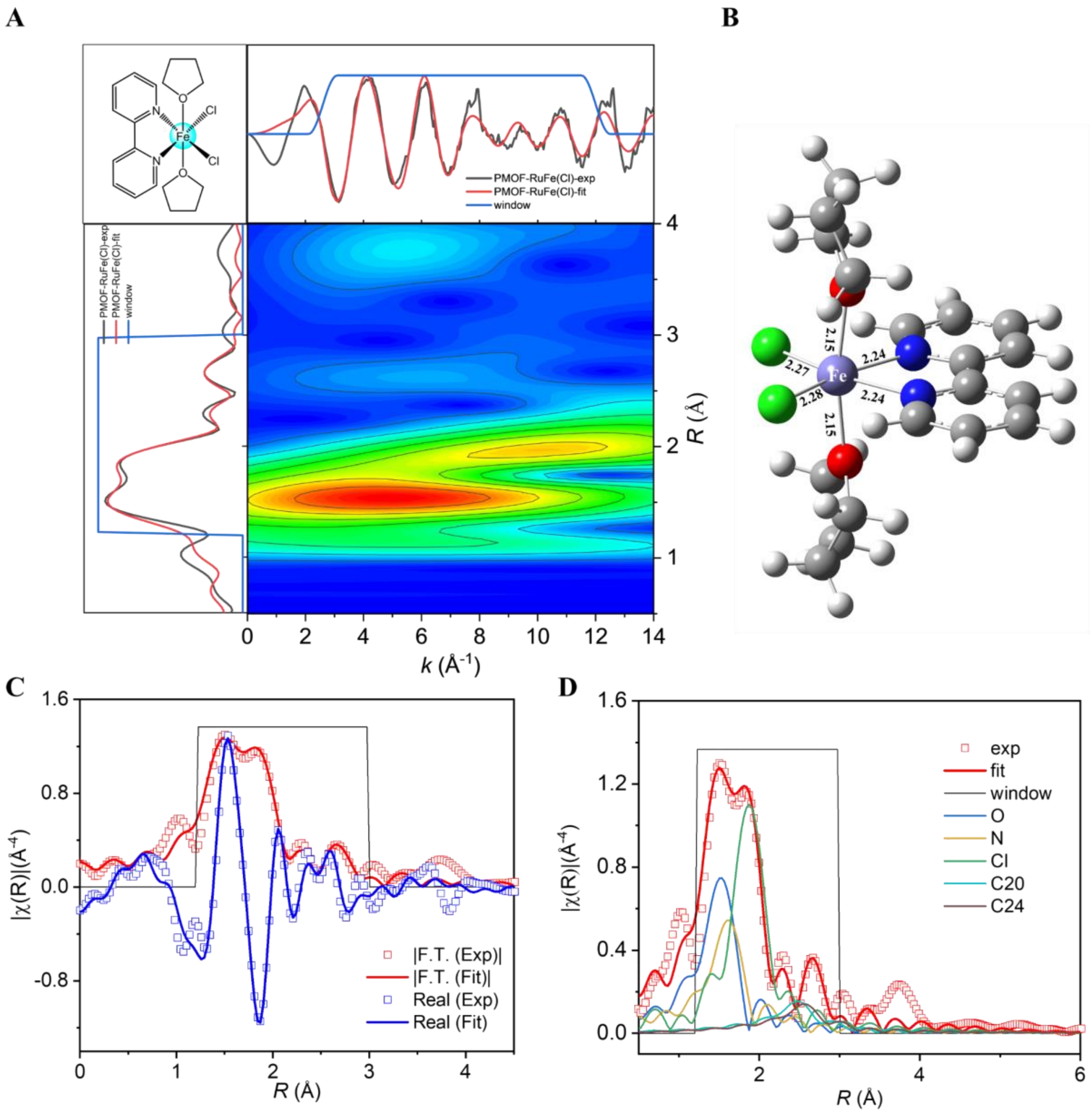


Figure S4. EXAFS spectra at the Fe K-edge absorption of PMOF-RuFe(Cl). The fitting range is 1.2 – 3.0 Å in *R* space (within the black line). (A) Wavelet transform spectra displaying the first and second coordination shell plotted as a function of k ($Å^{-1}$) on the *x*-axis and R (Å) on the *y*-axis. (B) The structural model, (bpy)$FeCl_2(THF)_2$, used for fitting was optimized by DFT calculations. (C) Experimental EXAFS spectra and fitting in *R* space showing the magnitude of Fourier transform (red hollow squares, red line) and real components (blue hollow squares, blue line). The EXAFS fitting agrees well with DFT-calculated bond distances with average Fe–Cl bond lengths of 2.28 ± 0.02 Å (×2), Fe–N_{bpy} bond lengths of 2.24 ± 0.08 Å (×2) and Fe–O_{THF} bond lengths of 2.15 ± 0.09 Å (×2). (D) The scattering paths of Fe to THF-oxygen (blue), bpy-nitrogen (khaki), and chloride (green) from the first shell, and bpy-carbons, C20 (cyan) and C24 (brown), from the second shell.

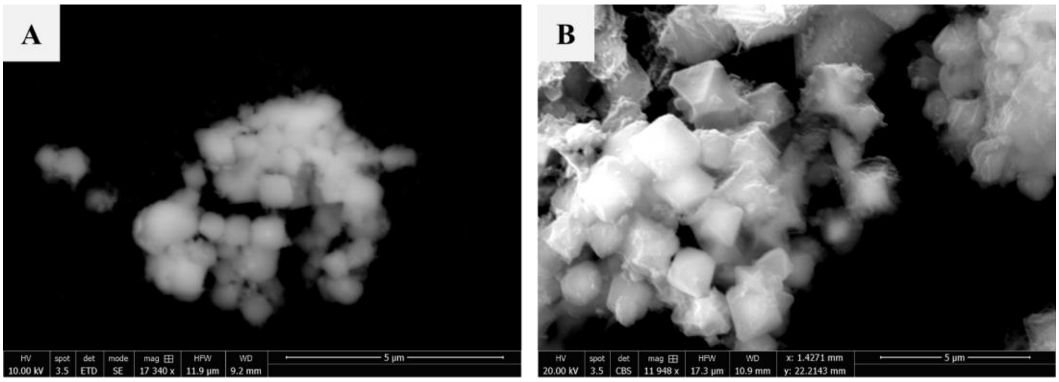


Figure S5. SEM images of (A) PMOF-RuFe(Cl) and (B) PMOF-RuFe(OH). All the MOF materials adopt a distorted octahedral morphology.

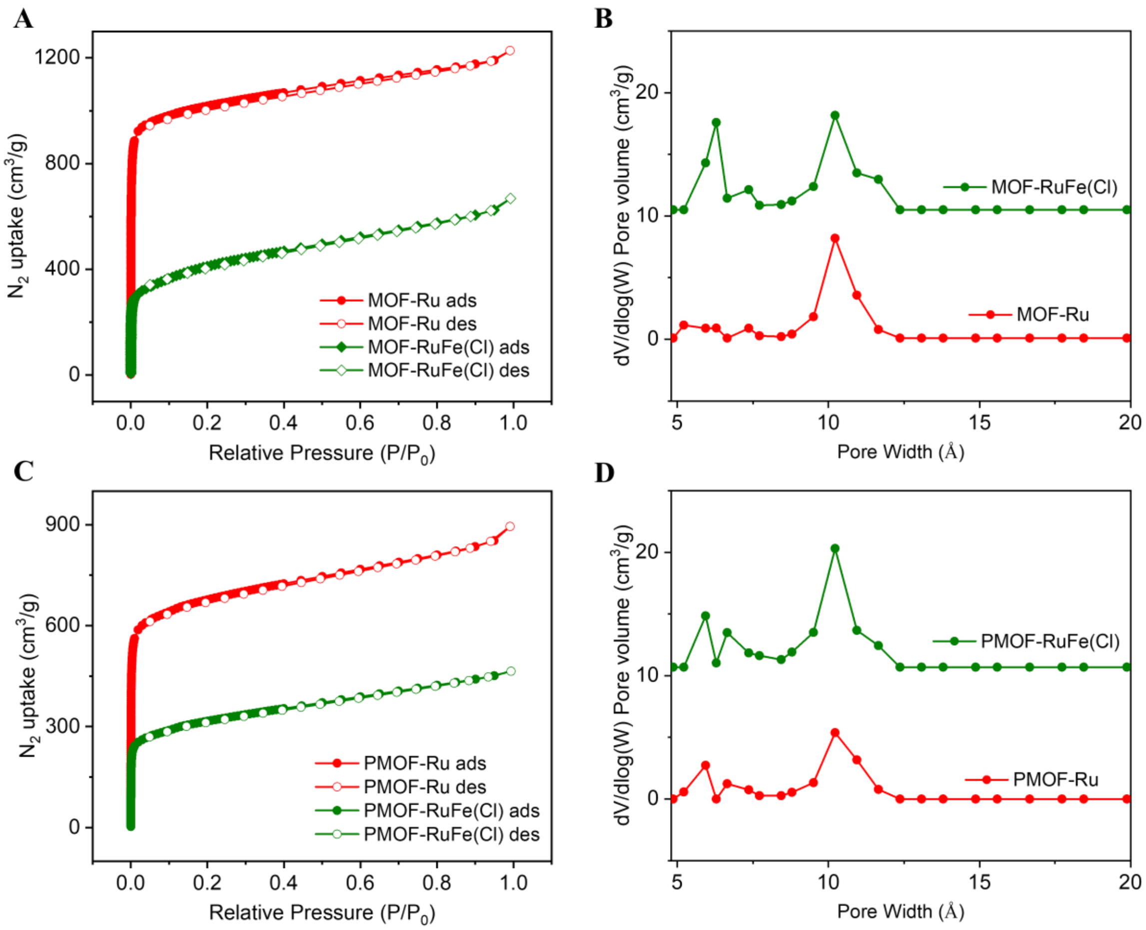


Figure S6. N_2 sorption isotherms and pore size distribution. (A) Isotherms for MOF-Ru and MOF-RuFe(Cl). (B) Pore size distribution for MOF-Ru and MOF-RuFe(Cl). (C) Isotherms for PMOF-Ru and PMOF-RuFe(Cl). (D) Pore size distribution for PMOF-Ru and PMOF-RuFe(Cl). Pore size distribution for the samples were calculated using the Horvath-Kawazoe method. MOF-RuFe(Cl) and PMOF-RuFe(Cl) show BET surface areas of 1378 and 1034 m^2 g^{-1}, respectively, lower than for the bare MOF, indicative of the successful introduction of functional groups within the pores. The pore size distribution confirms the presence of open channels after modification. The detailed analysis is summarized in Table S3.

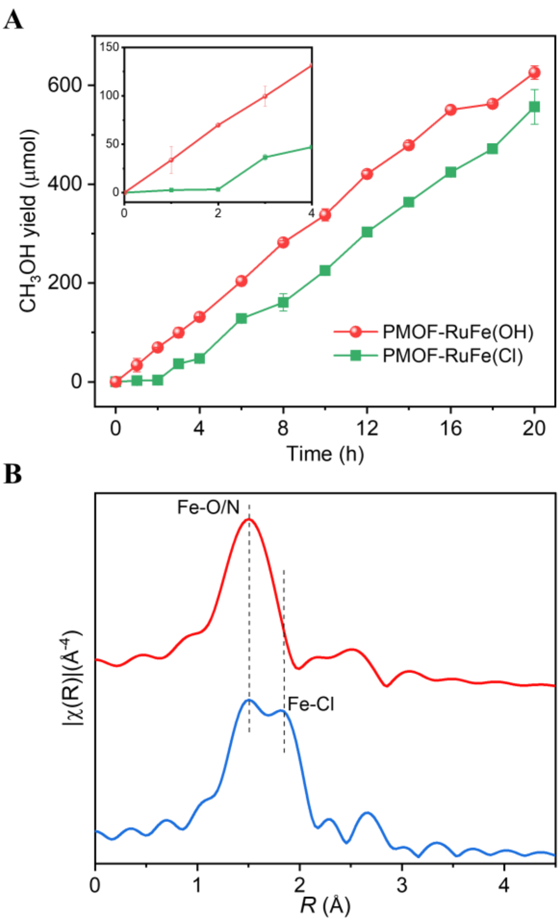


Figure S7. Transformation of active sites**.** (A) Yield of CH_3OH with error bars in batch mode as a function of reaction time using PMOF-Ru(Cl) and PMOF-RuFe(OH) as catalysts. The induction period is due to the transformation of Fe-Cl to the active Fe-OH sites. Note that the difference in the observed yield of CH_3OH between the two catalysts is due to the presence of an induction period of 2 h for PMOF-RuFe(Cl) to allow its transformation to active PMOF-RuFe(OH). (B) k^2-Weighted Fe K-edge EXAFS spectra of PMOF-RuFe(OH) and PMOF-RuFe(Cl) showing a change in the first-shell coordination of the Fe center.

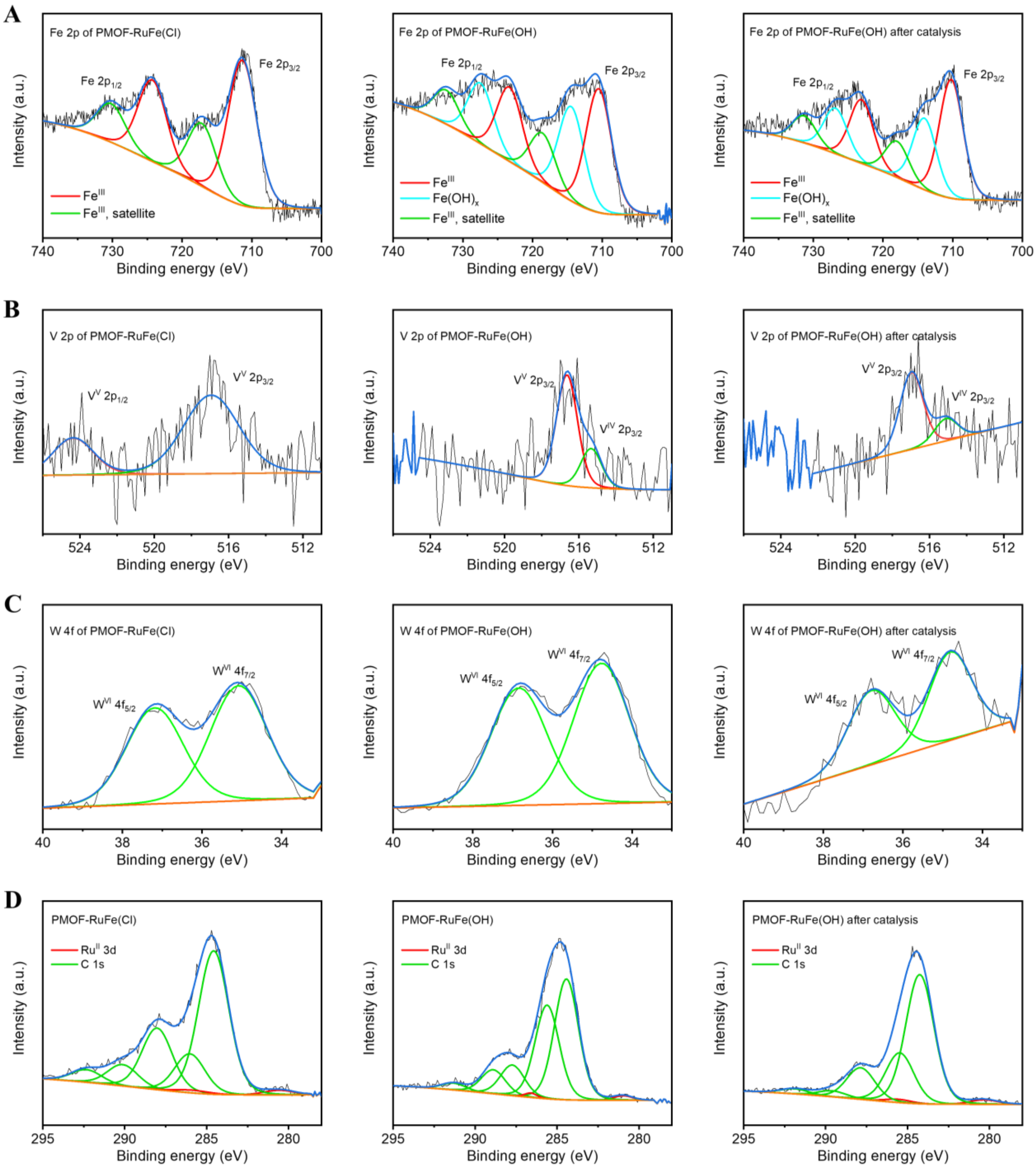


Figure S8. XPS data of as-synthesized and used PMOF-RuFe(Cl) and PMOF-RuFe(OH) catalysts. (A) Fe 2p. (B) V 2p. (C) W 4f. (D) Ru 3d and C 1s. These analyses were referenced to the XPS database. **NOTE:** XPS spectra of the activated PMOF-RuFe(OH) catalyst exhibit typical features for Fe^{III} species with broadening of peaks at 710 and 724 eV. Two spin-orbit splitting doublets are assigned to strong Fe^{III} 2p3/2 and 2p1/2 at 710 and 723 eV, respectively, and weak satellite peaks at 718 and 731 eV, respectively.[30,31] The additional peaks at 714 and 727 eV are related to the effects of hydroxyl groups, which can be found in the XPS spectra of α-Fe^{III}OOH, but not in PMOF-RuFe(Cl).[31] This is also consistent with the absorption at 590 nm in UV-vis spectra (Figure 2B).[32] The fingerprint satellites at 8.0 eV are above the main binding energy, and the main core line is asymmetrical at the high energy side, both of which indicate a *pseudo*-octahedral geometry for the Fe center bound to different ligands.

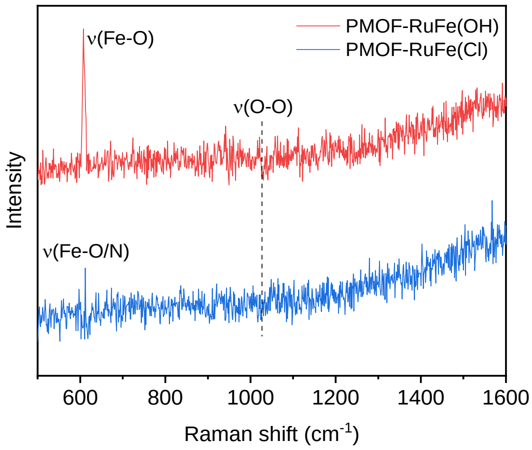


Figure S9. Raman spectra of PMOF-RuFe(Cl) and PMOF-RuFe(OH). Raman spectra confirm the absence of a ν(O-O) stretching vibration as would be expected in Fe^{III}-OOH (at 1000-1200 cm-1) or Fe-O_2 (700-880 cm^{-1}) species. This is consistent with UV-vis spectroscopic results where no LMCT (ligand-to-metal charge-transfer) transitions at 550 nm and 755 nm for Fe^{III}-OOH and Fe-O_2 species, respectively, are observed (Figure 2B).[33]

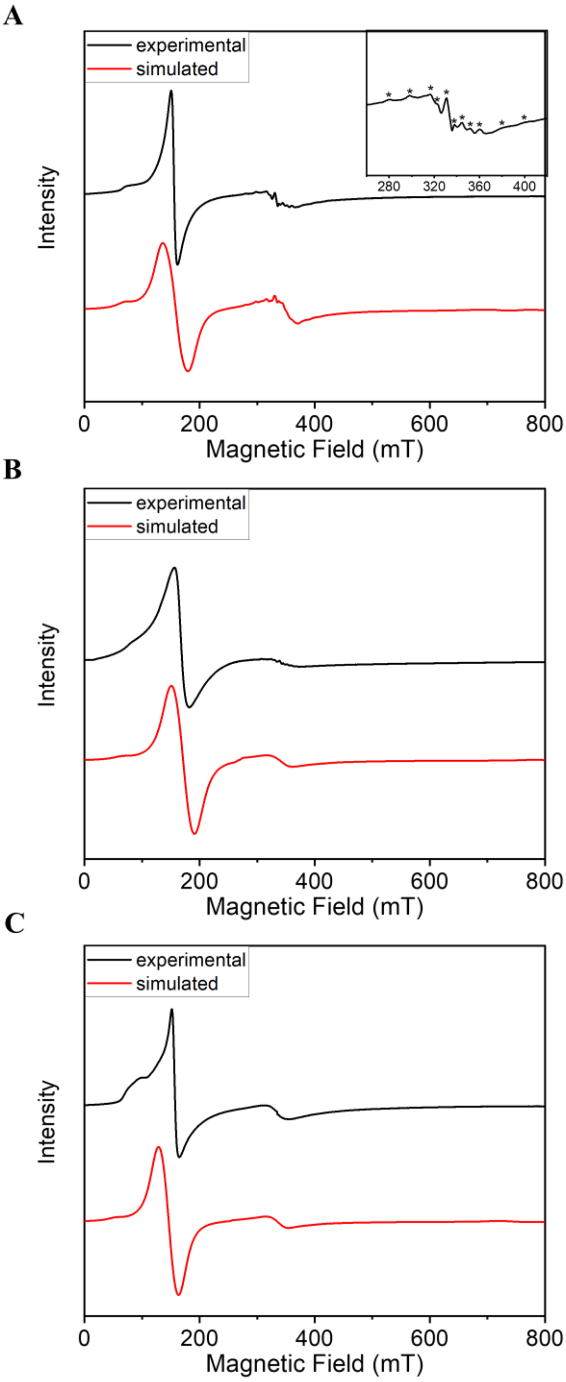


Figure S10. Experimental (at 5 K) and simulated X-band EPR spectra of (A) PMOF-RuFe(OH), (B) PMOF-RuFe(Cl) and (C) MOF-RuFe(OH). Most PW_9V_3 in PMOF-RuFe(OH) is diamagnetic V^{V} with a small amount of V^{IV} signals (inset spectrum shows V^{IV} signal labeled with *).[34] All the parameters for simulation are given in Table S4.

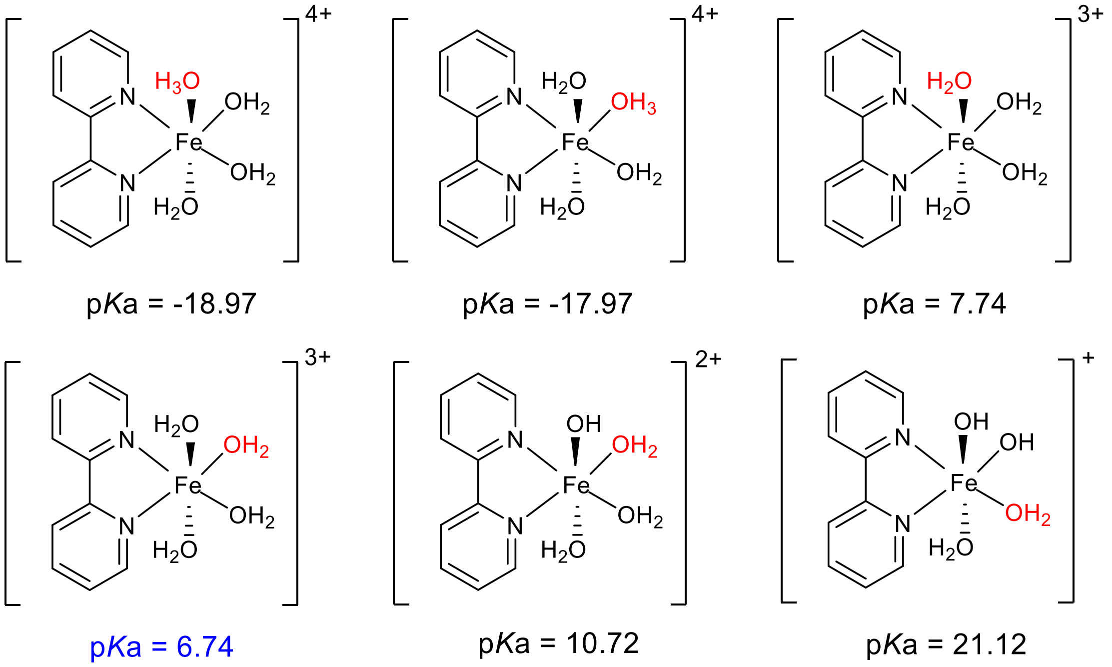


Figure S11. Calculated values of pK_a for bound water in possible Fe coordination sites in PMOF-RuFe(OH). As the reaction is conducted at pH ≈ 7, the initial structure of the iron species is assigned to $[(bpy)Fe(OH)(H_2O)_3]^{2+}$.

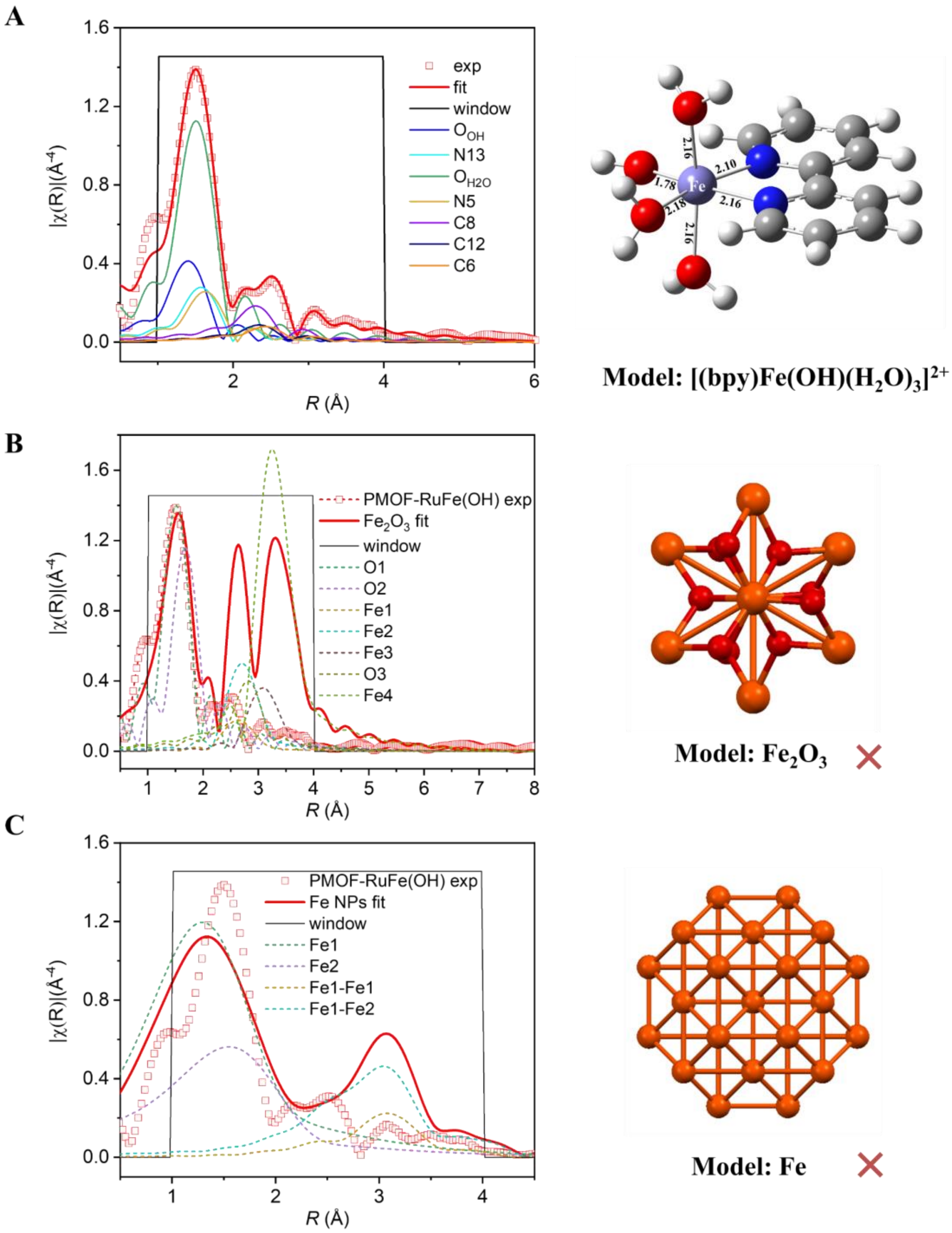


Figure S12. Fitting of EXAFs data for PMOF-RuFe(OH) in *R*-space and the scattering paths using different structural models: (A) $[(bpy)Fe(OH)(H_2O)]^{2+}$, (B) Fe_2O_3 and (C) metallic Fe. Although the distances of Fe-C8, Fe-C12 and Fe-N5 in the bipyridyl ring are similar to that of Fe···Fe in metallic Fe and Fe-O-Fe in Fe_2O_3, such fittings show notable differences to the experimental spectra (B and C).

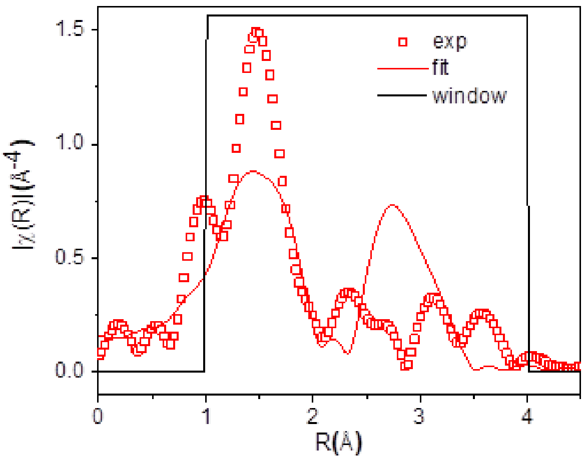
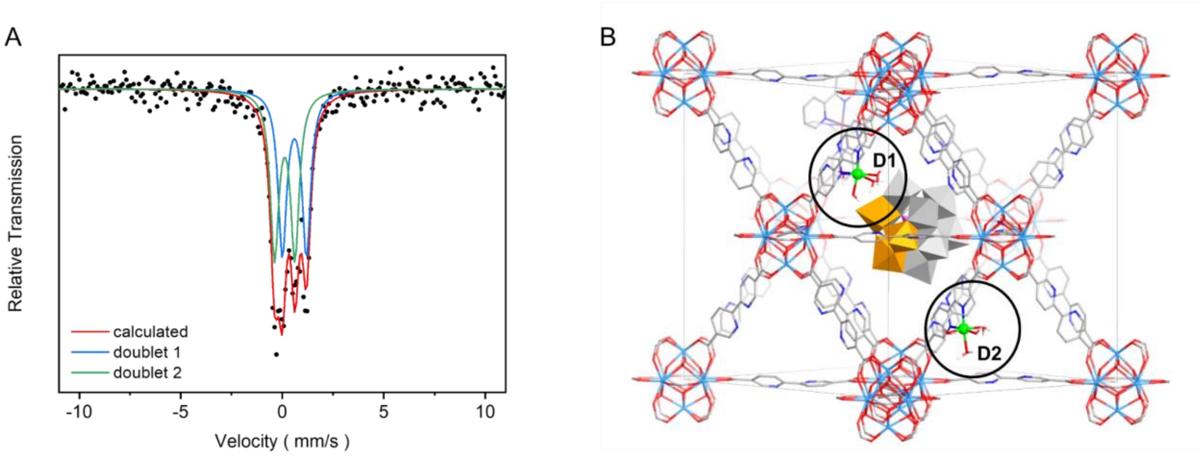


Figure S13. Fitting of EXAFs data for PMOF-RuFe(OH) in *R*-space using the combined models of $[(bpy)Fe(OH)(H_2O)]^{2+}$ and Fe_2O_3 in a ratio of ~1:1. The mismatch with R = 0.42853 excludes the presence of Fe_2O_3 due to a large discrepancy in the scattering of Fe-O-Fe at ~3 Å in *R*-space.



C

Component	Assignment	IS (mm/s)	QS (mm/s)	LW (mm/s)	Area (%)
Doublet 1	D1	0.11	1.18	0.51	50.5
Doublet 2	D2	0.59	0.19	0.50	49.5

Figure S14. (A) ^{57}Fe Mössbauer transmission spectrum measured at 293 K for PMOF-RuFe(OH) and the fitting to spectral components, $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ surrounded by POM (D1) and confined in the open cavity (D2) shown in (B). (C) Mössbauer parameters derived from the fitting of the data to give isomer shift (IS), quadrupole splitting (QS), line width (LW) and relative spectral area % for each component.

The ^{57}Fe Mössbauer spectrum of PMOF-RuFe(OH) was measured at room temperature to further characterise the active site. The spectrum was fitted to two doublets, and, based on the values of isomer shift (IS) and quadrupole splitting (QS) obtained, both are assigned to the Fe(III) species in $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ in an octahedral coordination sphere.[35,36] The similar values for QS suggest the same iron species with slightly different binding interactions, and a relatively low value for IS indicates a high *s* electron density at the ^{57}Fe nuclei.[37,38] Thus, the doublets, 1 and 2, are assigned, respectively in an approximate 1:1 ratio, to $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ close to $[PW_9V_3O_{40}]^{6-}$ POM (D1) and to $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ confined in an open cavity without a neighbouring POM (D2). No sextets or singlets were observed in the Mössbauer spectra suggesting the absence of zero-valent (metallic) iron.[39] No signal for iron oxides was detected either.[36] The Mössbauer spectroscopic analysis is in excellent agreement with EXAFS (Figures 2D) and XANES analyses (Figure S15).

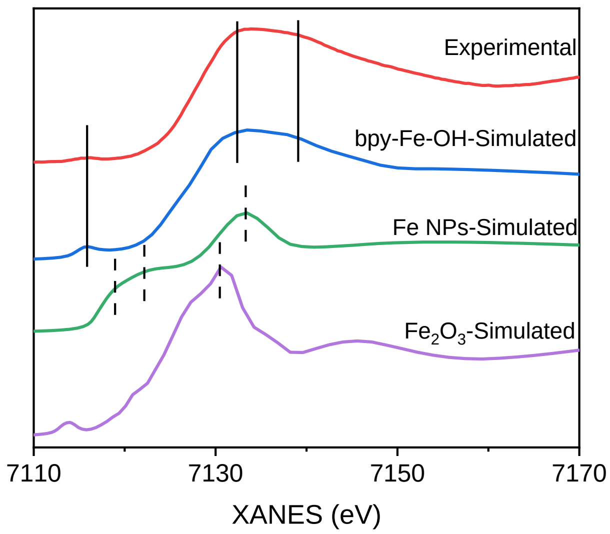


Figure S15. Fitting plots for XANES data using FEFF 10 code[40,41] for different Fe species compared to experimental results.

The XANES spectra are sensitive to the symmetry and the local coordination environment of the Fe(III) center. As shown in Figure S15, the simulated XANES spectra of $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ matches well with the experimental data, giving one pre-edge around 7116 eV and two peaks after the edge (7131-7139 eV). This is also consistent with the fitted EXAFS results shown in Figure 2D. In comparison, the characteristic absorption peaks of Fe nanoparticles (7116-7126 eV, 7133 eV) or Fe_2O_3 (7113 eV and 7130 eV) are absent in the experimental spectra, thus ruling out the presence of metallic iron or Fe_2O_3.

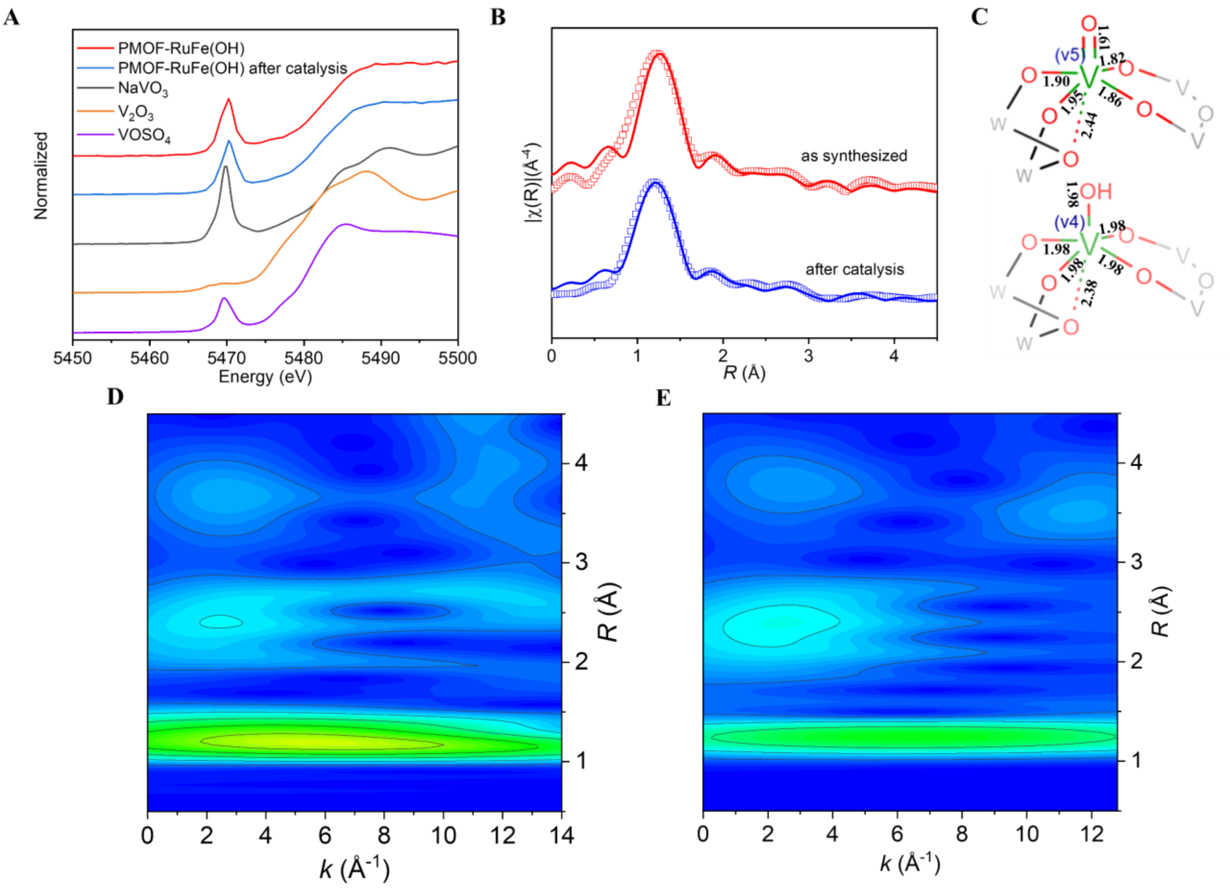


Figure S16. V K-edge XAS of PMOF-RuFe(OH). (A) XANES spectra confirming mixed states of V^{V} and V^{IV}. (B) EXAFS spectra and fitting in *R*-space (1-4 Å) showing the magnitude of Fourier transform (hollow squares) and real components (lines). Red: as synthesized; blue: after catalysis. The spectrum fits well with models for VO_6 (red line: 83.7% $PW_9V^{V}_3$ and 16.3% $PW9V^{IV}_3$; blue line: 78.1% $PW_9V^{V}_3$ and 21.9% $PW_9V^{IV}_3$). (C) Structural models for $PW_9V^{V}_3$ (top) and $PW_9V^{IV}_3$ (bottom). The models were optimized by DFT calculations, and are entirely consistent with the CIF file 168924 from the Cambridge Crystallographic Database. Wavelet spectra of PMOF-RuFe(OH) (D) as synthesized and (E) after catalysis.

NOTE: The most striking feature is the presence of an intense pre-edge (1s→3d) transition at 5469 eV in PMOF-RuFe(OH) due to the terminal oxo group (V=O). This oxo-ligand causes a distorted octahedral symmetry around V centers in PW_9V_3. Thus, the usually dipole-forbidden 1s→3d transition for a site with the inversion symmetry becomes dipole-allowed due to a combination of stronger 3d→4p mixing and overlap of the metal 3d orbitals with the 2p orbitals of the oxygen ligand gaining substantial intensity.[42] Since electronic quadrupole transitions are much weaker than electronic dipole transitions, the intensity of the pre-edge feature depends on the site symmetry of V. When 3d–2p mixing is forbidden for a local symmetry with an inversion center (e.g. V_2O_3 for regular octahedral VO_6 units), the lowest intensity is expected, whereas for tetrahedral geometry, the intensity can be comparable with an adsorption edge. Therefore, V-XANES spectrum of PMOF-RuFe(OH) exhibits a pre-edge absorption with moderate-intensity.

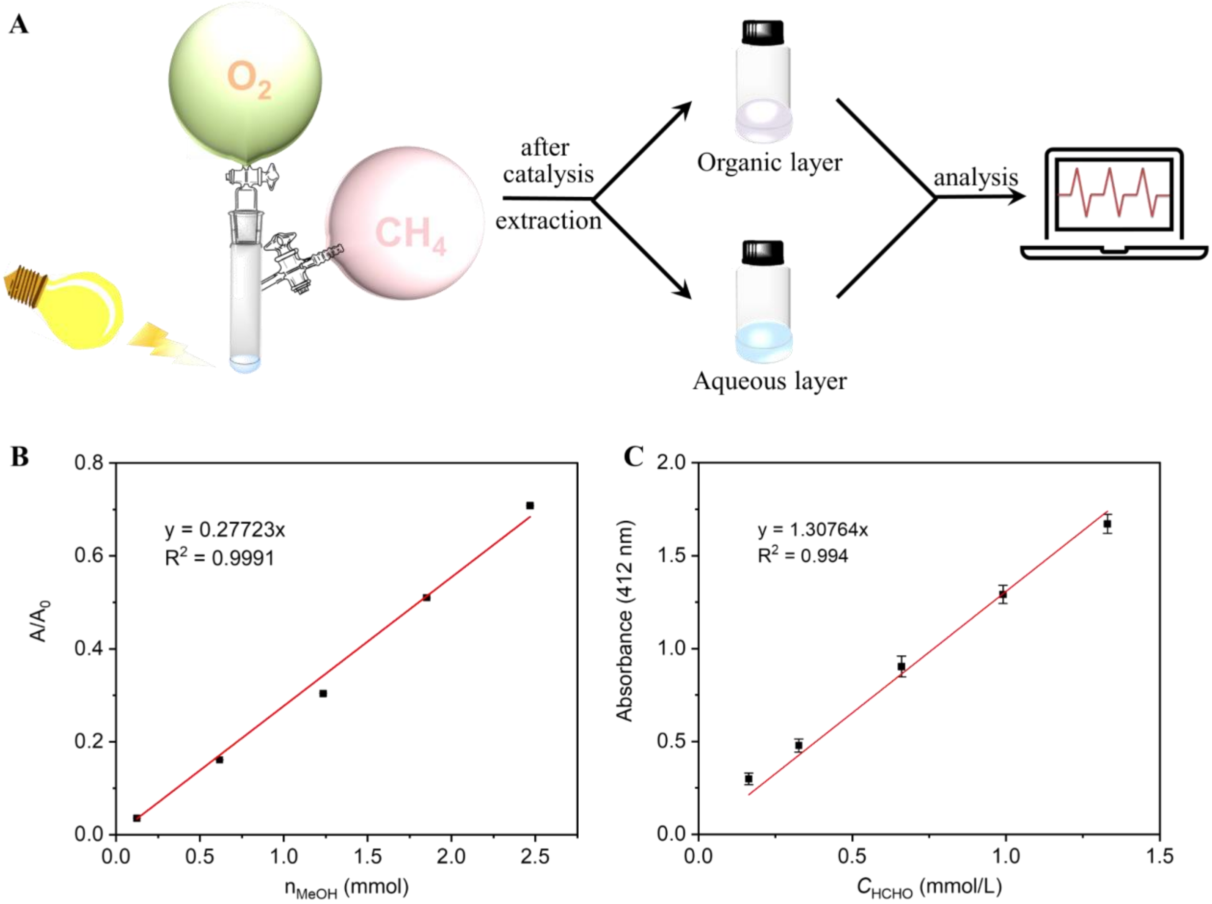


Figure S17. Photocatalytic system. (A) Schematic view of the experimental setup. (B) Standard curves for GC quantification for production of CH_3OH using 100 μL of 1,4-dioxane as the internal standard. (C) Calibration curve for the quantification of HCHO by colorimetric method.[43] All error bars represent the standard deviation from three independent tests. Pentane-2,4-dione reacts with HCHO to form the corresponding C=N Schiff base compound, which shows characteristic adsorption band in the Uv-vis spectrum. Typically, 100 mL a solution was prepared by dissolving 15 g of ammonium acetate, 0.3 mL of HOAc, and 0.2 mL of pentane-2,4-dione in water. Then, 0.5 mL of liquid product was mixed with 0.5 mL of this chromogenic agent solution and 2.0 mL of water. The mixed solution was maintained at 35 °C in a water bath and the UV-vis absorption spectrum measured. In the experiments described herein, HCHO was not detected. However, in parallel experiments using other POMs other than PW_9V_3 (for example, PW_{12}, $PW_{11}V$, $PW_{10}V_2$, PW_9V_3 and PW_8V_4 and PW_8V_4) HCHO and other oxidation products were detected depending on the precise POM used.

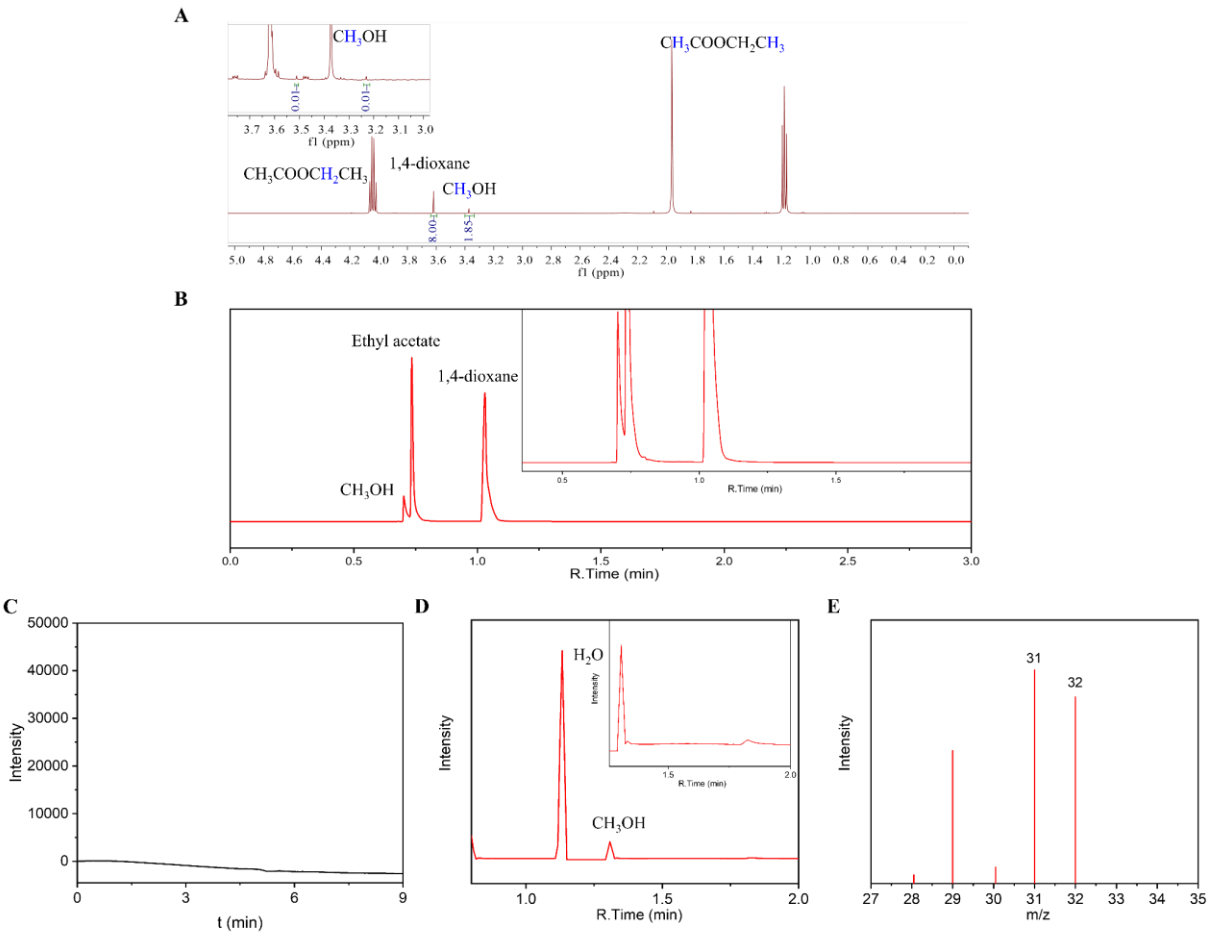


Figure S18. Analysis of the liquid phase from the photo-oxidation of CH_4 for 20 h over PMOF-RuFe(OH) in batch mode. (A) ^{1}H NMR spectrum and (B) GC (FID) trace of the liquid product extracted with ethyl acetate. 1,4-Dioxane was used as internal standard; FID, column: HP-5. (C) The HPLC trace of the solution in the aqueous layer as monitored at 210 nm by a UV-Vis detector with 5% H_2SO_4 (aq) as the mobile phase (HCO_2H, t = 7.7 min), confirming the absence of formic acid. (D) GC-MS analysis of the raw liquid solution and (E) the corresponding mass spectrum of CH_3OH product. A combination of GC (FID and MS), NMR and HPLC analyses of the liquid solution confirms a selectivity for CH_3OH of >99% with only trace amounts of other oxygenates and no formic acid detected.

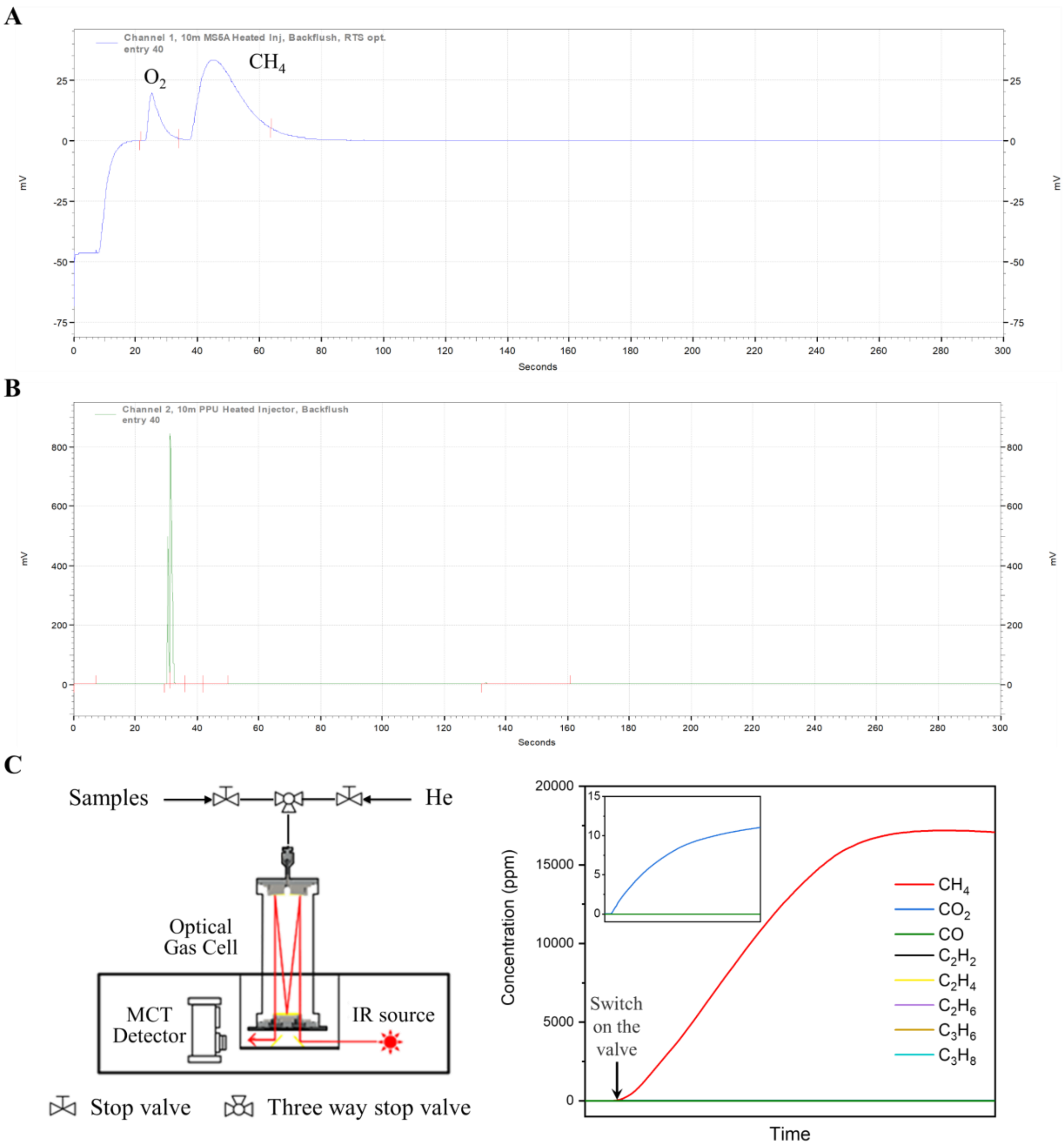
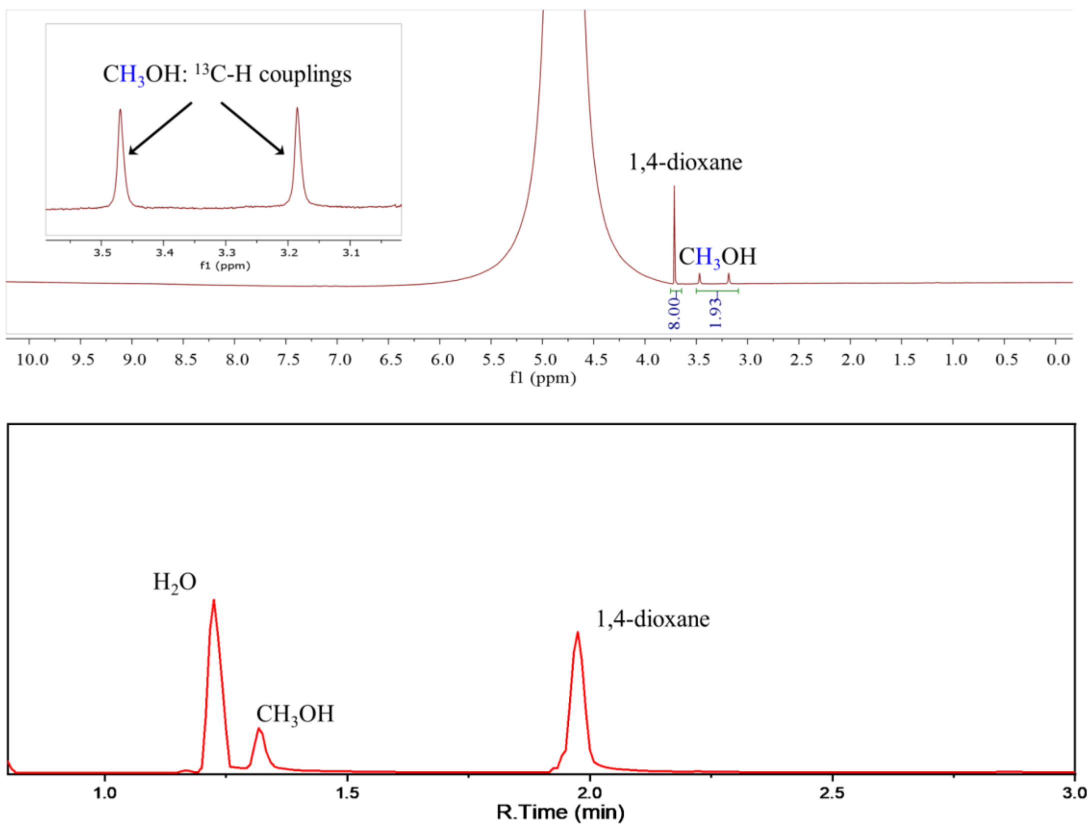


Figure S19. Analysis of the gaseous product for photo-oxidation of CH_4 after a 20h run using PMOF-RuFe(OH) as catalyst in batch mode. GC trace of the headspace gas detected by TCD (A) Channel 1, column: molecular sieve 5A and (B) Channel 2, column: PLOT-U. The peaks were identified as the reactants O_2 and CH_4. (C) FTIR analysis equipment and concentration of the headspace gas monitored by FTIR. All these confirm that only trace amounts of CO_2 were generated as a byproduct.

A

B



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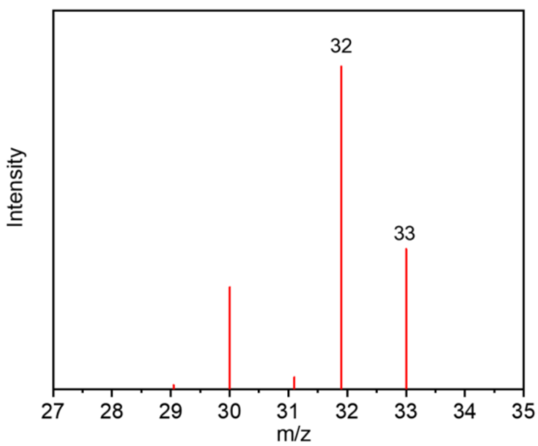


D

Assignments of fragment peaks	
m/z	Assignments
33	$^{13}CH_3OH^+$
32	$^{13}CH_3O^+$ or $^{13}CH_2OH^+$
31	$^{13}CH_2O^+$
30	$^{13}CHO^+$
29	$^{13}CO^+$

Figure S20. Analysis of the liquid composition for photo-oxidation of labeled $^{13}CH_4$ using PMOF-RuFe(OH) as catalyst in batch mode. (A) ^{1}H NMR spectrum showing the yield of CH_3OH and (B) GC-MS trace of the liquid product. (C,D) The corresponding mass spectrum and the assignments of peaks for the product.

NOTE: The peak at m/z = 33 is attributed to $^{13}CH_3OH$, and the yield is the same as that using $^{13}CH_4$ as the feed gas, further demonstrating that CH_3OH is the sole product.

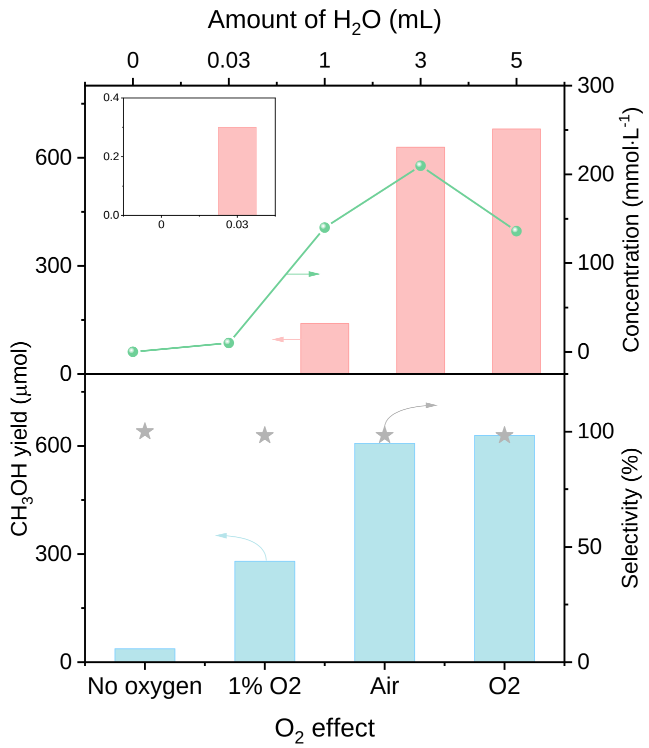


Figure S21. Dependences of yield of CH_3OH over PMOF-RuFe(OH) in batch mode on different amounts of water and different pressures of O_2.

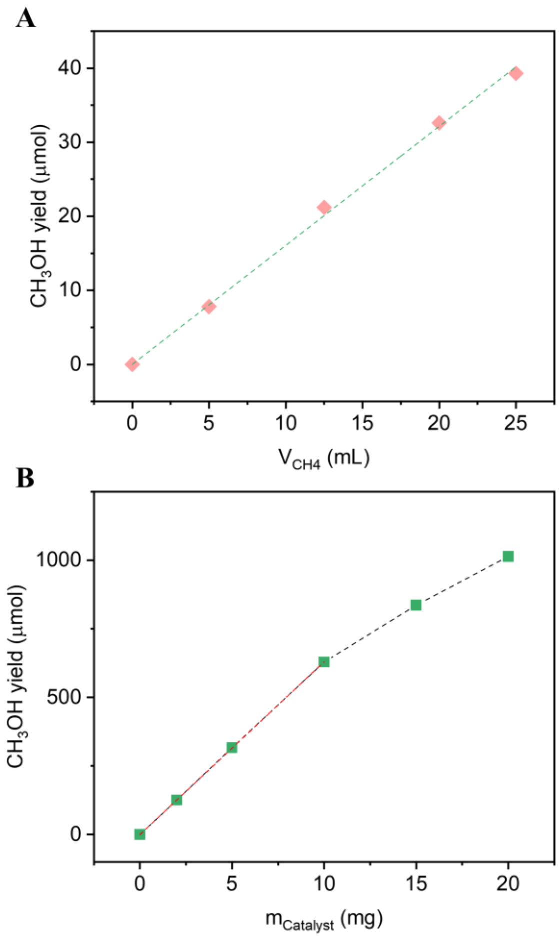
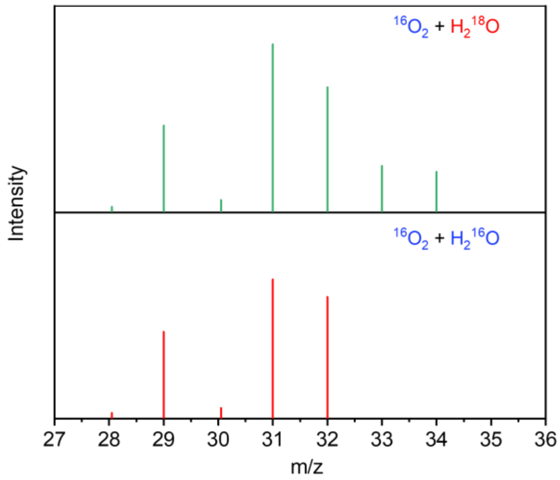


Figure S22. Dependence studies of the yield of formation of CH_3OH over PMOF-RuFe(OH) (A) versus the amount of CH_4, and (B) versus catalyst. Reaction conditions: 10.0 mg of catalyst, and mixed CH_4 (10%-50% by volume) and air (10% by volume containing 21% of O_2) balanced in N_2 were dosed into a 50 mL Schlenk bottle containing 3 mL of H_2O at 298 K. Visible light irradiation was used for 20 h. Ratios of feed gas were selected by considering the upper explosion limited. As the amount of input gases is decreased by a factor of >20, the corresponding yield of CH_3OH (0-40 μmol) is reduced compared to a typical batch experiment (629±68 μmol)]. The yield of CH_3OH deviates from the linear relationship when more than 10.0 mg of catalyst is used due to over-oxidation.

NOTE: Schlenk bottles were used in this specific test instead of balloons (which were used in the batch experiments) to quantify precisely the volume of input gas. Compared with a typical batch experiment, reduced amounts of reactant gases (CH_4 and O_2) by a factor of >20 were used to evaluate the dependence of yield of CH_3OH versus volume of CH_4 (10%-50%) and catalyst loading (0-20.0 mg). Compressed air (containing 21% of O_2) was used instead of pure O_2 to avoid the explosive limit of CH_4. N_2 acts as a balancing gas to retain the same initial pressure.



Assignments of fragment peaks	
m/z	Assignments
34	$CH_3{}^{18}OH^+$
33	$CH_3{}^{18}O^+$, $CH_2{}^{18}OH^+$
32	$CH_2{}^{18}O^+$, $CH_3{}^{16}OH^+$
31	$CH^{18}O^+$, $CH_3{}^{16}O^+$, $CH_2{}^{16}OH^+$
30	$C^{18}O^+$, $CH_2{}^{16}O^+$
29	$CH^{16}O^+$
28	$C^{16}O^+$

Figure S23. GC-MS spectra of CH_3OH generated over PMOF-RuFe(OH) with $^{16}O_2$ + $H_2{}^{18}O$ or $^{16}O_2$ + $H_2{}^{16}O$ in photocatalytic oxidation of CH_4 and the assignments of fragment peaks. Analysis of the spectra shows 75% yield of $CH_3{}^{16}OH$ (m/z = 32, 29) and 25% yield of $CH_3{}^{18}OH$ (m/z = 34,33), indicating both O_2 and H_2O are involved in the formation of CH_3OH product.

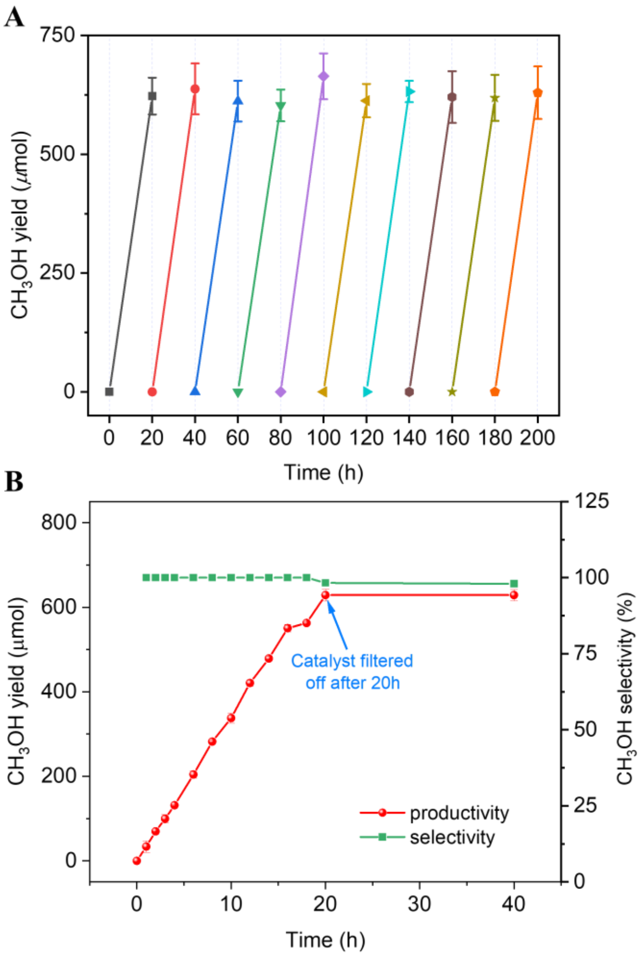


Figure S24. Stability tests of PMOF-RuFe(OH) catalyst for photo-oxidation of CH_4 to CH_3OH. (A) Recycling experiments in batch mode of photo-oxidation of CH_4 for 10 runs under standard reaction conditions. The experiments confirm that the production of CH_3OH is strongly light-dependent. (B) Removal of PMOF-RuFe(OH) during photo-oxidation of CH_4 to CH_3OH confirms heterogeneous catalysis and no measurable homogeneous activity. The catalyst was removed after 20 h and no further formation of CH_3OH was observed.

ICP-OES analysis suggests <0.1%, <0.3% and 0.05% leaching of Fe, Zr and W, respectively, into the solution after 20 h; these values are within the detection limits of the instrument. We have performed additional reactions at 50 h, 100 h and 200 h in batch mode, and the ICP-OES analysis of the leaching of Fe/Zr/W into the solution was 0.12%/0.30%/0.06%, 0.47%/0.31%/0.05% and 1.33%/0.34%/0.08%, respectively. While very low levels leaching was observed for the reaction over the first 100 h, more leaching of Fe (1.3%) was observed after 200 h. This is in contrast with most literature reports, where stability is studied typically over 2-4 h only; this therefore demonstrates the high stability of the present system.

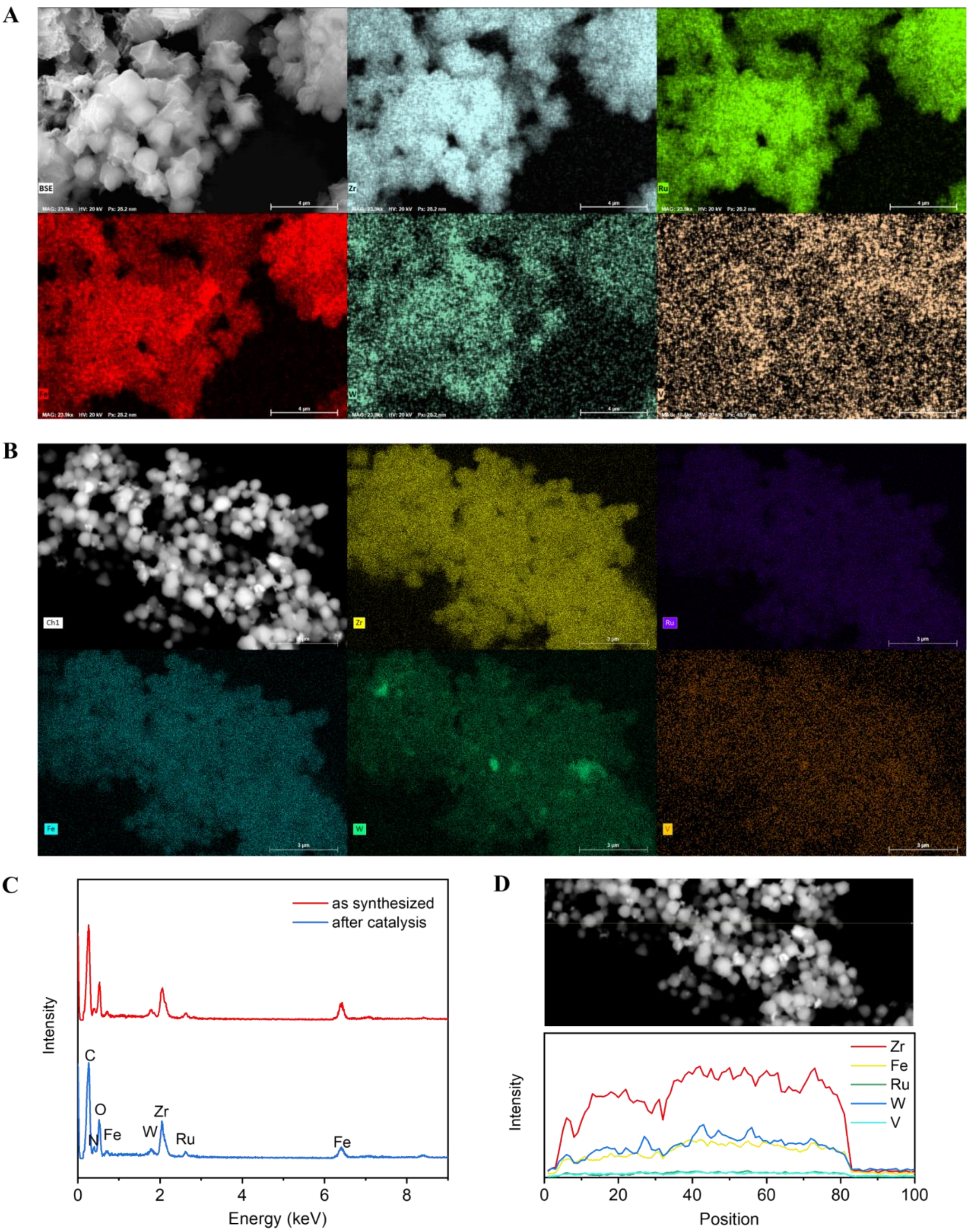


Figure S25. SEM and EDX images of (A) fresh and (B) used PMOF-RuFe(OH) catalysts after a 20 h run. (C) EDX spectra. (D) EDX line scans of the used catalyst. EDX studies indicate that Ru, Fe and PW_9V_3 are dispersed homogeneously throughout the MOF particles before and after the catalysis.

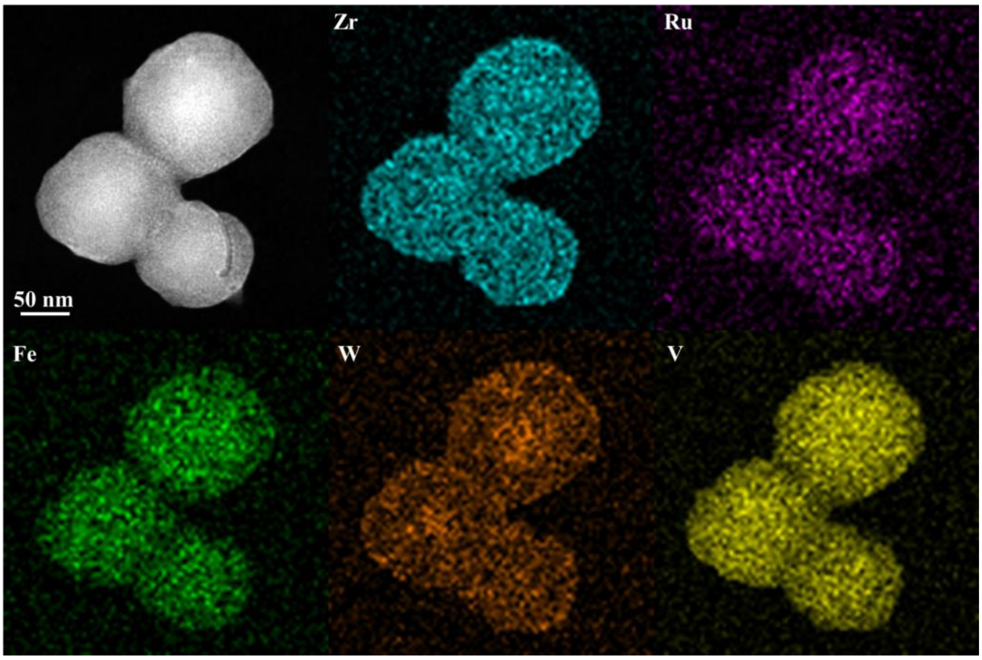


Figure S26. TEM (High-Angle Annular Dark-Field, HAADF) and EDX-mapping images (Zr, cyan; Ru, purple; Fe, green; W, orange and V, yellow) of used PMOF-RuFe(OH) catalyst after a 20 h run.

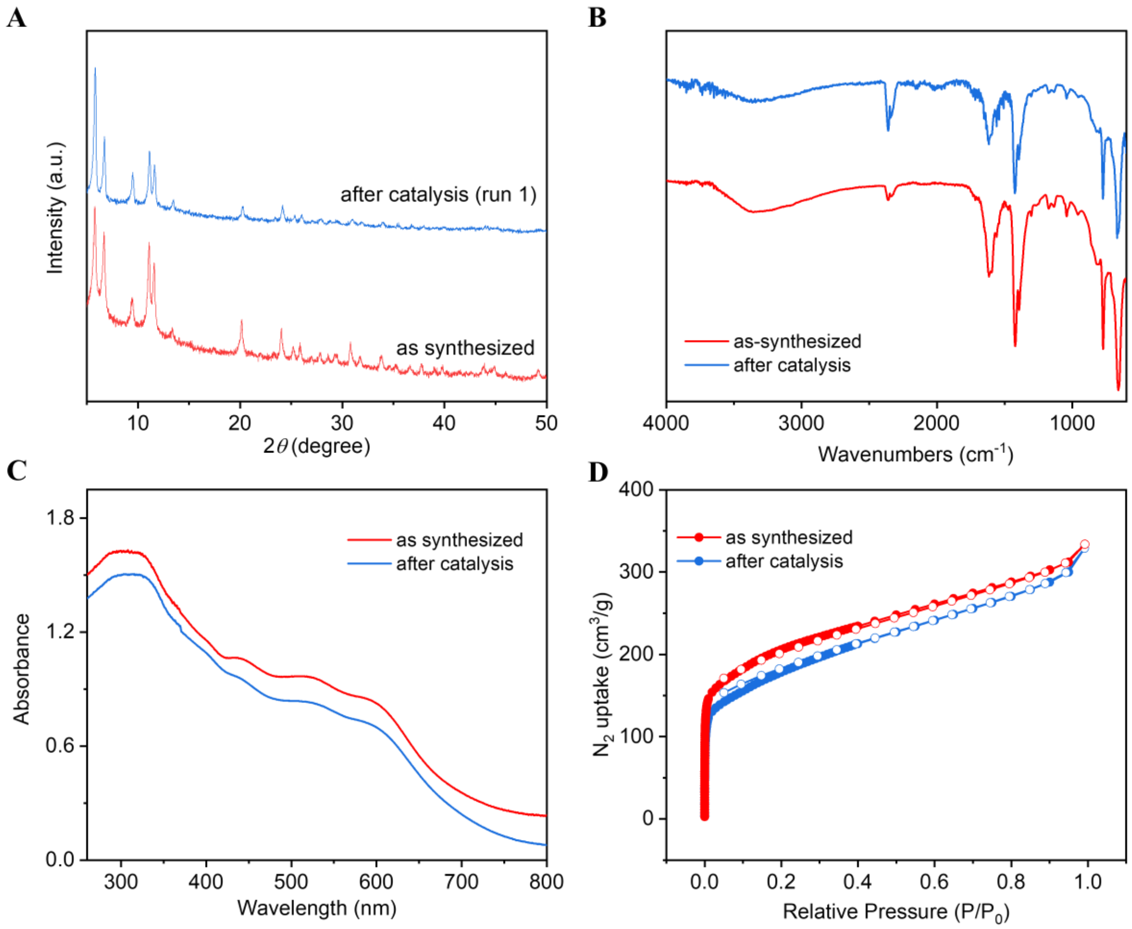


Figure S27. Characterisation of used PMOF-RuFe(OH) catalyst after a 20 h run. (A) PXRD patterns. (B) FTIR transmission spectra (spectra are offset vertically for clarity). (C) UV-vis spectra. (D) N_2 adsorption isotherms at 77 K. BET surface areas: as-synthesised sample 684 *vs* used sample 610 $m^2 \cdot g^{-1}$. The slight decrease in porosity is likely due to the incomplete elution of substrates and intermediates (*e.g.*, methyl and hydroxyl radicals) bound to the structure.

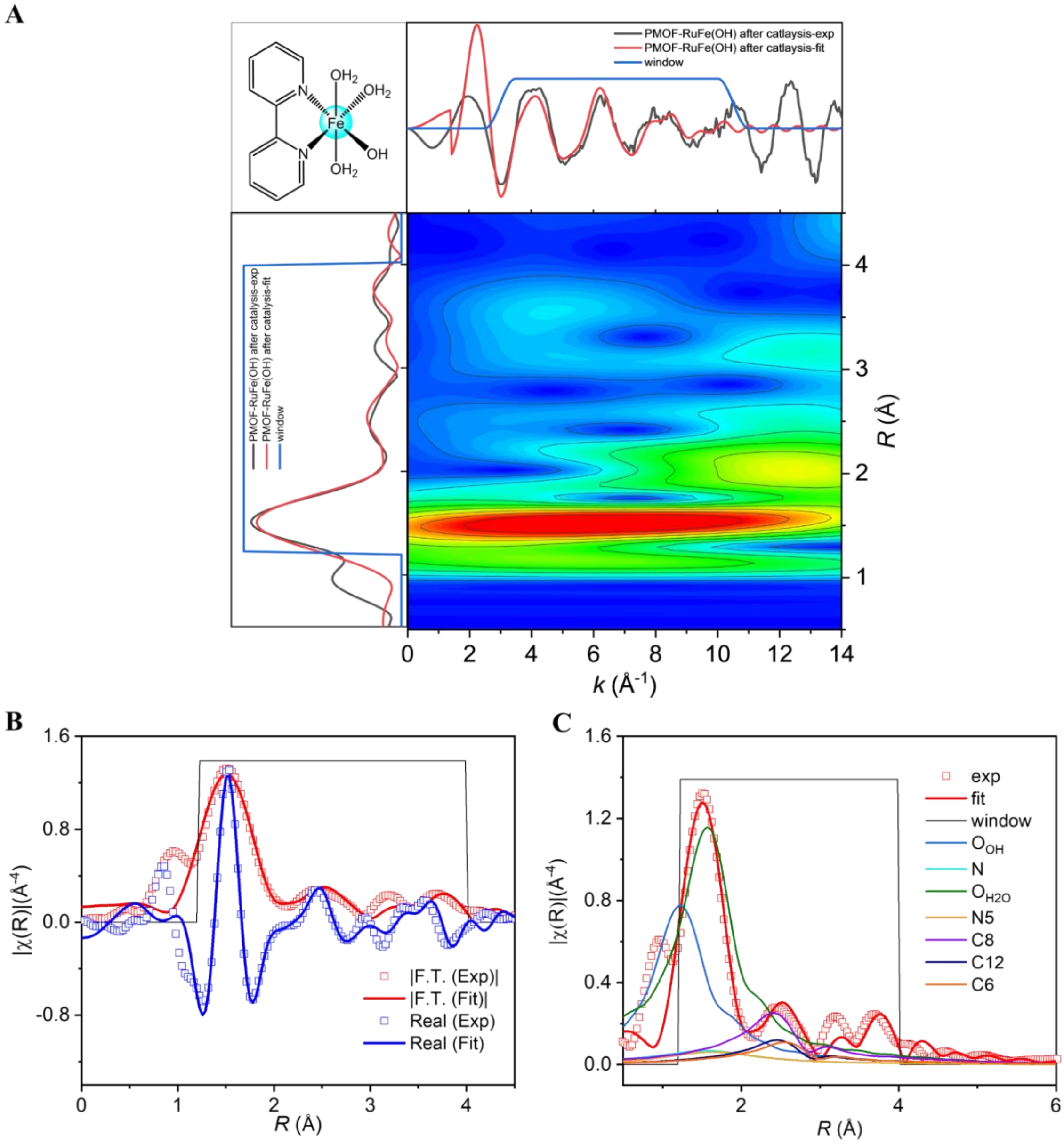


Figure S28. EXAFS analysis at the Fe K-edge adsorption of PMOF-RuFe(OH) after 20 h of catalysis. (A) Wavelet transform spectra, (B) experimental EXAFS spectra and fitting in *R*-space showing the magnitude of Fourier transform, and (C) the scattering paths. The spectrum was satisfactorily fitted to the $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ model and indicated the absence of metallic Fe species and/or metal oxides and nearly identical Fe environment to fresh PMOF-RuFe(OH).

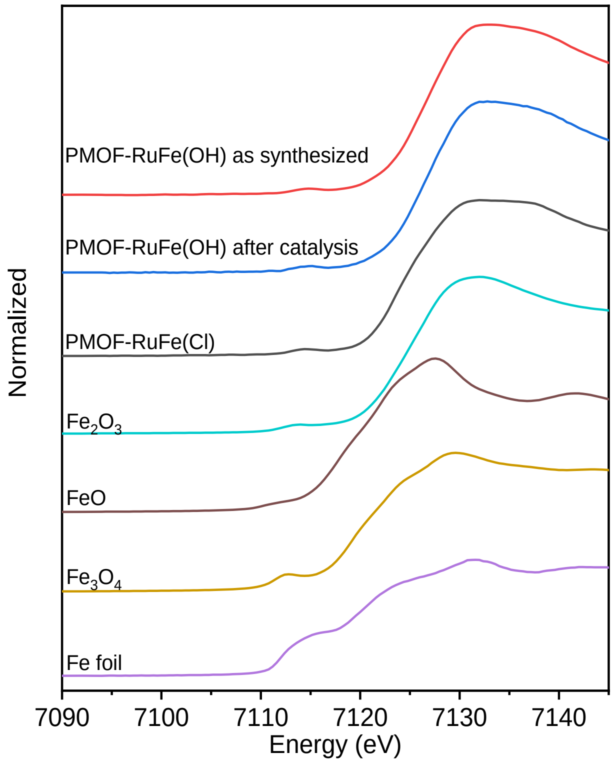


Figure S29. Normalized Fe K-edge XANES spectra of used PMOF-RuFe(OH) showing the presence of Fe in the +3 oxidation state by comparing the energies of the pre-edge peaks to reference spectra.

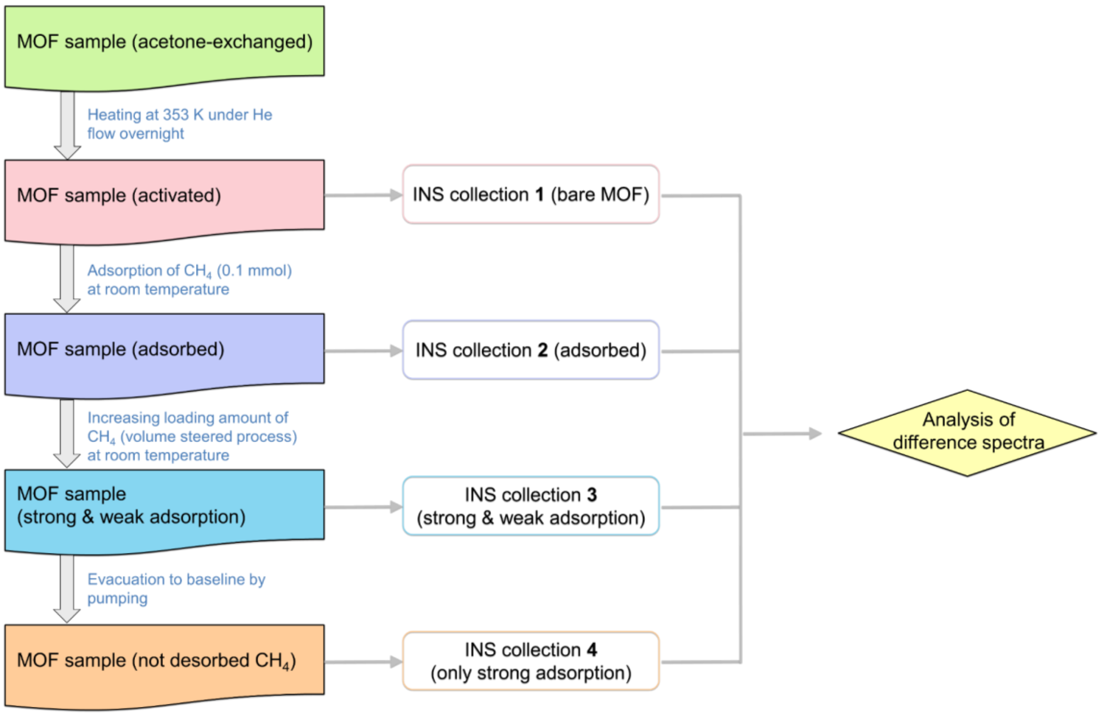


Figure S30. Schematic view of the *in situ* INS experiment and data collection.

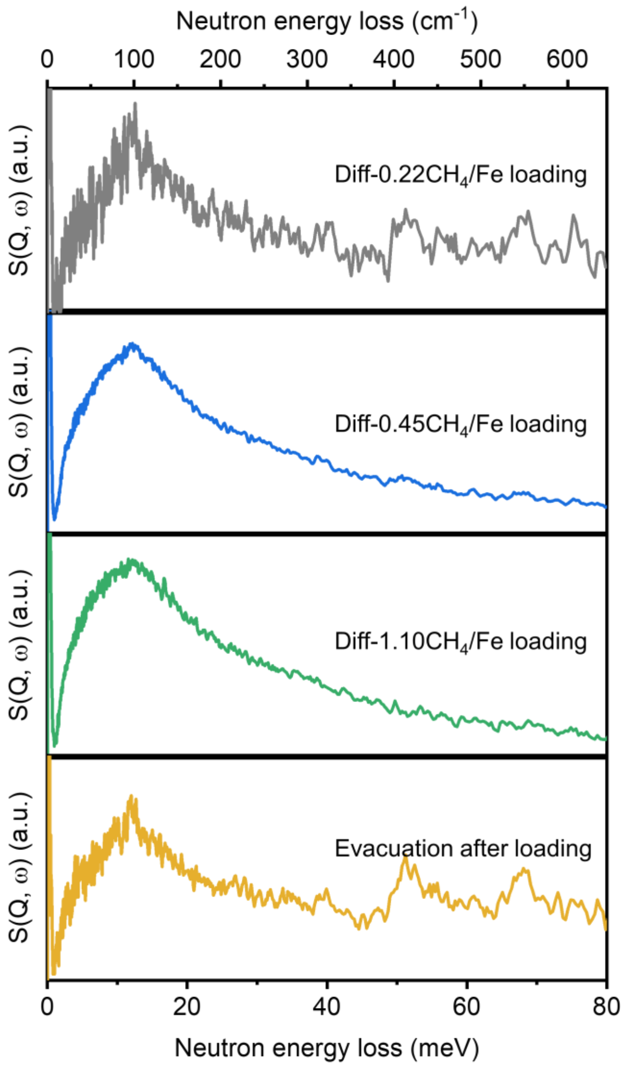


Figure S31. Difference INS spectra of different CH_4-loaded PMOF-RuFe(OH) (0.22 CH_4/Fe, 0.45 CH_4/Fe and 1.10 CH_4/Fe) and evacuation to baseline to remove free CH_4.

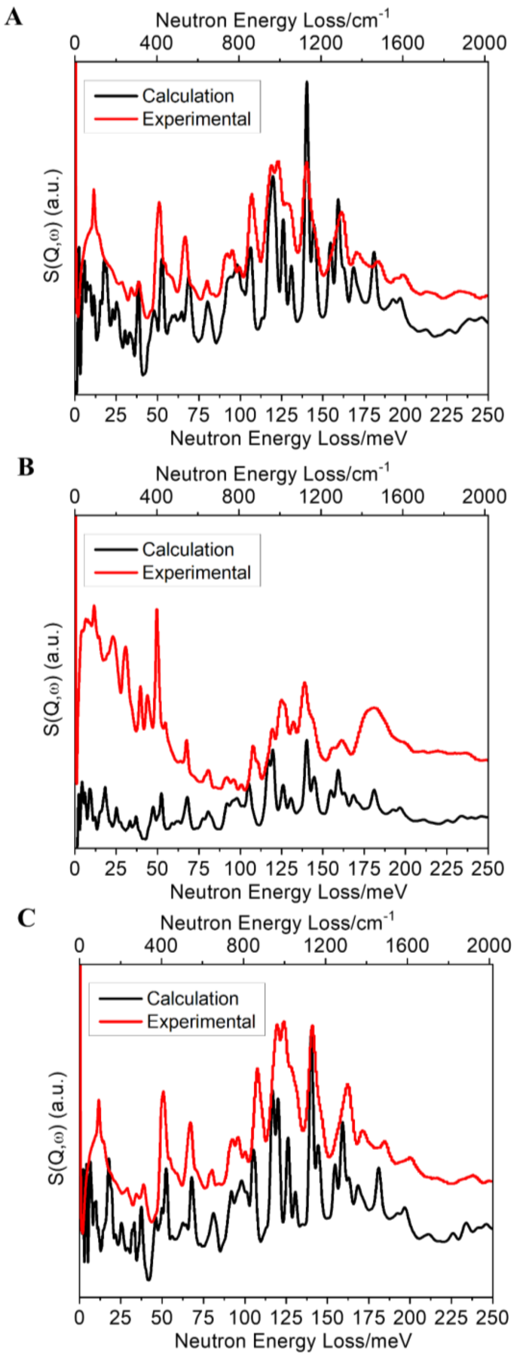


Figure S32. Comparison of the experimental and calculated INS spectra for the bare MOFs. (A) PMOF-RuFe(OH), (B) PMOF-Ru, and (C) MOF-RuFe(OH). The models derived from structural analysis were used for initial DFT modelling, where the positional disorder cannot be fully accounted for due to the periodic boundary conditions. This partially accounts for the discrepancies observed between calculated and experimental INS plots. However, these discrepancies do not affect the reliability of the key findings from the DFT modelling that gives assignments of Peak II at 50-55 meV and Peak III at 65-70 meV (Fig. 4b). In Fig. 4b - left panel of Peak II: vibration of OH with H moving perpendicular to O-H and out of the [(bpy)Fe-O_{OH}] plane; right panel of Peak II: wagging of H_2O located out of the [(bpy)Fe-O_{OH}] plane; Left panel of Peak III: wagging of H_2O located in the [(bpy)Fe-O_{OH}] plane plus vibration of OH with H moving perpendicular to O-H within the [(bpy)Fe-OOH] plane; right panel of Peak III: rocking of H_2O located out of the [(bpy)Fe-O_{OH}] plane. These peaks are clearly reproduced and can be unambiguously attributed to the modes assigned. Experimental spectra shown are after subtraction of the spectrum due to the empty cell.

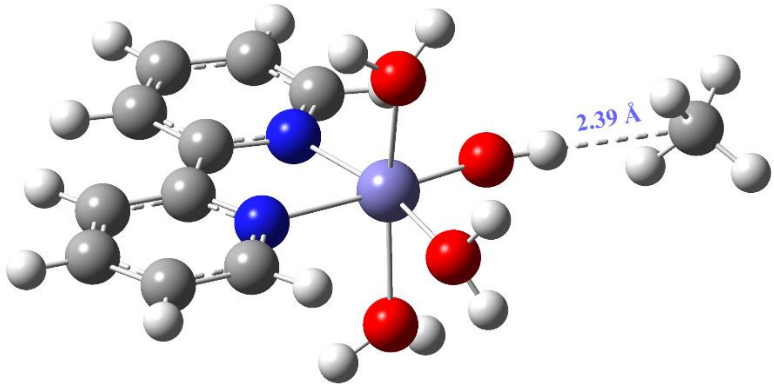


Figure S33. View of the structural model for CH_4-loaded $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ in PMOF-RuFe(OH). The binding site for CH_4 is in the form of [Fe-OH···CH_4] with a H_{OH}−C_{CH4} distance of 2.39 Å. The model was optimized by DFT calculations and is entirely consistent with the results from INS.

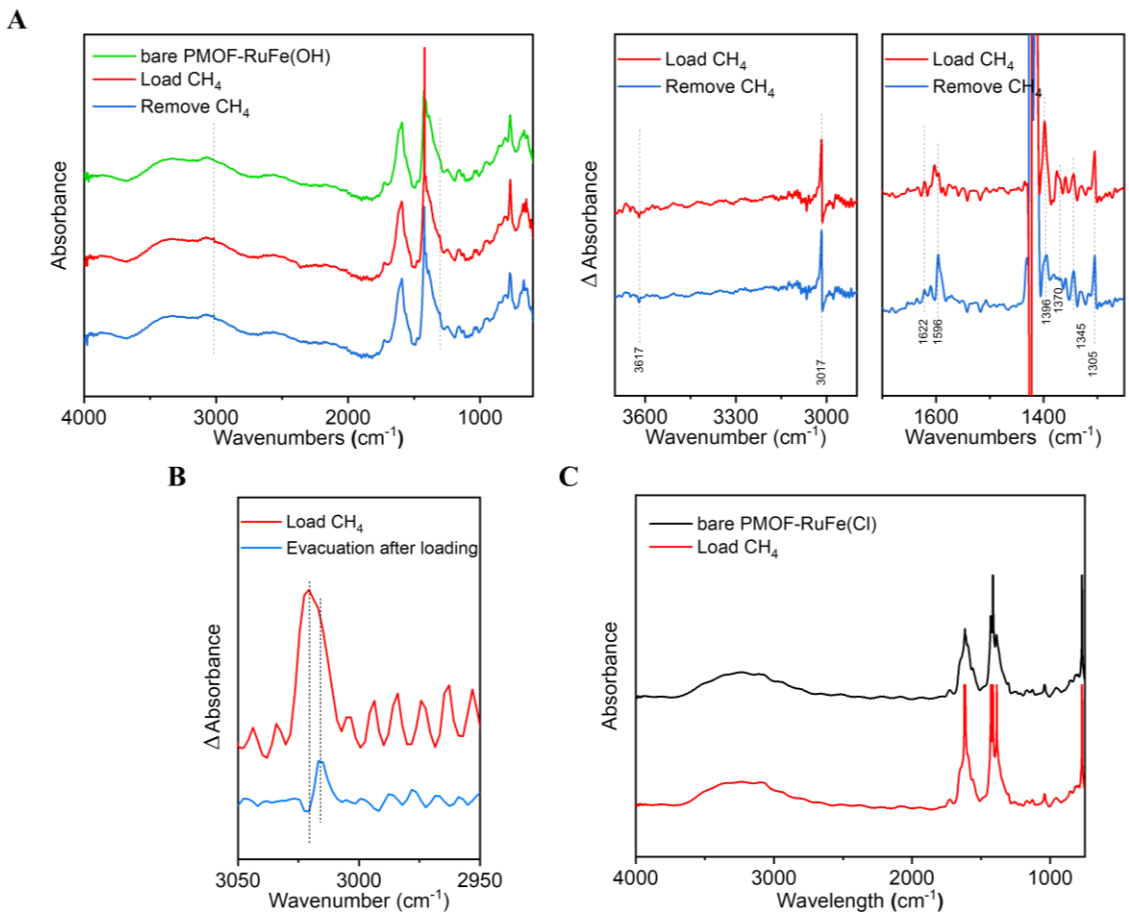


Figure S34. *In situ* IR spectra for (A,B) PMOF-RuFe(OH) and (C) PMOF-RuFe(Cl) as a function of CH_4-loading at room temperature. The difference spectra for PMOF-RuFe(OH) upon loading and removal of CH_4 obtained by purging with N_2 are shown on A (right). The red curve of ΔAbsorbance is the spectrum of CH_4-loaded PMOF-RuFe(OH) minus the spectrum of bare PMOF-RuFe(OH).The blue curve of ΔAbsorbance is the spectrum of CH_4-loaded PMOF-RuFe(OH) minus the spectrum of free CH_4 and minus the spectrum of the bare PMOF-RuFe(OH). The difference spectra on evacuation of the CH_4-loaded PMOF-RuFe(OH) are shown on B. The IR band at 3017 cm^{-1} is assigned to adsorbed CH_4 retained upon removal of excess CH_4, confirming the interaction between {Fe-OH} sites and adsorbed CH_4 molecules.

NOTE: Generally, the characteristic absorption band of C–H bond occurs in three regions: 3010–2700 cm^{-1}, 1475–1300 cm^{-1} and 1000–650 cm^{-1}. Since 1000–650 cm^{-1} falls into the fingerprint region, there are many influencing factors that are too complex to give an unequivocal identification. Thus, this study focused on the absorption band in the other two regions.

On loading of CH_4 into PMOF-RuFe(OH), the characteristic C-H stretching and bending modes at 3017, 1412 and 1305 cm^{-1} appear immediately. These bands are slightly shifted compared with that of gaseous CH_4 (3020 and 1302 cm^{-1}).[44] The red-shift of the C−H stretching peak indicates a decrease in C−H bond strength, resulting from the formation of hydrogen bonding.[45] For the bands at 1475–1300 cm^{-1}, although the vibrations of the framework of MOF (aromatic ring stretching vibrations) and $-CH_3/CH_2$ groups overlap, the difference spectra show new peaks at 1345, 1370 and 1396 cm^{-1} which can be assigned to the scissoring vibration, symmetrical vibration and deformational mode of $-CH_3$ groups, respectively.[46] Moreover, a negative peak at 3617 cm^{-1}, assigned to the O-H stretch of the hydroxyl group, and several positive peaks in the region of 1587-1617 cm^{-1}, assigned to H−O−H bending modes, are observed. This suggests that the adsorbed CH_4 forms hydrogen bonds to the hydroxyl group, leading to a decrease in concentration and strength of the O-H bond. Additionally, the absence of vibrational modes at 1000-1200 cm^{-1} that could be ascribed to C−O stretching vibration from a Fe-OH…CH_4 moiety or to a C−H symmetric rocking of OCH_3 (formed *via* direct cleavage of CH_4 to form Fe-OCH_3), indicates that the adsorption of CH_4 is not through a direct C···O interaction. Interestingly, upon flowing dry N_2 over the CH_4-loaded PMOF-RuFe(OH), all bands assigned to adsorbed CH_4 molecules are retained, in excellent agreement with the relatively strong host-guest interactions as observed in INS and the isotherm experiments.

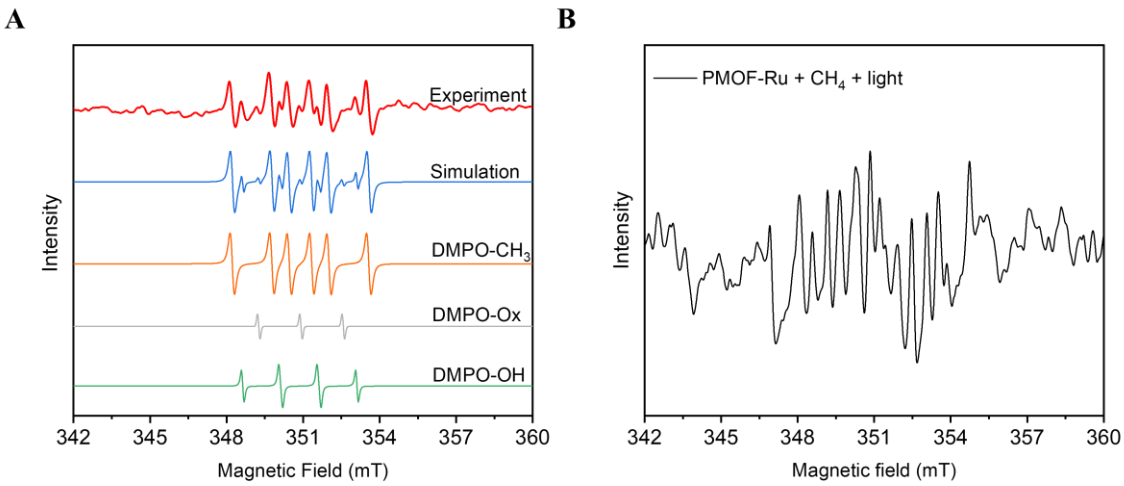


Figure S35. X-band (9.85 GHz) EPR spectroscopic spin-trapping experiments. (A) EPR spectrum from DMPO spin-trapping experiment of PMOF-RuFe(OH) in the presence of CH_4 and O_2 after irradiation; simulated spectra for individual DMPO-radical adducts detected and the weighted sum (parameters in Table S12). The absence of hydrogen (•H) adducts (DMPO-H) eliminates the possibility that dissociation of the C−H bond proceeds *via* a simple homolytic cleavage process. (B) EPR-trapping spectrum of PMOF-Ru with DMPO, CH_4 and light. The signal for •DMPO-CH_3 signals are not observed confirming that Fe(OH) sites are responsible for the activation of CH_4.

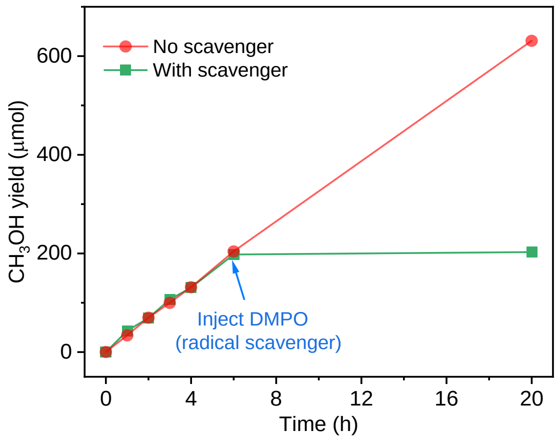


Figure S36. Study of the effect on catalysis of radical scavenger (DMPO) on the yield of CH_3OH over PMOF-RuFe(OH). On injection of excess of DMPO, no CH_3OH product was formed indicating a radical mechanism during catalysis.

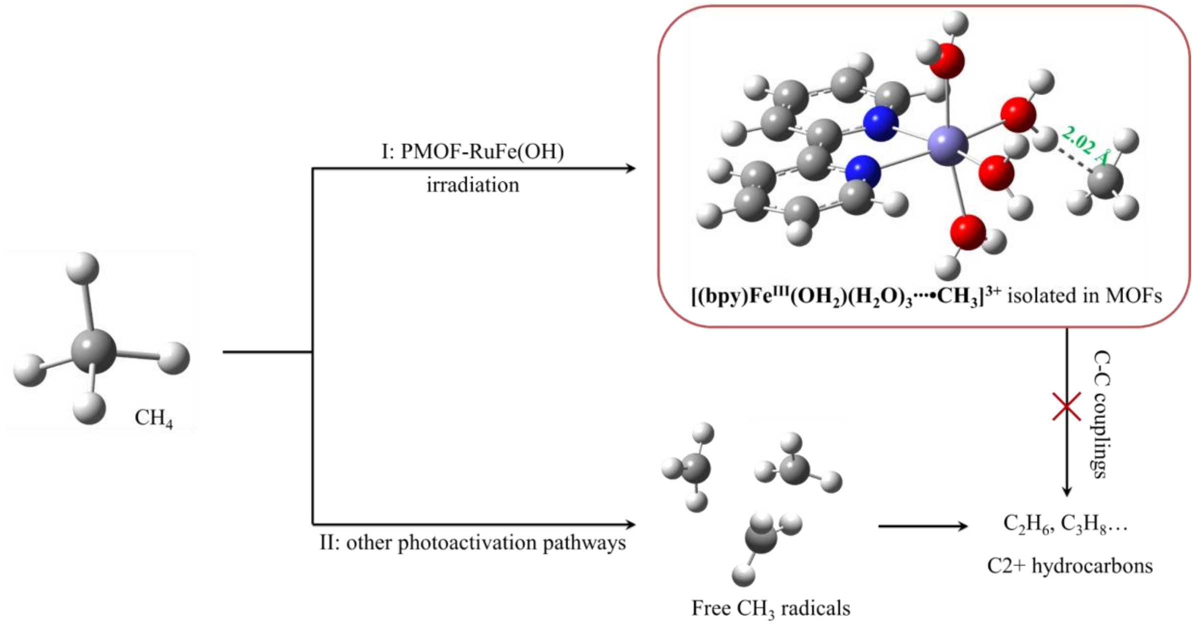


Figure S37. Comparison of the photo-activation of CH_4 to methyl radicals over PMOF-RuFe(OH) (I) and conventional catalysts (II).

NOTE: In pathway I catalysed by PMOF-RuFe(OH), the CH_4 molecules are first adsorbed onto iron-hydroxyl sites to form [Fe-OH···CH_4] intermediates and then photo-activated to [(bpy)Fe(OH_2)(H_2O)$_3$···•CH_3]$^{3+}$ ([Fe(H_2O) ···•CH_3]) (shown in red box) with H_{H2O}···C_{CH3} of 2.02 Å (based upon DFT calculations). In this photo-activated process, electron-transfer retains the charge of the [(bpy)Fe(OH_2)(H_2O)$_3$] species, as confirmed by EPR spectroscopy that shows no change in Fe^{III} signals (Figure S40). Also, [Fe(H_2O) ···•CH_3] species are isolated due to immobilization within the MOF platform, thus inhibiting two or more radicals coupled to give C_2+ hydrocarbons, which is observed in most photocatalytic systems (pathway II). The mono-iron-hydroxyl sites that bind CH_4 and •CH_3 intermediates are remote from one another and confined within PMOF-RuFe(OH) and therefore give a higher selectivity of C_1 products.

DFT calculations were used to study the spin population of [Fe-OH···CH_4] and [Fe(H_2O) ···•CH_3]. The initial spin density of CH_4 is 0, and upon activation, an increase of spin distribution is observed on the C-centre of CH_3 species (0.951), indicating formation of a •CH_3 radical. The spin density on the Fe^{III} centre shows little change before and after activation (Tables S13-S16). This is consistent with the *in situ* EPR spectroscopic results, which did not detect any Fe^{IV} species (Figure S40).

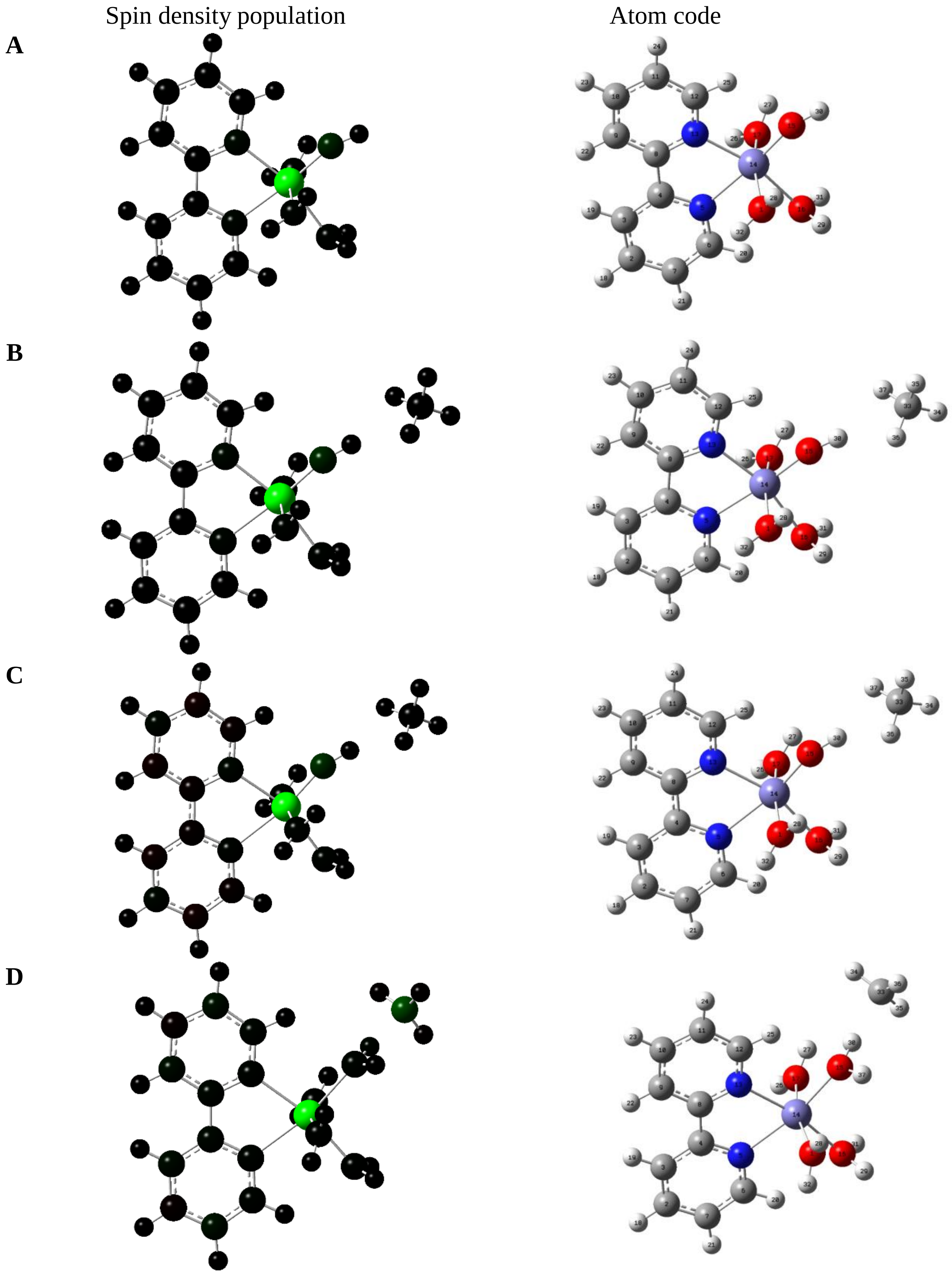
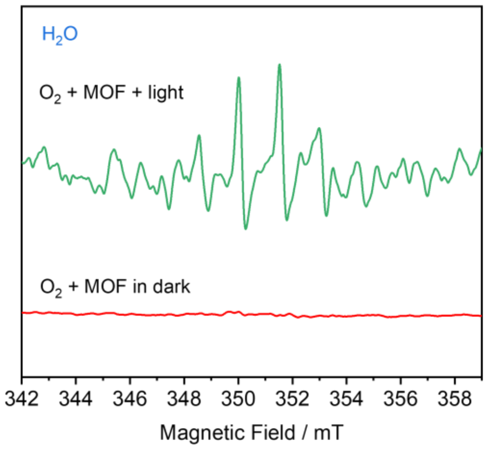


Figure S38. Spin density population of the (A) $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ (**1**), (B) $[(bpy)Fe(OH)(H_2O)_3\cdots CH_4]^{2+}$ (**2**), (C) $[(bpy)Fe(OH)(H_2O)_3\cdots CH_4]^{3+}$ (removal of one electron on irradiation) (**3**), and (D) $[(bpy)Fe(H_2O)_4\cdots {}^{\bullet}CH_3]^{3+}$ (**4**). High and low spin atoms are encoded by green and black, respectively. Fe, purple; O, red; N, blue; C, grey; H, white. The summary of natural population analysis (NPA) charges and spin densities are given in Tables S13-16.

A



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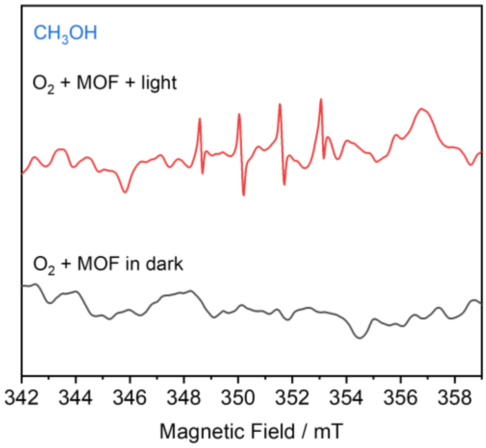


Figure S39. EPR spectra for reactive oxygen species over PMOF-RuFe(OH). X-band (9.85 GHz) spin-trapping spectra under O_2 atmosphere with and without irradiation in (A) water and (B) CH_3OH. Only signals attributed to the DMPO-OH adducts were detected.

NOTE: The intermediate of photoreduction of O_2 was also measured in methanolic media using the EPR spin-trapping technique for the following reasons: i. the EPR signals of DMPO-O_2•$^-$ rapidly converts to DMPO-OH in pure H_2O solvent[47]; ii. the DMPO-OOH adduct is difficult to detect in aqueous solution because of the slow reaction between •OOH and DMPO[48]. Thus, we can rule out O_2•$^-$ and •OOH as reactive oxygen species.

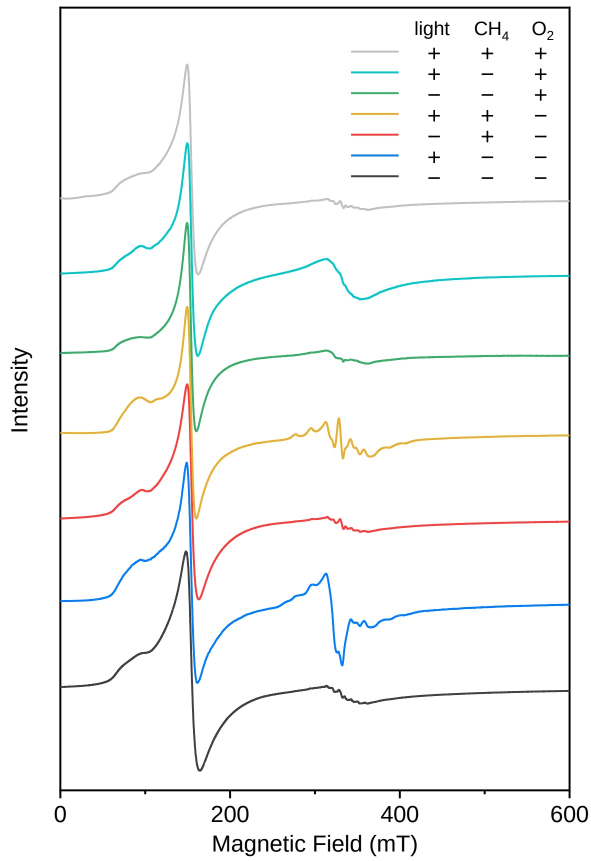
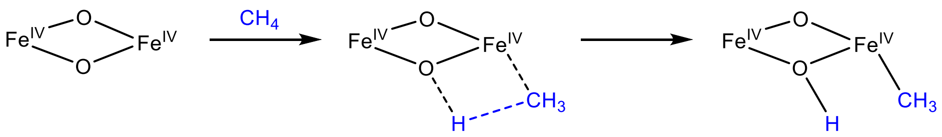


Figure S40. EPR spectra at 5 K of bare PMOF-RuFe(OH) suspended in water (——), after irradiation under light but without CH_4 and O_2 (——), with CH_4 but without O_2 in the dark (——), after irradiation with CH_4 but without O_2 (——), with O_2 but without CH_4 in the dark (——), after irradiation with O_2 but without CH_4 (——), and after irradiation with CH_4 and O_2 (——).

NOTE: *In situ* EPR spectra were measured to unveil the mechanism of electron-transfer (ET). Note that the "native" $PW^{VI}_9V^V_3$ is diamagnetic and EPR silent. When illuminated without any reactants (blue line), characteristic ^{51}V hyperfine structure for the formation of V^{IV} appears, accompanied by the appearance of a signal with $g \approx 2$, which can be assigned to PS^+, corresponding to the unpaired electron being predominantly on the ligand π* orbitals rather than in a metal d orbital.[49,50] This suggests photo-induced ET from PS* to produce reduced $PW_9V^{IV/V}_3$. In the presence of CH_4, V^{IV} peaks are found upon irradiation, but no EPR signal for PS^+ was detected (orange line), indicative of PS^+ conversion back to PS accompanying the generation of $\bullet CH_3$. The signals for Fe^{III} show almost no change with CH_4 after irradiation, ruling out a Fe^{IV}-oxo activation pathway [adopted by soluble methane monooxygenase (sMMO)], where the C−H bond cleavage is promoted through the direct interaction with the metal at the catalytic site, resulting in the formation of a σ M−C bond.[51]

M−C formation over sMMO:



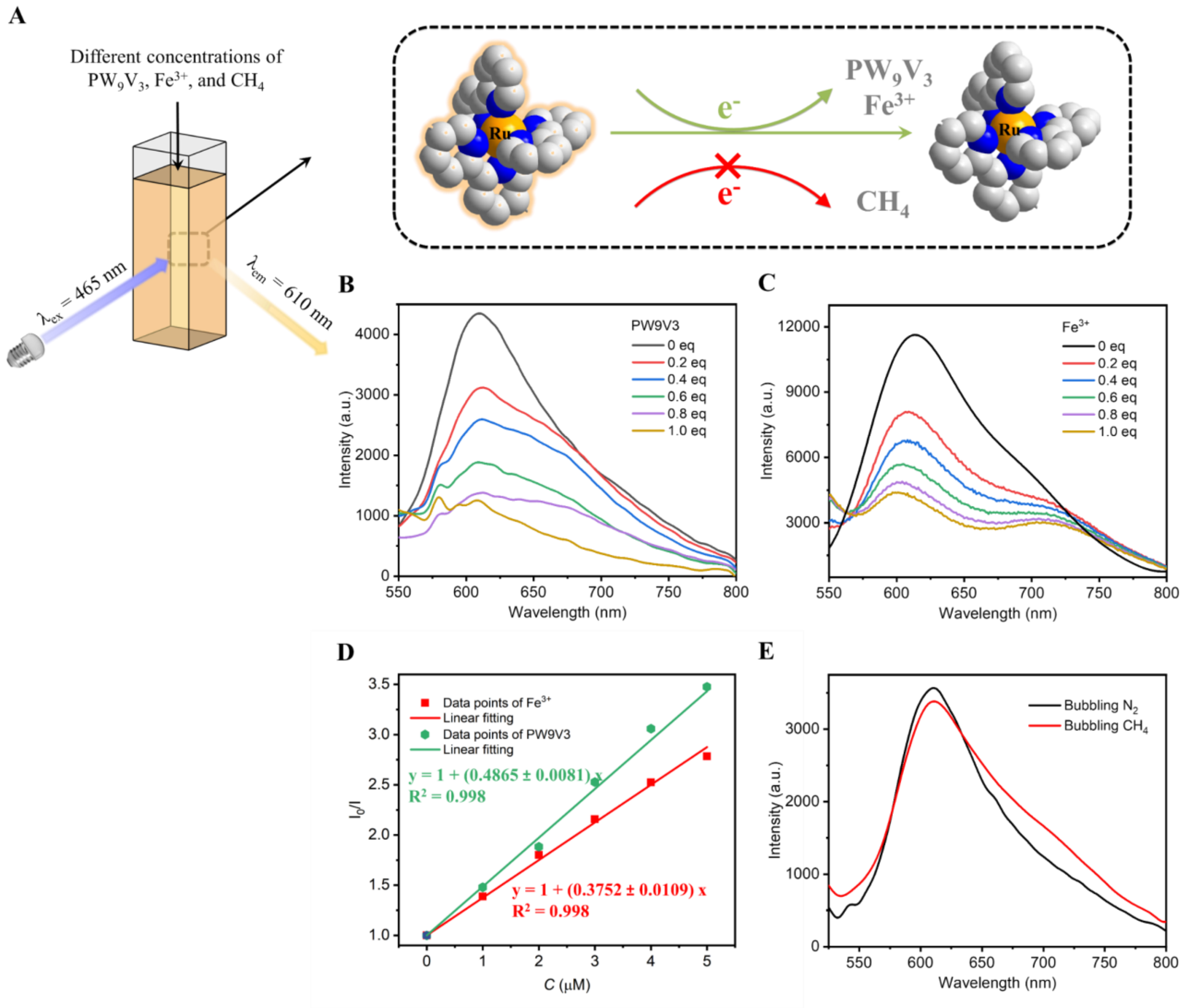


Figure S41. Studies of electron-transfer (ET) by luminescence (static) quenching measurements. (A) Schematic showing luminescence quenching measurements of PS in MOF-Ru by PW_9V_3, Fe^{3+}, and CH_4. (B,C) Luminescence spectra upon excitation at 465 nm of a suspended MOF-Ru solution in H_2O (containing 5 μM of PS) with varying concentrations of (B) PW_9V_3 and (C) $FeCl_3$. (D) Stern-Volmer fitting of I_0/I to the concentrations of PW_9V_3 and Fe^{3+}. (E) Emission spectra of PS (5 μM) in MOF-Ru after bubbling N_2 and CH_4.

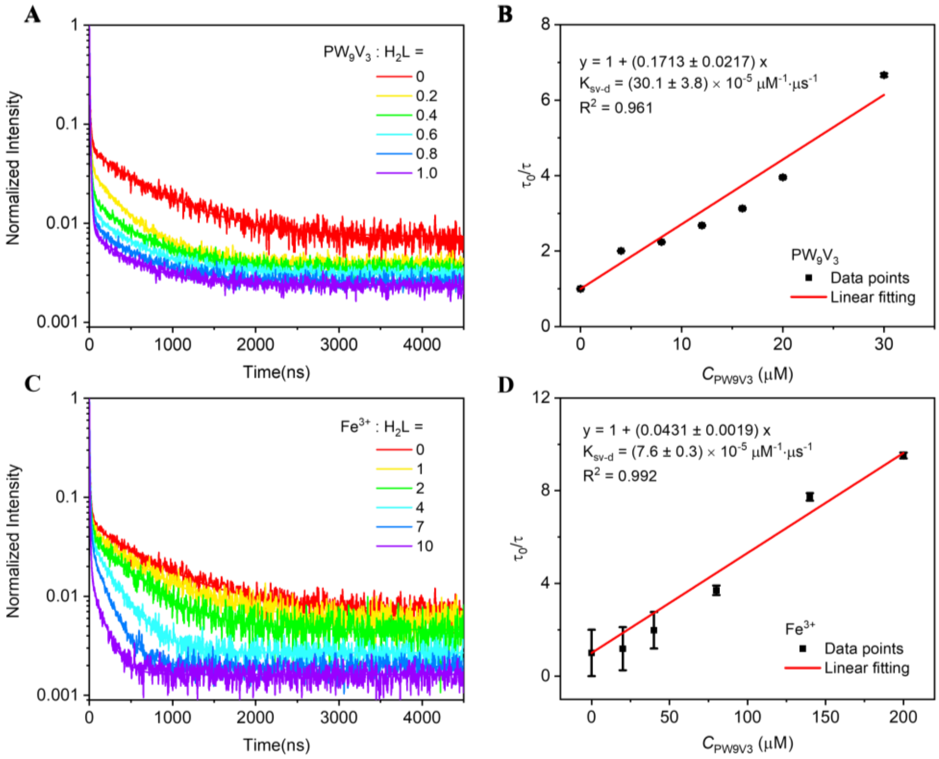


Figure S42. Normalized luminescence decay trace of PS (H_2L) after addition of different amounts of PW_9V_3 and Fe^{3+} measured at the 610 nm emission peak (with 405 nm excitation). The emission decays were fitted to exponential expression $A = A_0+A_1e^{-t/\tau 1} + A_2e^{-t/\tau 2} + A_3e^{-t/\tau 3}$.

NOTE: The process of electron transfer by luminescence quenching has been measured. The $Ru(bpy)_3$-derived chromophore, $[Ru(bpy)_2(H_2bpydc)]^{2+}$ (PS, H_2L) can be excited to the ^{1}MLCT excited state under irradiation, which efficiently transfers to the ^{3}MLCT state through intersystem crossing. The long-lived ^{3}MLCT state decays to the ground-state, which is accompanied with phosphorescence.[52]

The excited Ru^{II}-polypyridyl chromophores (PS^*) in MOF-Ru can be efficiently quenched by PW_9V_3 and/or Fe-species as electron acceptors, but not quenched by CH_4 as electron donors. This indicates that the PS^* was initially oxidatively quenched to generate PS^+ (Figure S41). PW_9V_3 and $FeCl_3$ were added incrementally to a suspended MOF-Ru solution in H_2O (containing 5 μM of PS) to afford the different molar ratios of PW_9V_3/Fe and PS. A luminescence spectrum was collected for each molar ratio by a fluorimeter with an excitation wavelength at 465 nm. Such oxidative quenching was supported by fitting obtained intensities at 610 nm to the concentration using the Stern-Völmer equation:

$$\frac{I_0}{I} = 1 + K_{SV}C$$

where K_{SV} is the Stern-Völmer constant and I_0/I is the ratio of luminescence intensity of PS at 610 nm in the absence and presence of PW_9V_3 and Fe^{3+}. The fitting of this equation confirmed that I_0/I has a linear relationship with respect to the concentration of PW_9V_3 (C_{PW9V3}) with $R^2 = 0.998$ and $K_{SV} = 0.4865 \pm 0.0081$, and Fe^{3+} ($C_{Fe^{3+}}$) with $R^2 = 0.998$ and $K_{SV} = 0.3752 \pm 0.0109$, indicating a more efficient quenching by PW_9V_3.

The oxidative quenching mechanism is further supported by time-resolved photoluminescence measurements (Figure S42).

$$\frac{\tau_0}{\tau} = 1 + K_{\mathrm{SV-d}}\tau_0 C$$

where $K_{\mathrm{SV\text{-}d}}$ is the dynamic Stern-Völmer constant and τ_0/τ is the ratio of luminescence decay intensity of PS after the addition of different amounts of PW_9V_3 and Fe^{3+} measured at the 610 nm emission peak (with 405 nm excitation). The Ru–polypyridyl ^{3}MLCT emissions exhibited linear quenching with different amounts of PW_9V_3 (C_{PW9V3}) with $R^2 = 0.961$ and $K_{\mathrm{SV\text{-}d}} = (30.1 \pm 3.8) \times 10^{-5}$ $\mu M^{-1} \cdot$ μs^{-1} and Fe^{3+} ($C_{Fe^{3+}}$) with $R^2 = 0.992$ and $K_{\mathrm{SV\text{-}d}} = (7.8 \pm 0.3) \times 10^{-5}$ $\mu M^{-1} \cdot$ μs^{-1}. This indicates that both static quenching (formation of quencher/$Ru(bpy)_3^{2+}$ ion pairs) and dynamic quenching (eg. electron/charge transfers) occurs in this system.

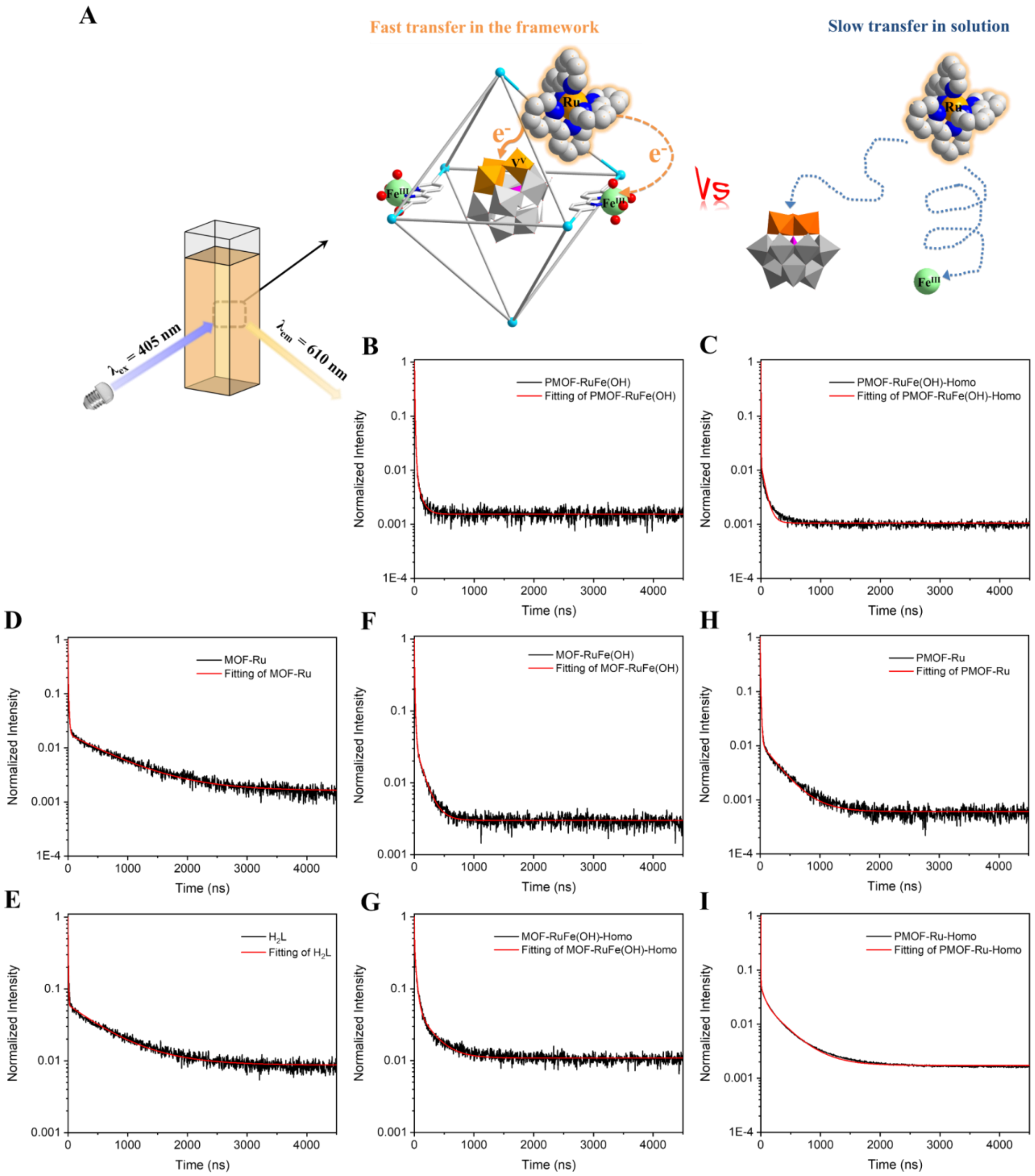


Figure S43. Time-resolved photoluminescence spectra. (A) Schematic view showing time-resolved photoluminescence measurements. (B-I) Normalized decay transients at 610 nm emission wavelength (with excitation at 405 nm) in aqueous solution for (B) PMOF-RuFe(OH), (D) MOF-Ru, (F) MOF-RuFe(OH) and (H) PMOF-Ru, and their homogeneous controls (C, E, G, I, respectively). The emission decays were fitted to the exponential expression $A = A_0+A_1e^{-t/\tau 1} + A_2e^{-t/\tau 2} + A_3e^{-t/\tau 3}$.

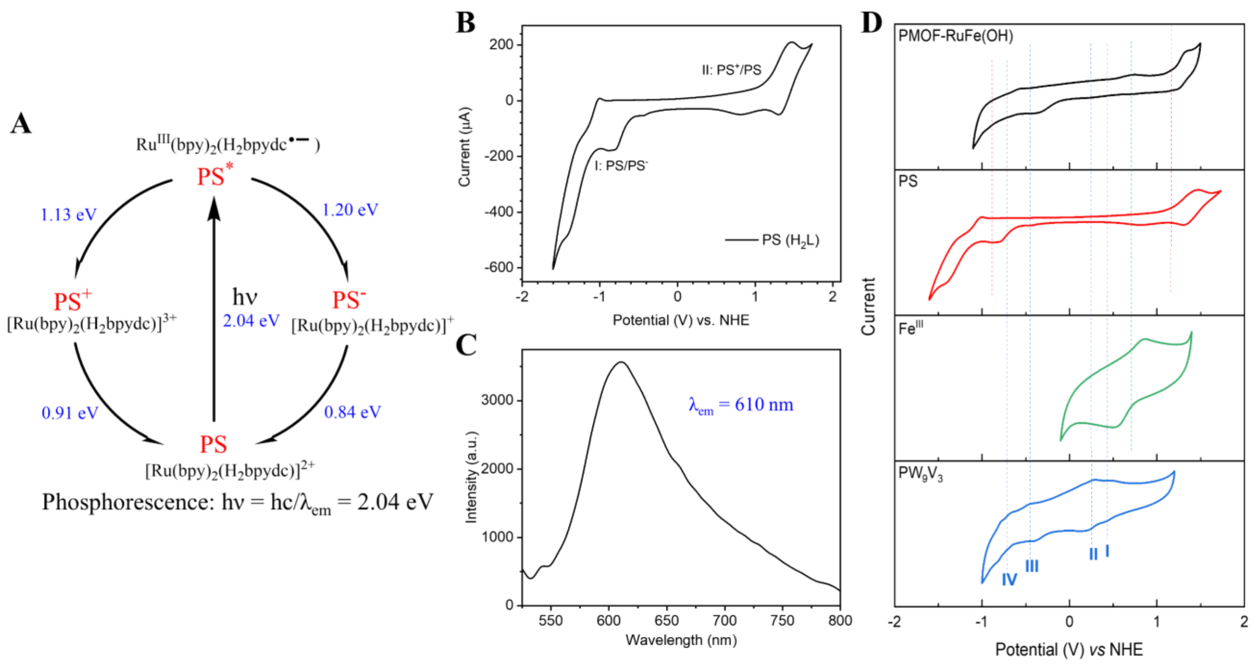


Figure S44. Electrochemical experiments for the study of electron-transfer. (A) Schematic showing photo-redox cycle for $[Ru(bpy)_2(H_2bpydc)]^{2+}$ (PS). (B) Cyclic voltammetry (CV) scan of $[Ru(bpy)_2(H_2bpydc)]^{2+}$ in 0.5 M Na_2SO_4 (aq) at a scan rate of 100 mV·s^{-1}. (C) Emission spectra of $[Ru(bpy)_2(H_2bpydc)]^{2+}$ in aqueous solution with 460 nm excitation. (D) CV scans of PMOF-RuFe(OH) (coated on a glassy carbon), $[Ru(bpy)_2(H_2bpydc)]^{2+}$, $FeCl_3$ and PW_9V_3 in 0.5 M Na_2SO_4 (aq) with a scan rate of 100 mV·s^{-1}. Working, reference, and counter electrodes are glassy carbon (3 mm diameter disk), Ag/AgCl, and Pt, respectively. Four redox couples were observed in the CV curve of PW_9V_3. I, III and IV redox couples correspond to a one-electron redox processes, and II corresponds to a two-electron redox process:

(I) $[PW_9V^V{}_3O_{40}]^{6-} + e^- = [PW_9V^V{}_2V^{IV}O_{40}]^{7-}$ E = 0.61 eV

(II) $[PW_9V^V{}_2V^{IV}O_{40}]^{7-} + 2e^- = [PW_9V^{IV}{}_3O_{40}]^{9-}$ E = 0.20 eV

(III) $[PW_9V^{IV}{}_3O_{40}]^{9-} + e^- = [PW^{VI}{}_8\,W^V\,V^{IV}{}_3O_{40}]^{10-}$ E = -0.38 eV

(IV) $[PW^{VI}{}_8\,W^V\,V^{IV}{}_3O_{40}]^{10-} + e^- + H^+ = [HPW^{VI}{}_8\,W^V\,V^{IV}{}_3O_{40}]^{10-}$ E = -0.80 eV

NOTE: CV scan shows that PMOF-RuFe(OH) has a similar oxidation potential to $[Ru(bpy)_2(H_2bpydc)]^{2+}$ (H_2L) ($E^{ox}_{1/2}$ = 0.91 V *vs* NHE). Both exhibit a luminescence emission peak at 610 nm, corresponding to energy the gap (2.04 eV) between PS* and PS. The energy difference ΔE between PS* and PS$^+$ is 0.91–2.04 = –1.13 eV, which is sufficiently negative to reduce both PW_9V_3 ($R_{POM,1}$, 0.61 V; $R_{POM,2}$, 0.20 V; $R_{POM,3}$, –0.38 V; $R_{POM,4}$, –0.80 *vs* NHE; POM = polyoxometallate) and Fe^{III} species (R_{Fe}, 0.70 V *vs* NHE), suggesting that the V^V in PW_9V_3 and/or Fe^{III} in $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ moieties can be reduced to V^{IV} and Fe^{II}, respectively. EPR confirmed the existence of V^{IV} centers upon irradiation in the absence of CH_4 and O_2: the typical V^{IV} (d^1, S = 1/2) spectra with resolution of the ^{51}V (I = 7/2) hyperfine interaction was observed (Figure S40). No changes in the signals for Fe^{III} have been observed by EPR spectroscopy (Figure S40, blue line). This could be due to (i) Fe^{II} being prone to oxidation to Fe^{III} by PS$^+$ or PW_9V_3 (even reduced to V^{IV}); and (ii) $R_{POM,1/2}$ (0.61 V/0.20eV) < R_{Fe} (0.70 V) suggesting more facile electron-transfer to PW_9V_3 than to Fe.

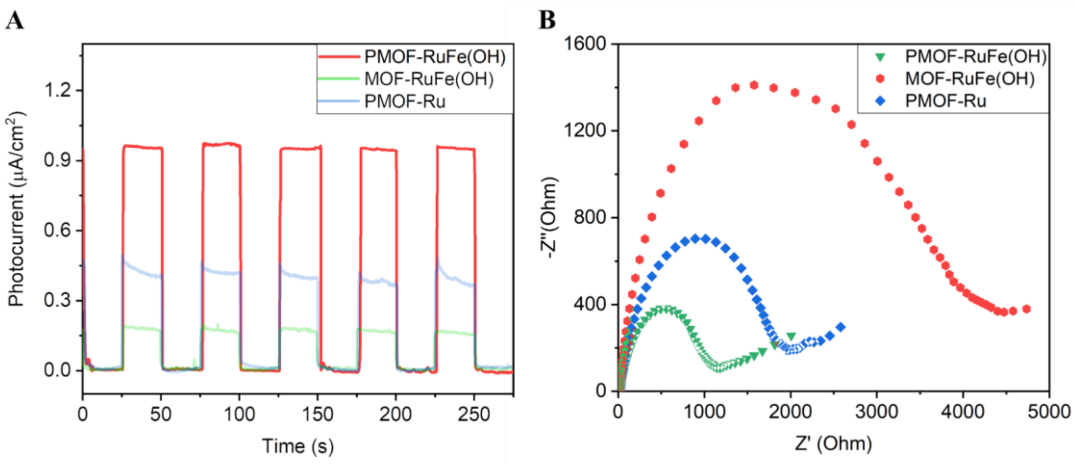


Figure S45. (A) Photocurrent and (B) EIS Nyquist plots at 5000 Hz and a bias of 1.5 V in a 0.5 M Na_2SO_4 aqueous solution of PMOF-RuFe(OH), MOF-RuFe(OH) (without PW_9V_3) and PMOF-Ru (without Fe). The higher photocurrent responses rely upon the higher photo-absorbance and higher charge-transfer efficiency of PMOF-RuFe(OH) under visible light. The apparent smaller radius of PMOF-RuFe(OH) implies a lower charge-transfer resistance, consistent with the higher photocurrent responses which rely on the higher photo-absorbance and higher charge-transfer efficiency.

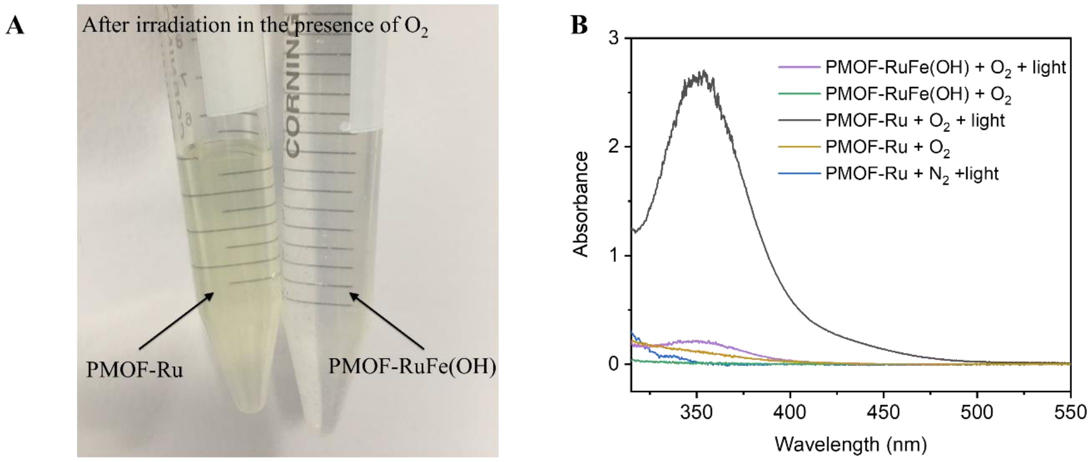


Figure S46. Photocatalytic production of H_2O_2 by the potassium iodide spectrophotometric method over PMOF-RuFe(OH), and PMOF-Ru in the presence of O_2 or N_2 after irradiation or in the dark. The absorbance at ~350 nm after 15 min standing was detected indicating the formation of H_2O_2.

NOTE: The amount of H_2O_2 was determined by the potassium iodide spectrophotometric method. The higher the intensity, the more H_2O_2 was produced. With the introduction of O_2 in presence of water (black line), more H_2O_2 was formed over PMOF-Ru than with N_2 bubbling (blue line), indicating that O_2 is responsible for H_2O_2 production. In addition, over PMOF-RuFe(OH) less H_2O_2 was detected indicating that PMOF-RuFe(OH) catalyzes the decomposition of H_2O_2.

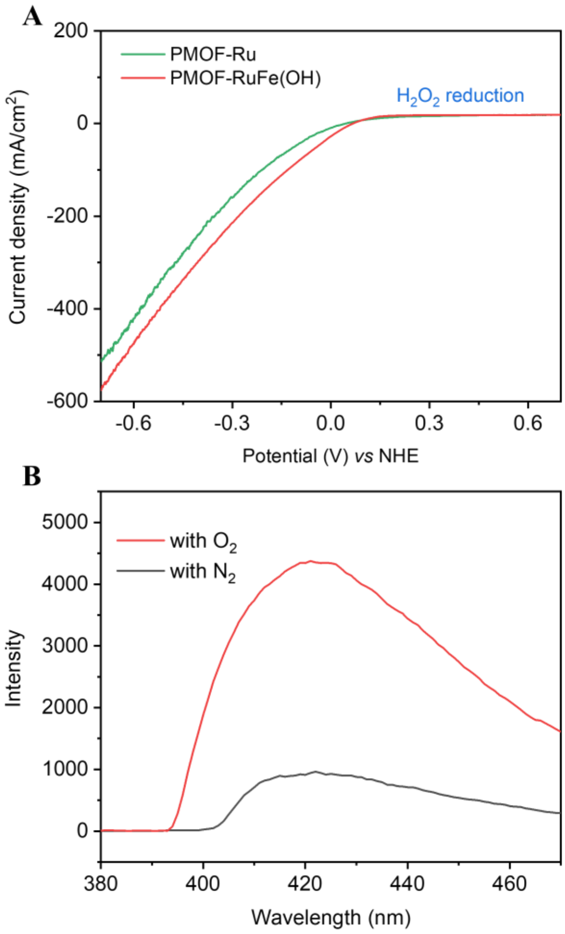
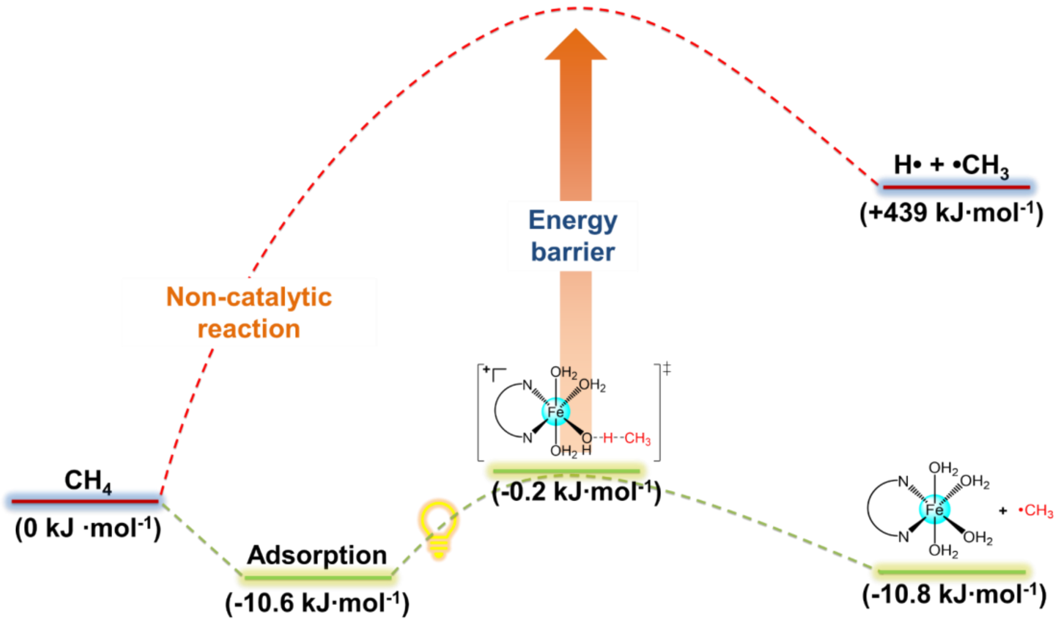


Figure S47. H_2O_2 reduction. (A) Linear sweep voltammetry (LSV) recorded with PMOF-RuFe(OH) and PMOF-Ru in 0.1 M H_2O_2 and 0.5 M Na_2SO_4 aqueous solution. (B) Fluorescent spectra for detection of •OH over PMOF-RuFe(OH) after irradiation under a O_2 and N_2 atmosphere in the presence of terephthalic acid added as a probe. The peak at ~420 nm corresponds to the formation 2-hydroxyterephthalic acid, confirming the production of •OH radical.[53] A stronger fluorescent signal with O_2 than that with N_2 in DMF-H_2O solution over PMOF-RuFe(OH) shows that both O_2 and H_2O are responsible for generating •OH radical.

A



B

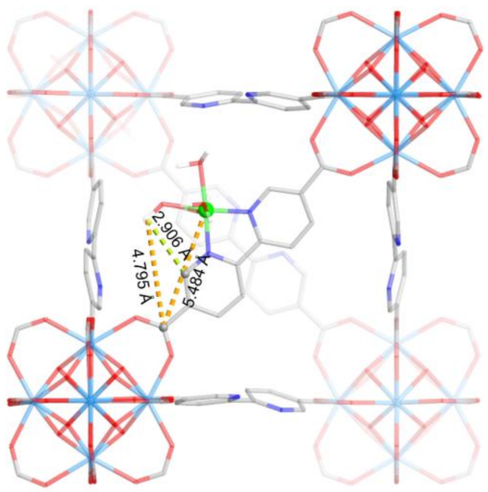


Figure S48. (A) Reaction profiles for adsorption and photo-activation of CH_4 to $\bullet CH_3$ obtained from DFT calculations. Adsorption of CH_4 can lower the dissociation energy of the C-H bond with an activation energy of -0.2 kJ·mol^{-1}, and this reduces the energy barrier significantly compared to the non-catalytic route. Binding of CH_4 to the active mono-iron-hydroxyl sites lowers the activation barrier for the C-H bonds in comparison to those on the bpy ligands. (B) View of the distances of Fe-$C_{C\text{-}O/C=O}$ and Fe-OH···$C_{C\text{-}O/C=O}$ in PMOF-RuFe(OH). Because both the [Fe-OH] site and the ligand are immobilised within the structure, the closest Fe···$C_{C\text{-}O/C=O}$ and Fe-OH···$C_{C\text{-}O/C=O}$ distances are greater than 4.7 Å, thus preventing the C-O and C=O bonds interacting with the active site. The distance between the [Fe-OH] site and C-H groups in the ligand (Fe−OH···C_{bpy} = 2.906 Å) is notably longer than that of adsorbed CH_4 on [Fe-OH] sites (Fe−OH···CH_4 = 2.39 Å).Thus, adsorption of CH_4 on [Fe-OH] sites significantly reduces the activation barrier and promotes oxidation.

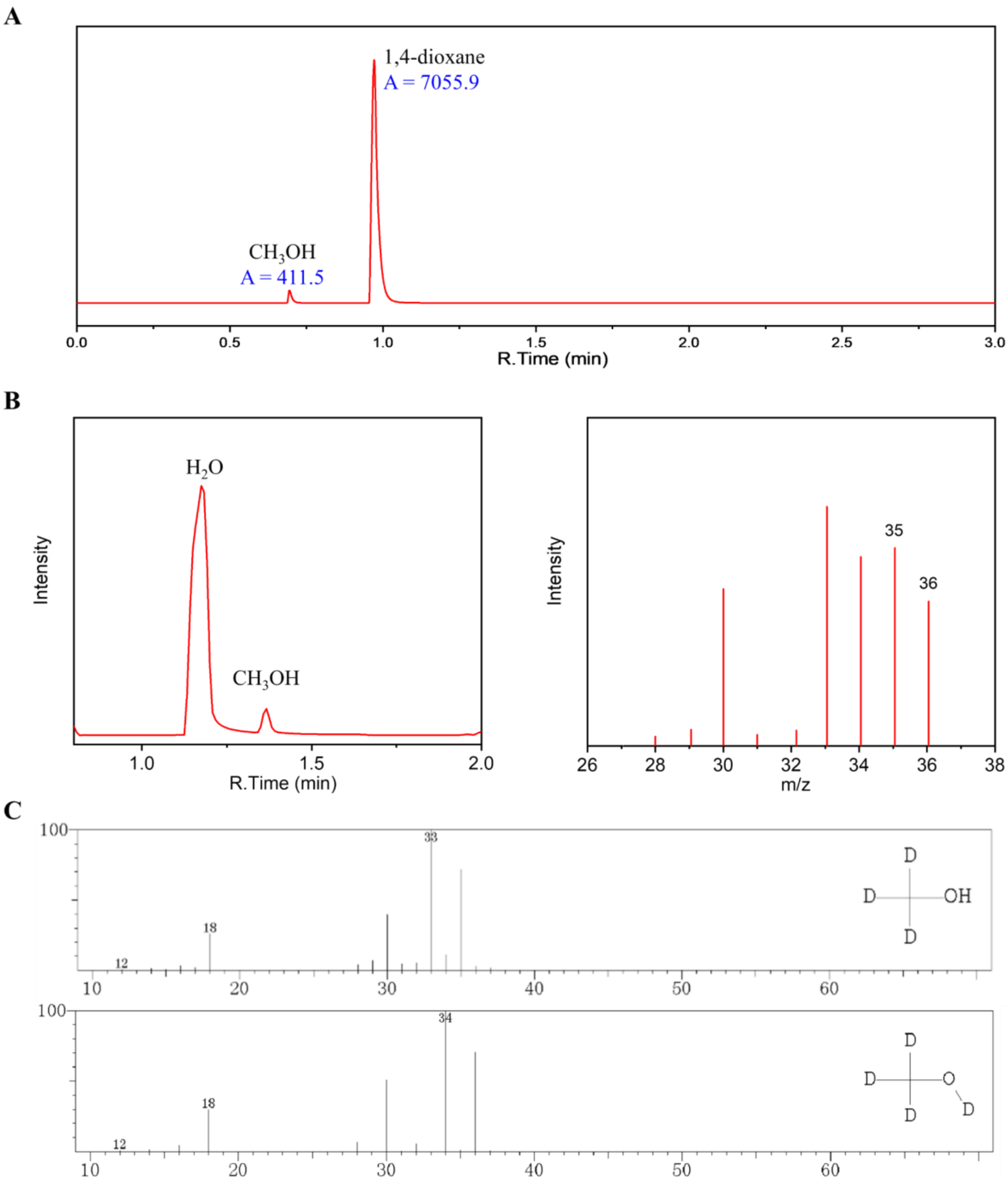


Figure S49. Photo-oxidation of CD_4 over PMOF-RuFe(OH) in batch mode. The solution was analyzed by (A) GC (FID) trace of the liquid product (without extraction; 1,4-dioxane used as an internal standard; FID, column: HP-5). (B) GC-MS of the original liquid solution and the corresponding mass spectrum. (C) Mass spectra of CD_3OH and CD_3OD from the database. The co-existence of CD_3OH and CD_3OD is due to H/D exchange in water solvent. Total amount of deuterated methanol products is 210 μmol, showing a value for the kinetic isotope effect of 3.0.

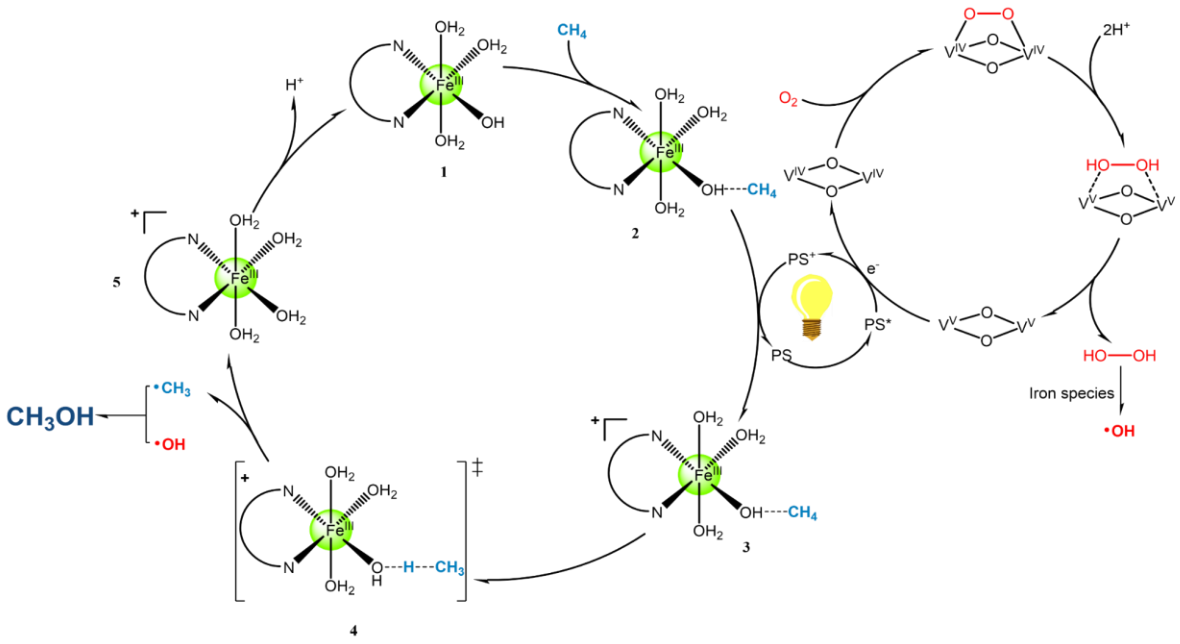


Figure S50. Proposed mechanism for the production of CH_3OH by photo-oxidation of CH_4 over PMOF-RuFe(OH) catalyst.

NOTE: The cycle for activation of CH_4 activation is proposed by experiments combined with theoretical calculations. $[(bpy)Fe(OH)(H_2O)_3{\cdots}CH_4]^{2+}$ (**2**) is oxidised to $[(bpy)Fe(OH)(H_2O)_3{\cdots}CH_4]^{3+}$ **(3)** upon the transfer of one-electron to PS^+. According to the spin population analysis of **3**, the oxidation process likely occurs on the {bpy} fragment to form bpy^+ rather than oxidation of Fe^{3+} to Fe^{4+}. This is consistent with the *in situ* EPR experiments that do not detect any Fe^{4+} species. In addition, for the transition state (TS, **4**), the Fe centre is in +3 oxidation state. Calculation of the barrier to C-H activation on the [Fe(III)-OH] site without any electron transfer (i.e. under dark conditions) does not reach any meaningful result, indicating that, according to DFT, a one-electron oxidation is required for the activation of CH_4 to proceed.

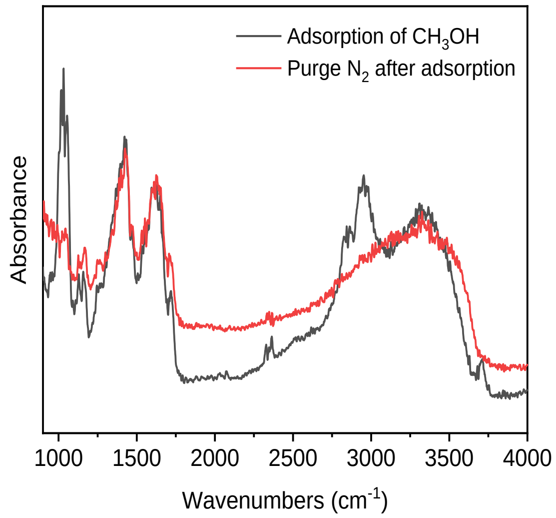


Figure S51. *In operando* diffuse reflectance infrared Fourier transform spectroscopy (DRIFTs) spectra of PMOF-RuFe(OH) loaded with CH_3OH vapor (black curve) and then purged N_2 to remove free and weakly adsorbed CH_3OH at room temperature. The absence of C-H stretching after N_2 purging indicates the absence of strong binding sites of CH_3OH on PMOF-RuFe(OH).

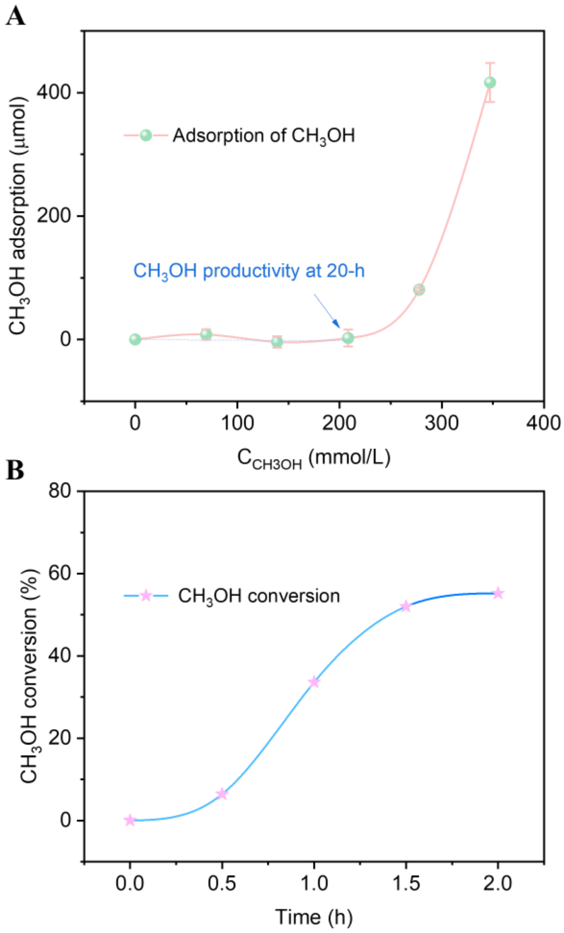
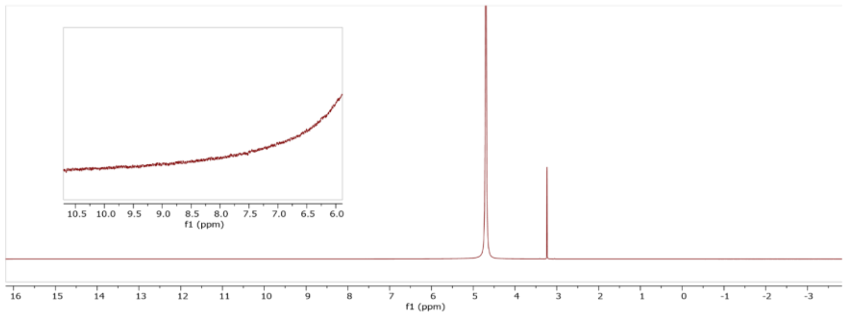


Figure S52. Adsorption and oxidation of CH_3OH on PMOF-RuFe(OH). (A) Adsorption of CH_3OH at room temperature. Conditions: 10.0 mg of sample was soaked in 3 mL of H_2O containing different amounts of CH_3OH. (B) Dependence of oxidation of CH_3OH oxidation on reaction time. Reaction conditions: 10.0 mg of PMOF-RuFe(OH), 500 μL of CH_3OH, 3 mL of H_2O, a balloon filled with O_2 (~1 atm), at 298 K under irradiation of visible light. All data were analyzed by GC with 100 μL of 1,4-dioxane as standard.

NOTE: It is well-known that CH_3OH is more reactive than CH_4 due to its lower C−H bond energy. As seen in Figure S52B, at high concentrations, CH_3OH converts rapidly to CO_2 owing to its rapid adsorption within the porous MOF scaffold at less than 1.5 h. However, after that the conversion is decreased because of the poor adsorption potential at low concentrations (Figure S52A), thus suppressing the CH_3OH oxidation. This is also the reason that over-oxidation occurs when more CH_3OH is produced. Therefore, the total amount of CH_3OH in the reaction system has a remarkable influence on the reaction pathway. Selective oxidation of CH_4 relies upon competitive adsorption of CH_4 vs CH_3OH, and particularly upon the rapid desorption of CH_3OH from the MOF catalyst.

To further study the possible presence of formic acid, we used a high concentration of CH_3OH (2.5 mol/L) instead of CH_4 as the initial reactant, and no formic acid was observed (Figure S53A) HCO_2H, s, $\delta = 8.4$ ppm) and CO_2 was the main product (Figure S53B).

A



B

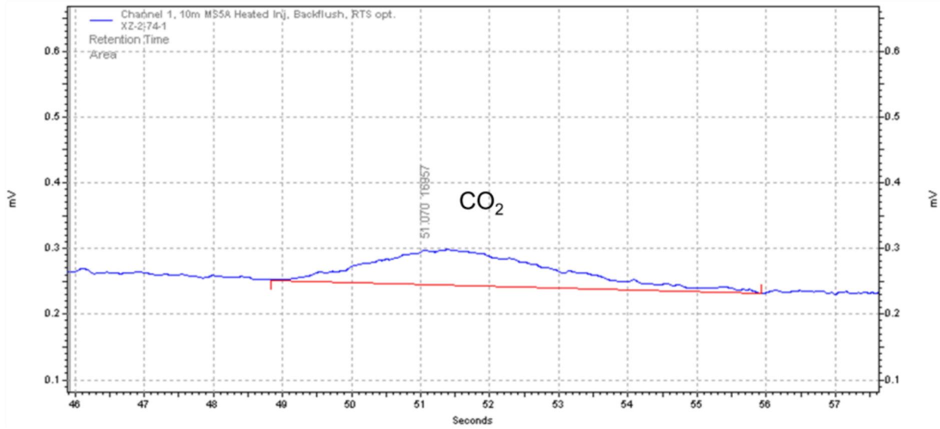


Figure S53. A) [1]H NMR spectrum of the liquid phase for the photo-oxidation of CH_3OH (2.5 mol/L) over PMOF-RuFe(OH) in batch mode. B) GC trace of the headspace gas detected by TCD showing CO_2 as the main product.

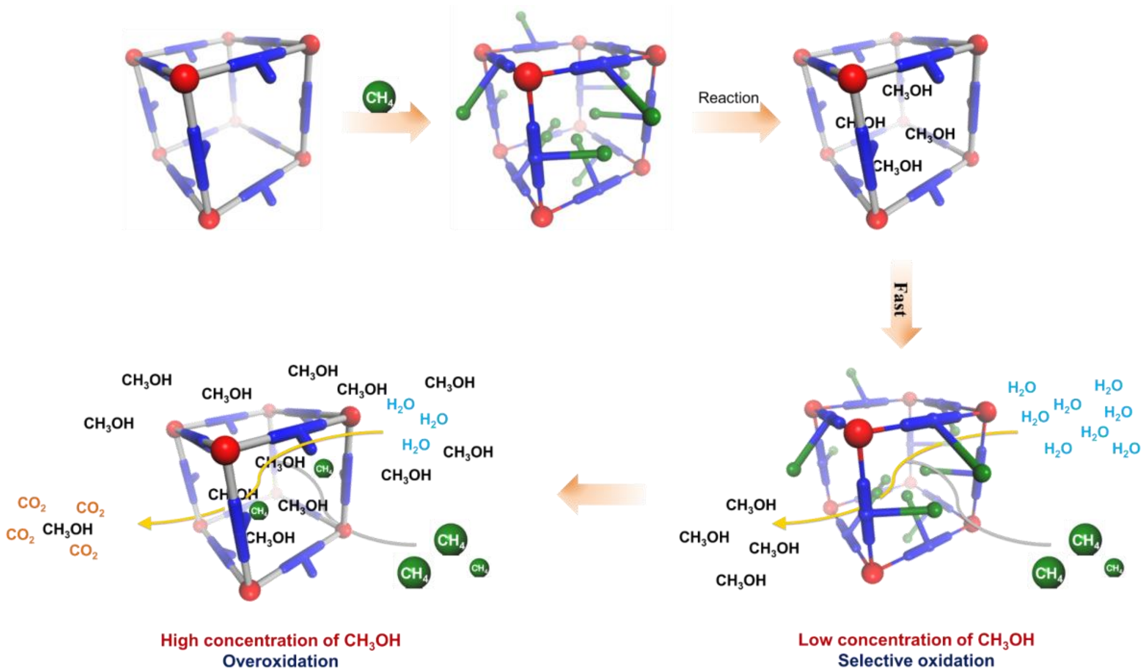


Figure S54. Scheme showing the molecular diffusion into the MOF catalyst in the photo-oxidation process. The water molecules accelerate desorption of CH_3OH product from the MOF, but at high concentrations retention of CH_3OH in the MOF is higher and results in over-oxidation. Thus, catalysis under flow is the most appropriate methodology.

NOTE: With increasing concentations of water, both the selectivity and yield of CH_3OH increases. CH_3OH can be readily eluted from the pores of the MOF by aqueous solution. Also, CH_4 can diffuse through the pores and bind to the iron species, displacing CH_3OH from the MOF catalyst. However, when more CH_3OH is produced, competitive sorption of CH_4 *vs* CH_3OH occurs and this is followed by over-oxidation. Low concentrations of CH_3OH minimize the over-oxidation of CH_4 to CO_2.[54,55]

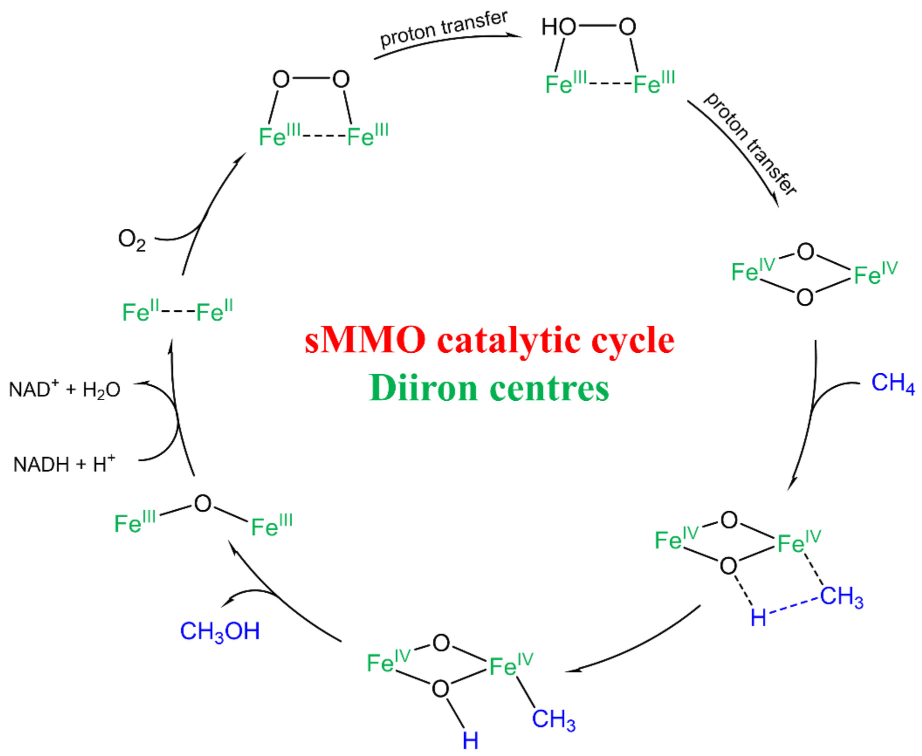


Figure S55. Proposed mechanisms for the conversion of CH_4 to CH_3OH over di-iron centers in soluble methane monooxygenase **(**sMMO) and its model catalysts based on literature examples.[56-58]

Supplementary Tables

Table S1. Elemental analyses of the MOF samples.

Samples	Zr [wt%]	Fe [wt%]	Ru [wt%]	W [wt%]	V [wt%]	C [wt%]	H [wt%]	N [wt%]
PMOF-RuFe(OH)								
[Calculated]	13.5	2.6	0.7	4.1	0.4	22.6	5.1	4.4
[Found]	13.6	2.6	0.7	4.6	0.4	22.9	5.0	4.8
Formulae	$(PW_9V_3)_{0.1}@Zr_6O_{6.1}(OH)_{1.9}(H_2O)_{0.24}(bpydc)_{3.7}[(bpydc)Fe(OH)(H_2O)_3]_{1.9}L_{0.28}$							
PMOF-Ru								
[Calculated]	16.4	-	0.9	5.0	0.5	44.6	3.4	5.4
[Found]	16.7	-	0.9	5.1	0.5	44.7	3.8	5.7
Formulae	$(PW_9V_3)_{0.1}@Zr_6O_{4.2}(OH)_{3.8}(H_2O)_{0.24}(bpydc)_{5.6}\mathbf{L}_{0.28}$							
PMOF-RuFe(Cl)								
[Calculated]	14.7	2.8	0.7	4.5	0.4	43.6	4.5	4.9
[Found]	14.6	2.8	0.7	4.5	0.4	44.0	4.7	4.7
Formulae	$(PW_9V_3)_{0.1}@Zr_6O_{6.1}(OH)_{1.9}(H_2O)_{0.24}(bpydc)_{3.7}[(bpydc)FeCl_2(THF)_2]_{1.9}L_{0.28}$							
MOF-Ru								
[Calculated]	18.5	-	1.0	-	-	36.5	4.2	6.2
[Found]	18.7	-	1.0	-	-	36.4	4.1	6.9
Formulae	$Zr_6O_{4.6}(OH)_{3.4}(bpydc)_{5.7}\mathbf{L}_{0.3}$							
MOF-RuFe(Cl)								
[Calculated]	13.8	4.2	0.8	-	-	46.0	4.8	4.7
[Found]	13.3	4.2	0.7	-	-	46.5	5.1	4.3
Formulae	$Zr_6O_{7.3}(OH)_{0.7}(bpydc)_{2.7}[(bpydc)FeCl_2(THF)_2]_3L_{0.3}$							
MOF-RuFe(OH)								
[Calculated]	12.6	3.9	0.7	-	-	21.5	5.8	4.3
[Found]	12.3	3.9	0.6	-	-	21.0	6.0	4.3
Formulae	$Zr_6O_8(bpydc)_{2.7}[(bpydc)Fe(OH)(H_2O)_3]_3L_{0.3}$							

Table S2. Comparison of ICP-OES results and elemental analyses by EDX of MOF products.

MOFs	n(Zr/W)			n(Zr/Ru)			n(Zr/Fe)		
	EDX	ICP	Calc.	EDX	ICP	Calc.	EDX	ICP	Calc.
POMF-RuFe(OH)	6.5±0.9	6.6±0.6	6.6	23.4±1.8	22.7±0.9	21.4	3.1±0.8	3.1±0.2	3.1
PMOF-Ru	6.7±1.3	6.6±0.8	6.6	22.2±1.4	21.4±0.8	20.6	-	-	-
PMOF-RuFe(Cl)	6.5±0.7	6.7±0.4	6.6	21.4±1.4	21.2±0.6	21.3	3.4±0.4	3.1±0.1	3.2
MOF-Ru	-	-	-	19.6±0.7	20.1±0.5	20.4	-	-	-
MOF-RuFe(Cl)	-	-	-	20.0±0.4	19.9±0.8	19.5	1.7±0.3	1.9±0.1	2.0
MOF-RuFe(OH)	-	-	-	20.6±0.6	21.0±0.6	20.7	1.9±0.3	1.9±0.1	1.9

Table S3. Specific surface areas of MOF catalysts determined from N_2 sorption isotherms.

Samples	S_{BET}[a] ($m^2 \cdot g^{-1}$)	S_{micro}[b] ($m^2 \cdot g^{-1}$)	S_{ext}[c] ($m^2 \cdot g^{-1}$)	V_{micro}[d] ($cm^3 \cdot g^{-1}$)	V_{meso}[e] ($cm^3 \cdot g^{-1}$)	D_{meso}[e] (nm)
PMOF-RuFe(OH)	684.3	220.6	468.7	0.33	0.11	4.0
PMOF-RuFe(OH) after catalysis	610.0	123.9	485.7	0.29	0.05	4.1
PMOF-Ru	2074	1374	698.3	0.73	0.54	4.3
PMOF-RuFe(Cl)	1034	483.6	550.7	0.25	0.39	3.9
MOF-Ru	3181	2432	749.2	1.25	0.54	4.0
MOF-RuFe(Cl)	1379	441.5	937.3	0.66	0.22	4.0
MOF-RuFe(OH)	1081	711.7	361.8	0.27	0.27	4.3

[a]BET surface area.
[b]Microporous surface area.
[c]External surface area.
[d]Micropore volume.
[e]Mesopore volume and mean pore diameter evaluated by the BJH method from the adsorption branches of isotherms.

Table S4. EPR spectroscopic parameters for PMOF-RuFe(OH), PMOF-RuFe(Cl) and MOF-RuFe(OH).[a]

PMOF-RuFe(OH)			
Fe signal		V signal	
Electron Spin	5/2	Electron Spin	1/2
g_x	2.15	$g_{x,y}$	1.986
g_y	2.06	g_z	1.949
g_x	2.05	A_{xy} (G)	73.0
E/D[b]	0.282	A_z (G)	185.5
Linewidth (mT)	30	Linewidth (mT)	6
PMOF-RuFe(Cl)			
Fe signal		V signal (Silent)	
Electron Spin	5/2	Electron Spin	-
g_x	2.10	$g_{x,y}$	-
g_y	2.03	g_z	-
g_x	1.90	A_{xy} (G)	-
E/D[b]	0.230	A_z (G)	-
Linewidth (mT)	30	Linewidth (mT)	-
MOF-RuFe(OH)			
Fe signal		V signal (Silent)	
Electron Spin	5/2	Electron Spin	-
g_x	2.10	$g_{x,y}$	-
g_y	2.08	g_z	-
g_x	1.93	A_{xy} (G)	-
E/D[b]	0.230	A_z (G)	-
Linewidth (mT)	30	Linewidth (mT)	-

[a]Linewidth is the homogeneous Lorentzian linewidth, *g* and A are the principle values of the *g* and ^{51}V hyperfine interaction tensors, respectively.

[b]The zero-field splitting parameters D_x , D_y , and D_z, where $D_x^2 + D_y^2 + D_z^2 = 0$, can be expressed as an axial (*D*) and rhombic (*E*) term: $D = 3D_z/2$ and $E = (D_x – D_y)/2$. The form of the high-spin Fe(III), $S = 5/2$ EPR spectra are such that $|D|$ must be much larger than the microwave energy. In this regime, the spectra are still very sensitive to the ratio *E*/*D*, also known as the rhombicity parameter $\eta = E/D$ with $0 \leq \eta \leq 1/3$.

NOTE: EPR spectra show signals for V^{IV} generated in axially distorted tetragonally octahedral $V^{IV}O_6$ (C_{4v}) units in PMOF-RuFe(OH). PMOF-RuFe(OH) is obtained after photo-activation of PMOF-RuFe(Cl) in water.[59] The presence of V^{IV} is also confirmed by XPS (Figure S8B) and V K-edge XAS results (Figure S16). The absence of any other signals in the EPR spectra of PMOF-RuFe(OH) is consistent with W^{VI} in PW_9V_3 and low-spin d^6 Ru^{II} in $[Ru(bpy)_2(H_2bpydc)]^{2+}$ being EPR-silent. XPS spectra confirming the presence of these species (Figures S8C,D).

Table S5. Summary of EXAFS fitting parameters on the Fe site of PMOF-RuFe(OH).

Sample	PMOF-RuFe(OH) as synthesized	PMOF-RuFe(OH) after catalysis
Fragment	(bpy)Fe(OH)$(H_2O)_3$	(bpy)Fe(OH)$(H_2O)_3$
Fitting range	k 2.6–9.8 Å^{-1} R 1.0–4.0 Å	k 2.6–9.8 Å^{-1} R 1.2–4.0 Å
Independent points	13.5	13.1
Variables	8	8
R-factor	0.010	0.025
S_0^2	0.595	0.841
ΔE_0 (eV)	0.91	7.98
R (Fe–O_{OH}) (Å)	1.78±0.09 (×1^a)	1.78±0.14 (×1^a)
σ^2 (Fe–O_{OH}) (Å^2)	0.0023±0.0008	0.0013±0.0005
R (Fe–O_{H2O}) (Å)	2.17±0.09 (×3^a)	2.17±0.14 (×3^a)
σ^2 (Fe–O_{H2O}) (Å^2)	0.0023±0.0008	0.0013±0.0005
R (Fe–N_{bpy}) (Å)	2.10±0.10 (×1^a)	2.10±0.03 (×1^a)
σ^2 (Fe–N_{bpy}) (Å^2)	0.003±0.002	0.0045±0.0008
R (Fe–N_{bpy}) (Å)	2.16±0.10 (×1^a)	2.16±0.03 (×1^a)
σ^2 (Fe–N_{bpy}) (Å^2)	0.003±0.002	0.0045±0.0008
R (Fe–C_{bpy}) (Å)	2.99±0.04 (×2^a)	2.99±0.01 (×2^a)
σ^2 (Fe–C_{bpy}) (Å^2)	0.008±0.005	0.012±0.001
R (Fe–C_{bpy}) (Å)	3.05±0.04 (×1^a)	3.04 ±0.01 (×1^a)
σ^2 (Fe–C_{bpy}) (Å^2)	0.008±0.005	0.012±0.001
R (Fe–C_{bpy}) (Å)	3.15±0.04 (×1^a)	3.15±0.01 (×1^a)
σ^2 (Fe–C_{bpy}) (Å^2)	0.008±0.005	0.012±0.001

^{a}The number in bracket is the coordination number according to the structural model.

Table S6. Summary of EXAFS fitting parameters on the V site of PMOF-RuFe(OH).

Sample		as synthesized		**Sample**		after catalysis	
Fitting range		*k* 3.0–10.0 Å^{-1}, *R* 1.0–4.0 Å		Fitting range		*k* 3.0–10.0 Å^{-1}, *R* 1.0–4.0 Å	
Independent points		13		Independent points		13	
Variables		7		Variables		7	
R-factor		0.016		R-factor		0.016	
S_0^2		1[a]		S_0^2		1[a]	
ΔE_0 (eV)		−1.61		ΔE_0 (eV)		−5.50	
V^{V} site in PW_9V_3		**V^{IV} site in PW_9V_3**		**V^{V} site in PW_9V_3**		**V^{IV} site in PW_9V_3**	
Percentage	83.7%	Percentage	16.3%	Percentage	78.1%	Percentage	219%
R (V–O5) (Å)	1.61±0.26 (×1[b])	R (V–O5) (Å)	1.98±0.26 (×5[b])	R (V–O5) (Å)	1.61±0.29 (×1[b])	R (V–O5) (Å)	1.98±0.29 (×5[b])
R (V–O17) (Å)	1.82±0.26 (×1[b])	R (V–O1) (Å)	2.38±0.26 (×1[b])	R (V–O17) (Å)	1.82±0.29 (×1[b])	R (V–O1) (Å)	2.38±0.29 (×1[b])
R (V–O18) (Å)	1.86±0.26 (×1[b])	R (V–V) (Å)	2.79±0.10 (×2[b])	R (V–O18) (Å)	1.86±0.29 (×1[b])	R (V–V) (Å)	2.79±0.09 (×2[b])
R (V–O19) (Å)	1.90±0.26 (×1[b])	R (V–W) (Å)	3.38±0.06 (×2[b])	R (V–O19) (Å)	1.90±0.29 (×1[b])	R (V–W) (Å)	3.38±0.05 (×2[b])
R (V–O20) (Å)	1.95±0.26 (×1[b])			R (V–O20) (Å)	1.95±0.29 (×1[b])		
R (V–O1) (Å)	2.44±0.26 (×1[b])			R (V–O1) (Å)	2.44±0.29 (×1[b])		
R (V–V) (Å)	2.76±0.10 (×2[b])			R (V–V) (Å)	2.76±0.09 (×2[b])		
R (V–W) (Å)	3.39±0.06 (×2[b])			R (V–W) (Å)	3.39±0.05 (×2[b])		
σ^2 (V–O) (Å^2)		0.004±0.0007		0.006±0.0009			
σ^2 (V–V) (Å^2)		0.002±0.0005		0.002±0.0007			
σ^2 (V–W) (Å^2)		0.009±0.004		0.009±0.004			

[a]The amplitude factor S_0^2 was set to 1.
[b]The number in bracket is the coordination number according to the structural model.

Table S7. Summary of catalytic data on photo-oxidation of CH_4.[a]

Entry	Catalyst	Selectivity (%)				Yield of CH_3OH (μmol)[b]	Fe-TON[c]
		CH_3OH	CO	CO_2	C_xH_y		
1	No catalyst	0	0	0	0	0	-
2	PMOF-RuFe(Cl)[d]	>97	0	0	0	557±52	111±10
3	PMOF-RuFe(OH)	98	0	2	0	629±68	135±16
4	PMOF-RuFe(OH) (without CH_4)[e]	0	0	0	0	0	-
5	PMOF-RuFe(OH) (in dark)[f]	0	0	0	0	0	-
6	PMOF-RuFe(OH) (at 100 °C)[g]	NA[h]	-	-	-	< 0.2	-
7	UiO-mixRu (MOF-Ru)	NA	-	-	0	0.5±0.1	-
8	UiO-mixRuFe(OH) (MOF-RuFe(OH))	17	51	32	0	7.6±0.6	1.1±0.1
9	PW_9V_3@UiO-mixRu (PMOF-Ru)[i]	54	12	-	0	6.2±1.3	-
10	PW_9V_3@UiO-mixFe(OH) (PMOF-Fe(OH))	34	-	-	66	8.7	1.6
11	PW_9V_3, H_2L, $FeCl_3$[j]	75	5	8	2	51±15	11±3
12	UiO-mixRuFe(OH), PW_9V_3[j]	90	-	3	0	20±1	2.8±0.2
13	$FeCl_3$, PW_9V_3@UiO-mixRu[j]	90	5	2	3	81±4	18±1
14	PW_9V_3@UiO-mixFe(OH), H_2L	82	0	18	0	10±1	1.7±0.2
15	Fe_2O_3	14[e]	-	86[e]	0	<0.1	-
16	Fe_2O_3/PMOF-Ru	19	-	81	0	7.8±0.4	1.6±0.1

[a]Reaction conditions: 10.0 mg of catalysts, 3 mL of H_2O, a mixture of CH_4 and O_2 (~1 atm), 298 K, visible light irradiation for 20 h. All error bars represent the standard deviation from three independent tests.
[b]Yield of CH_3OH was determined by GC with 1,4-dioxane as the internal standard.
[c]Fe-TON is defined as the number of molecules of methanol (n_{MeOH}) produced per iron active site.
[d]CH_3Cl can be detected.
[e]Without CH_4.
[f]Without light.
[g]Heating at 100°C without light.
[h]NA = not accurate. No product was detected due to the low activity.
[i]Formaldehyde was detected in the liquid product.
[j]4.7 μmol of $FeCl_3$, 0.6 μmol of H_2L and 0.3 μmol of PW_9V_3 were loaded as catalysts, consistent with content of PMOF-RuFe(OH).

Table S8. Comparison of the reported catalysts for photo-oxidation of CH_4 to CH_3OH.

Catalysts	Oxidant	P (bar)	T (°C)	Time (h)	CH_3OH time yield ($\mu mol \cdot g_{cat}^{-1} \cdot h^{-1}$)	CH_3OH Select. (%)	Ref
PMOF-RuFe(OH) (batch)	$O_2 + H_2O$	~1	r.t.	2	3485 ± 120	100	This work
	$O_2 + H_2O$	~1	r.t.	20	3145 ± 340	98	
PMOF-RuFe(OH) (flow)	$O_2 + H_2O$	~1	r.t.	120	8810 ± 340	100	This work
0.1 wt % Au/ZnO	O_2	20	r.t.	2	2060[a]	15.7	54
0.1 wt % Pd/ZnO	O_2	20	r.t.	2	3035[a]	26.2	
0.1 wt % Pt/ZnO	O_2	20	r.t.	2	2225[a]	19.1	
30CW (C_3N_4-$Cs_{0.33}WO_3$)	O_2	1	r.t.	4	4.375	38.1	60
4.51%VOx/SBA-15	O_2	-	220	5	4.4[b]	1.4	61
0.23wt% Mo/TiO_2	O_2	1	60	0.5	12.5	1.4	62
Bi_2WO_6	H_2O	~1	55	2	15	27.6	63
q-$BiVO_4$ (10 mL of H_2O)	O_2	10	25	3	971	62.8	55
q-$BiVO_4$ (10 mL of H_2O)	O_2	10	25	3	370	96.6	
$BiVO_4$	H_2O	~1	55	2	21	48	
$BiVO_4$ thick platelet	H_2O	~1	r.t.	2	79.2	85.7	64
$BiVO_4$ bipyramids	H_2O	~1	r.t.	2	111.9	85	
commercial V_2O_5	H_2O	~1	70	2	2.0	100	65
V-HBEA	H_2O	~1	70	2	11.3	85.4	
Bi-V-HBEA	H_2O	~1	70	2	10.7	100	
Cu-0.5/PCN	H_2O	~1	r.t.	1	5.5[c]	12.8	66
WO_3/La	H_2O	1	55	2	29.0	46	67
WO_3/Fe^{3+} (2mM)	H_2O	1	55	2	55.5	37.4	68
WO_3/Cu^{2+} (0.1mM)	H_2O	1	55	2	46.0	38.4	
WO_3/Ag^{+} (2mM)	H_2O	1	55	2	16.8	11.8	
WO_3	H_2O	1	55	2	5.1	22	
WO_3-2	H_2O	1	55	2	27.5	46	
Au1/BP	O_2	30	90	2	56.8	99	69
Au-Pd/TiO_2	O_2	40	52.3	1	12600	42.3	70
0.12_{metal} wt%AuO/TiO_2	H_2O_2	~1	r.t.	3	53.3	32	31
0.33_{metal} wt%Cu_2O/TiO_2	H_2O_2	~1	r.t.	3	6.7	3	
0.47wt%Fe_2O_3/TiO_2	H_2O_2	~1	r.t.	3	357	89	
0.33_{metal} wt%FeO_x /TiO_2	H_2O_2	~1	r.t.	3	360	79	
0.6 umol $FeCl_2$	H_2O_2	~1	r.t.	3	1.3	20	
pristine m-WO_3	H_2O_2	~1	r.t.	4	19.8	25	71
1.98% FeOOH/m-WO_3	H_2O_2	~1	r.t.	4	211.2	91	
g-CN AM1.5G	H_2O_2	30	35	2	140	63	72
Imp-V/MCM-41(acid)	NO	0.005	22	3	11.3	88.4	73
Imp-V/MCM-41(base)	NO	0.005	22	3	3.0	74.7	
V-MCM-41	NO	0.005	22	3	9.5	87.6	

[a]The values of all liquid products are 12545, 10615 and 10610 $\mu mol \cdot g_{cat}^{-1} \cdot h^{-1}$ over 0.1 wt % Au/ZnO, 0.1 wt % Pd/ZnO and 0.1 wt % Pt/ZnO, respectively. [b]246 $\mu mol \cdot g_{cat}^{-1} \cdot h^{-1}$ for time yield of HCHO. [c]21 $\mu mol \cdot g_{cat}^{-1} \cdot h^{-1}$ for time yield of ethanol.

Table S9. Comparison of state-of-art thermocatalysts for oxidation of CH_4 to CH_3OH.

Catalysts	Oxidant	P(bar)	T (°C)	Condition	CH_3OH time yield ($\mu mol\cdot g_{cat}^{-1}\cdot h^{-1}$)	Ref
PMOF-RuFe(OH)	$O_2 + H_2O$	~1	r.t.	flow(light)	8810 ± 340	This work
Cu-NU-1000	O_2	1	200	loop	17.7	74
Cu-2.9-NU-1000	O_2	1	200	loop	1.5	75
Cu-2.9-NU-1000	O_2	40	200	loop	6.5	
MOF-808-Bzz-Cu	N_2O	1	150	loop	71.8 ± 23.4	76
MIL-53(Al,Fe)	H_2O_2	30.5	50	batch	450	77
UiO-66(2.5TFA)-Fe	H_2O_2	30	50	batch	259	78
pMMO	Duroquinol	1	45	batch	1380	79
FeN_4/GN[a]	H_2O_2	20	25	batch	236[b]	80
Cu-CHA	$O_2 + H_2O$	1	300	loop	195	81
Cu-Y	O_2	33	200	loop	0.8	82
Cu-SSZ-16	O_2	1	200	loop	32.5	83
Cu-ZSM-5	O_2	~110	210	loop	982	84
Cu-MOR	O_2	6	200	loop	24	85
Cu-MOR	H_2O	7	200	loop	35	86
FeZSM-5	N_2O	1	160	flow	53.3	87
Fe-ZSM-5	N_2O	0.5[c]	160	loop	116	88
0.3 wt% Rh/ ZrO_2	H_2O_2	30	70	batch	36	89
1 wt% Cr/TiO_2	H_2O_2	30	50	batch	4390	90
Cu-ERI	O_2	30	300	loop	294	91
Fe-ZSM-5	air	15	30	batch	880	92
AuPd@ZSM-5-C16	O_2	30	70	batch	4720[d]	93
Au-Pd colloids	O_2+H_2O_2	30	50	batch	152[d]	94
5wt%Au-Pd/TiO_2	O_2	30.5	50	batch	320	95
Ru_xCu_{1-x} ILs/ SiO_2-200	O_2	30	50	batch	75.3	96
IrO_2/CuO	air	20	150	batch	646	97
$FeSO_4$+1% Pd/C	H_2O_2	15	20	batch	94.3	98
$Fe_2(SO_4)_3$ +1% Pd/C	H_2O_2	15	20	batch	113	
$FeCl_2$+1% Pd/C	H_2O_2	15	20	batch	124	
$FeCl_3$+1% Pd/C	H_2O_2	15	20	batch	97	
0.5wt% Rh-ZSM-5	O_2 + CO	20	150	batch	1224	99
Fe-silicalite-1	H_2O_2	30.5	50	batch	1326	100
0.5wt%Fe^{3+}/ZSM-5	H_2O_2	30.5	50	batch	1481	
$[VO(EtO)(EtOH)]_2(L)$[e]	H_2O_2	20	50	batch	23.4	101
$[Cu^ICu^ICu^I(7\text{-N-Etppz})]^{1+}$	O_2	10	r.t.	batch	571.2	102
CNT@PNC/Ni SAs (~0.68wt%)	H_2O_2	20	50	batch	1063	103
MoO_3/SiO_2 sol/gel	O_2	1	500	flow	70	104

[a]GN = graphene nanosheets. [b]The value contains all oxygenate products. [c]P = 0.5 Torr. [d]The unit is $\mu mol\cdot g_{AuPd}^{-1}\cdot h^{-1}$ (AuPd as the active sites) rather than per gram of the catalyst. If converted for direct comparison, the time yield is 4720 and 152 $\mu mol\cdot g_{cat}^{-1}\cdot h^{-1}$ for references 93 and 94, respectively. [e]L = bis(2-hydroxybenzylidene)oxalohydrazonic acid.

Table S10. The effect of water on photocatalytic oxidation of CH_4 over PMOF-RuFe(OH).[a]

Entry	Solvents	Amount (mL)	Yield (μmol)[b]		CH_3OH selectivity (%)
			CH_3OH	CO_2	
1	No solvent	0	0	320	0
2	Anhydrous benzene[c] (without water)	3	0	296	0
3	Benzene (ACS reagent, containing ~10000 ppm of H_2O)	3	<0.3	222	<0.2%
4	H_2O	1	140	185	43
5	H_2O	3	629	11	>98
6[d]	H_2O	5	680	0	100

[a]Reaction conditions: 10.0 mg of catalyst, CH_4 and O_2 (~1 atm), 298 K, visible light irradiation for 20 h.
[b]The yield of liquid oxygenates was determined by GC using 1,4-dioxane as internal standard. Gaseous products were quantified by GC with TCD or FTIR.
[c]Anhydrous benzene used due to its C−H bond energy (460 kJ·mol^{-1}) being higher than that of CH_4 (439 kJ·mol^{-1}).
[d]The yield of CH_3OH shows no significant increase possibly due to fewer gas/solid/liquid interfaces compared with reaction with 3 mL of water.

Table S11. The effect of O_2 on photocatalytic oxidation of CH_4 over PMOF-RuFe(OH).[a]

Entry	Atmosphere	Yield (μmol)[b]		CH_3OH selectivity (%)
		CH_3OH	CO_2	
1	Only CH_4	37	0	100
2	CH_4 + Ar	35	0	100
3	CH_4 + 1% O_2/Ar	280	5	98.3
4	CH_4 + Air	607	10	98.4
5	CH_4 + O_2	629	11	98.3

[a]Reaction conditions: 10.0 mg of catalyst, CH_4 and O_2 (~1 atm), 3 mL of H_2O, 298 K, visible light irradiation for 20 h.
[b]Yield of liquid oxygenates was determined by GC using 1,4-dioxane as internal standard. Gaseous products were quantified by GC with TCD.

Table S12. Simulation parameters of EPR spectroscopic spin-trapping experiments for PMOF-RuFe(OH) suspended in DMPO solution with CH_4 and O_2 after irradiation.[a]

Adducts	*g* factor	$A^{14}N$ / G	$A^{\alpha}H$ / G	lw/mT[b]	Weighting
DMPO-CH_3	2.00555	15.7	22.4	0.12 (Gaussian) 0.20 (Lorentzian)	0.67
DMPO-OH	2.00570	15.2	14.5	0.07 (Gaussian) 0.09 (Lorentzian)	0.24
$DMPO_{degr}$[a]	2.00545	16.4		0 (Gaussian) 0.12 (Lorentzian)	0.09

[a]$DMPO_{degr}$ is derived from the decomposition of DMPO with O_2 under irradiation.
[b]The linewidth (lw, in mT) defines the FWHM (full width at half height) of the absorption Gaussian (first number) and Lorentzian (second number) broadening of the lines in the spectrum.

Table S13. Summary of NPA charges and spin densities of $[(bpy)Fe(OH)(H_2O)_3]^{2+}$ (**1** in Figure S38).

	Atom	Charge	Spin
1	O	-0.63167	0.033055
2	C	-0.06579	0.002544
3	C	-0.15923	0.00123
4	C	0.234405	0.001073
5	N	-0.28567	0.043947
6	C	-0.04645	-0.00011
7	C	-0.08991	0.002106
8	C	0.248574	0.002797
9	C	-0.16248	0.003365
10	C	-0.05713	0.004225
11	C	-0.08839	0.002421
12	C	-0.04349	-0.00259
13	N	-0.19387	0.075192
14	Fe	1.09098	4.451263
15	O	-0.7648	0.295495
16	O	-0.65762	0.029721
17	O	-0.63126	0.033094
18	H	0.199154	-0.00017
19	H	0.172899	0.000497
20	H	0.190374	0.000618
21	H	0.196335	0.00051
22	H	0.170852	0.000859
23	H	0.198412	-0.0003
24	H	0.196353	0.000867
25	H	0.193368	0.001001
26	H	0.397083	0.000872
27	H	0.40899	0.001498
28	H	0.409157	0.001483
29	H	0.403545	0.000833
30	H	0.366552	0.010882
31	H	0.403589	0.000846
32	H	0.397155	0.000873

Table S14. Summary of NPA charges and spin densities of the $[(bpy)Fe(OH)(H_2O)_3\cdots CH_4]^{2+}$(**2** in Figure S38).

	Atom	Charge	Spin
1	O	-0.77427	0.058069
2	C	-0.0915	0.004684
3	C	-0.14933	0.002032
4	C	0.272923	0.003088
5	N	-0.54517	0.083056
6	C	0.03623	-0.00086
7	C	-0.11574	0.003143
8	C	0.329648	0.002984
9	C	-0.15144	0.004223
10	C	-0.08809	0.009441
11	C	-0.12374	0.003451
12	C	0.058577	-0.00013
13	N	-0.5982	0.135123
14	Fe	1.019258	4.044079
15	O	-0.72576	0.511947
16	O	-0.76286	0.0491
17	O	-0.77407	0.058135
18	H	0.229727	-0.00023
19	H	0.208565	0.000909
20	H	0.218136	0.001633
21	H	0.222969	0.000778
22	H	0.208176	0.001473
23	H	0.230081	-0.00043
24	H	0.223096	0.001547
25	H	0.231016	0.002639
26	H	0.48547	0.002227
27	H	0.490897	0.002369
28	H	0.491144	0.002342
29	H	0.485956	0.00103
30	H	0.470652	0.007811
31	H	0.485968	0.001042
32	H	0.485569	0.002232
33	C	-0.65407	0.001127
34	H	0.184413	0.000184
35	H	0.141048	-0.00019

36	H	0.20852	0.000176
37	H	0.136181	-0.00023

Table S15. Summary of NPA charges and spin densities of $[(bpy)Fe(OH)(H_2O)_3\cdots CH_4]^{3+}$ (**3** in Figure S38)

	Atom	Charge	Spin
1	O	-0.77438	0.064204
2	C	-0.07834	0.102533
3	C	-0.10387	-0.18888
4	C	0.303434	-0.09756
5	N	-0.53642	0.076673
6	C	0.087788	-0.12653
7	C	-0.07148	-0.21848
8	C	0.358511	-0.09484
9	C	-0.10565	-0.19059
10	C	-0.0748	0.106942
11	C	-0.07623	-0.21875
12	C	0.117093	-0.129
13	N	-0.5923	0.110549
14	Fe	1.034772	4.013623
15	O	-0.69672	0.605571
16	O	-0.77092	0.059754
17	O	-0.77448	0.064128
18	H	0.288818	-0.00291
19	H	0.262102	0.008225
20	H	0.275928	0.007173
21	H	0.292534	0.009334
22	H	0.262051	0.008698
23	H	0.289318	-0.00312
24	H	0.293428	0.009838
25	H	0.270775	0.008238
26	H	0.487927	0.002015
27	H	0.504243	0.002041
28	H	0.504249	0.002034
29	H	0.500158	0.000746
30	H	0.467808	0.002448
31	H	0.50013	0.000737
32	H	0.488011	0.002001
33	C	-0.67002	0.002767

34	H	0.23054	0.000243
35	H	0.232848	0.000439
36	H	0.137505	-0.00014
37	H	0.135621	-0.00015

Table S16. Summary of NPA charges and spin densities of $[(bpy)Fe(H_2O)_4\cdots\bullet CH_3]^{3+}$ (**4** in Figure S38)

	Atom	Charge	Spin
1	O	-0.62754	0.026712
2	C	-0.05818	-0.0575
3	C	-0.09309	0.102506
4	C	0.262061	0.107665
5	N	-0.22318	0.058392
6	C	-0.03742	0.030883
7	C	-0.02486	0.19188
8	C	0.26516	0.086102
9	C	-0.08606	0.127928
10	C	-0.05878	-0.06093
11	C	-0.02805	0.182261
12	C	-0.02064	0.063534
13	N	-0.20426	0.050951
14	Fe	0.929101	4.001235
15	O	-0.62444	0.049349
16	O	-0.62891	0.032513
17	O	-0.62703	0.026357
18	H	0.224457	0.001604
19	H	0.193887	-0.00298
20	H	0.200743	-0.00033
21	H	0.228181	-0.00547
22	H	0.194874	-0.00354
23	H	0.224184	0.001563
24	H	0.22706	-0.00514
25	H	0.200305	-0.00193
26	H	0.393231	0.003011
27	H	0.39053	0.004635
28	H	0.395667	0.002303
29	H	0.399954	0.002223
30	H	0.329462	0.086383
31	H	0.398827	0.001876

32	H	0.394242	0.002212
33	C	-0.471	0.95113
34	H	0.182413	-0.02051
35	H	0.193967	-0.02034
36	H	0.19695	-0.01985
37	H	0.388153	0.003308

Table S17. Effects of different additives on the quenching of the photo-oxidation of CH_4 catalysed by PMOF-RuFe(OH).[a]

Entry	Additive	Selectivity (%)					Yield of CH_3OH (μmol)[b]	Fe-TON[c]
		CH_3OH	CH_3OOH	CO	CO_2	C_xH_y		
1	None	98	0	0	2	0	629±68	135±16
2[d]	Isopropanol	4	0	0	64	32	0.3 (Trace)	<0.1
3[e]	2-methylfuran	98	0	0	2	0	632	136
4[f]	Benzoquione	97	0	0	3	0	617	132
5[g]	UiO-66-NH_2	98	0	0	2	0	656	141
6[h]	H_2O_2	86	0	0	14	0	840	180
7[i]	H_2O_2	15	40	0	45	0	14	3.0
8[j]	H_2O_2	16	42	0	42	0	15	3.0
9[k]	H_2O_2 + 2,4-dihydroxybenzonic acid	70	30	0	0	0	62.3	13.4
10[l]	H_2O_2 + 2,4-dihydroxybenzonic acid	100	0	0	0	0	171	36.9

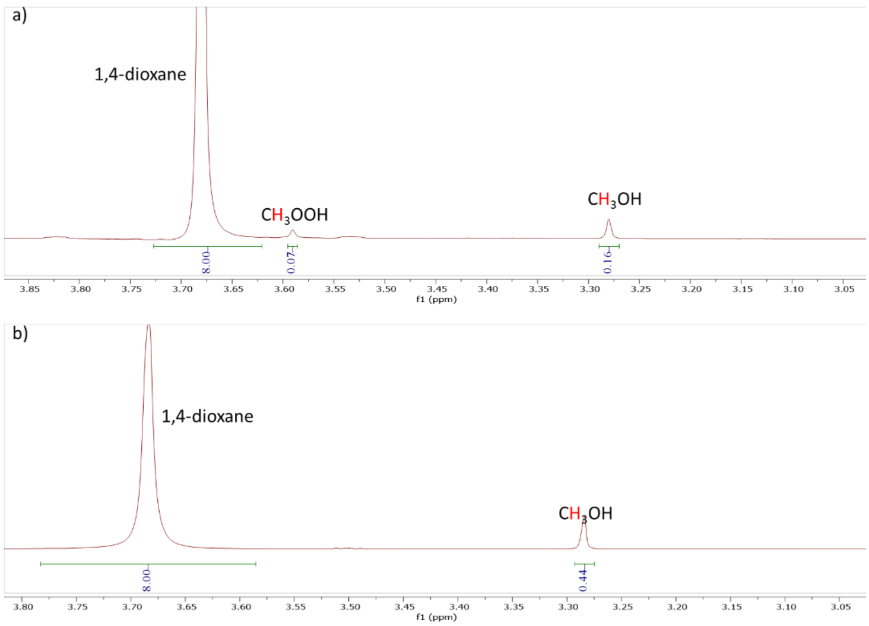
[a]Reaction conditions: 10.0 mg of catalyst, 3 mL of H_2O, CH_4 and O_2 (~1 atm), 298 K, visible light irradiation for 20 h.
[b] The yield of CH_3OH was determined by GC using 1,4-dioxane as internal standard.
[c]Fe-TON is defined as the number of molecules of CH_3OH produced (n_{MeOH}) per iron active site.
[d]Additive: 1 mL of isopropanol as •OH sacrificial agent.
[e]Additive: 20 mM of 2-methylfuran as $^1O_2$ quencher.
[f]Additive: 20 mM of benzoquinone as $O_2^{\bullet -}$ capture agent.
[g]Additive: 10.0 mg of UiO-66-NH_2 (a traditional $O_2^{\bullet -}$ generating photocatalyst) as co-catalyst.
[h]Additive: 100 μL of 30% H_2O_2 under irradiation.
[i]Additive: 100 μL of 30% H_2O_2 in dark. 40% of CH_3OOH product.
[j]Additive: 0.5 mL of 30% H_2O_2 as oxidant, CH_4 (1.0 atm), in dark. The productivity of total C1 liquid oxygenates (CH_3OH and CH_3OOH) is 55 μmol.
[k]Additive: 0.5 mL of 30% H_2O_2 as oxidant, CH_4 (1.0 atm), 20mM of 2,4-dihydroxybenzonic acid as •OOH radical scavenger in dark for 20 h.
[l]Additive: 0.5 mL of 30% H_2O_2 as oxidant, CH_4 (1.0 atm), 20mM of 2,4-dihydroxybenzonic acid as •OOH radical scavenger in dark for 20 h, and then reaction under irradiation for 2 h.

NOTE: The experiment with isopropanol, a scavenger of •OH, as the only solvent greatly reduced the yield of CH_3OH. This is in good agreement with the observed signals for •DMPO-OH after illumination, confirming tha •OH is an important intermediate. The introduction of 2-methylfuran as a $^1O_2$ quencher has little impact on the activity even though derivatives of $[Ru(bpy)_3]^{2+}$ have a well-characterized behavior for producing $^1O_2$.[105] This then excludes the role of $^1O_2$ in the catalysis. The addition of benzoquinone as a capute agent for $O_2\bullet^-$ or the use of UiO-66-NH_2 (a photocatalyst that generates $O_2\bullet^-$) as co-catalyst does not significantly alter the photocatalytic activity, suggesting that $O_2\bullet^-$ is not the critical oxygen-containing intermediate.

$$CH_4 + H_2O_2 \xrightarrow[\text{in dark, 20 h}]{\text{PMOF-RuFe(OH)}} \xrightarrow{\text{adding 100 } \mu L \text{ of 1,4-dioxane as internal standard}} CH_3OH\text{, 15 } \mu mol\text{, 16\%; } CH_3OOH\text{, 42\%; } CO_2\text{, 42\%}$$

$$CH_4 + H_2O_2 + \text{scavenger} \xrightarrow[\text{in dark, 20 h}]{\text{PMOF-RuFe(OH)}} \xrightarrow{\text{adding 100 } \mu L \text{ of 1,4-dioxane as internal standard}} CH_3OH\text{, 63.2 } \mu mol\text{, 70\%; } CH_3OOH\text{, 30\%} \xrightarrow[\text{2 h}]{\text{illumination}} CH_3OH\text{, 171.4 } \mu mol\text{, 100\%}$$

Experiments were performed in the dark in the presence of H_2O_2 and in the presence and absence of a radical scavenger (2,4-dihydroxybenzoic acid) over PMOF-RuFe(OH). 2,4-Dihydroxybenzoic acid was selected as radical scavenger since it does not scavenge H_2O_2.[106] It was found that in the presence of H_2O_2 CH_3OH productivity is over 4 times higher with radical scavenger than without (62 vs 15 µmol), and that selectivity for CH_3OH is increased from 16% to 70% (entries 8 and 9). No other bi-product was detected. In addition, the reaction mixture with scavenger was tested under illumination for a further 2 h, and on illumination CH_3OH is afforded as the only product (entry 10). This suggests that illumination promotes the conversion of CH_3OOH to CH_3OH and quenches the radicals and intermediates that would lead to over-oxidation of CH_3OH.



NMR spectra of liquid solutions for CH_4 oxidation in the presence of H_2O_2 and a radical scavenger (2,4-dihydroxybenzoic acid) a) in dark for 20 h and b) in dark for 20h and then under irradiation for another 2 h.

Table S18. The lifetimes of MOFs and their homogeneous controls in aqueous solution.[a]

Compounds	Lifetime (ns)	Homogeneous controls	Lifetime (ns)
PS (H_2L)	569.0	-	
MOF-Ru	526.2	-	
MOF-RuFe(OH)	33.9	MOF-RuFe(OH)-Homo (Fe^{3+}, H_2L)	135.3
PMOF-Ru	114.2	PMOF-Ru-Homo (PW_9V_3, H_2L)	274.3
PMOF-RuFe(OH)	14.6	PMOF-RuFe(OH)-Homo (PW_9V_3, Fe^{3+}, H_2L)	43.8

[a]The emission decays were fitted to the exponential expression $A = A_0+A_1e^{-t/\tau1} + A_2e^{-t/\tau2} + A_3e^{-t/\tau3}$, and the lifetime $(\tau_{avg}) = \frac{A_1\tau_1^2+A_2\tau_2^2+A_3\tau_3^2}{A_1\tau_1+A_1\tau_1+A_1\tau_1}$

NOTE: Effective quenching of the Ru–polypyridyl ^{3}MLCT (PS*) emission by the encapsulated PW_9V_3 and grafted bpyFeOH fraction in PMOF-RuFe(OH) can be attributed to rapid electron-transfer from the excited states (PS*) to V^V in PW_9V_3 or to Fe(III).

Supplementary Notes

Cost analysis of catalysts for CH_3OH production

We performed a simple calculation on the cost of the catalyst to gain a sense of its cost-effectiveness. Clearly, this cost analysis will be subject to change as external conditions, costs and supply chains vary and develop.

The total cost for the synthesis of PMOF-RuFe(OH) is ~\$4200/kg due mainly to the costs involving the preparation of the organic ligands, H_2bpydc and the photosensitizer. The current industrial route for the synthesis of CH_3OH from syngas ($CO/CO_2/H_2$) relies upon relatively high pressures (50-100 bar) and high temperature (220-240 ^oC) using a ternary $Cu/ZnO/Al_2O_3$ (CZA) catalyst which was first adopted by ICI in 1966. The CZA catalyst needs to be activated at high temperature under a flow of H_2, and thus, the total cost for activated catalyst is ~\$240/kg, less than for PMOF-RuFe(OH). However, CZA catalysts undergo relatively fast deactivation even in the absence of poisons, and more than one third of the activity, MTY (mass-time-yield, $mol \cdot kg_{cat}^{-1} \cdot h^{-1}$), is lost over the first 1000 h (Figure S56).[107] Assuming that (i) the CZA catalyst has a selectivity for CH_3OH of 55.7% with an average MTY_{CZA} for CH_3OH production of ~3.1 $mol \cdot kg_{cat}^{-1} \cdot h^{-1}$ from syngas,[108,109] and (ii) the cost of both methane and syngas is \$0.03/m^3, the Δcost (the difference in cost) (\$/kg_{CH3OH}) for CH_3OH synthesis using PMOF-RuFe(OH) from CH_4 against that of commercial $Cu/ZnO/Al_2O_3$ catalyst from syngas can be calculated using the following equation:

$$\begin{aligned}\Delta \mathrm{cost}(\$/kg_{CH3OH}) &= \frac{1\,kg}{0.032\,kg/mol} \times \Big[\frac{\$4200\,/kg}{MTY_{PMOF-RuFe(OH)}\,(mol/kg_{cat}/h) \times lifetime\,(h)} \\ &+ \frac{\$0.03/m^3 \times 0.0224\,m^3/mol}{100\%} \\ &- \frac{\$240\,/kg}{MTY_{CZA}\,(mol/kg_{cat}/h) \times lifetime\,(h) \times 55.7\%} \\ &- \frac{\$0.03/m^3 \times 0.0224\,m^3/mol}{55.7\%}\Big] = \frac{1.3125 \times 10^5}{MTY_{PMOF-RuFe(OH)} \times t} - \frac{7500}{3.1 \times t} - 0.0167\end{aligned}$$

As shown in Figure S56, only when the MTY is up to 10.7 $mol \cdot kg_{cat}^{-1} \cdot h^{-1}$, is PMOF-RuFe(OH) competitive against the commercial catalysts for CH_3OH synthesis. CZA performs at high pressure, while catalysis of oxidation of CH_4 to CH_3OH under the atmospheric condition has a competitive advantage, including the cost of reactor and safety issues. For example, one stainless steel autoclave reactor with a Teflon vessel and quartz window used for high-pressure photo-reaction costs at least \$2500 depending on different pressure-proof capacities, although costs will of course vary according to different suppliers. However, the simple glasses kit used for atmospheric photoreaction in our flow or batch system costs less than \$120, 20x cheaper. Thus, the activity of PMOF-RuFe(OH) in flow mode has reached 8.81 $mol \cdot kg_{cat}^{-1} \cdot h^{-1}$ which is only 18% less than the target MTY. Considering the mild conditions, the simple equipment and a 100% CH_3OH selectivity, PMOF-RuFe(OH) appears to be a viable candidate and has potential for further development leading to commercialisation.

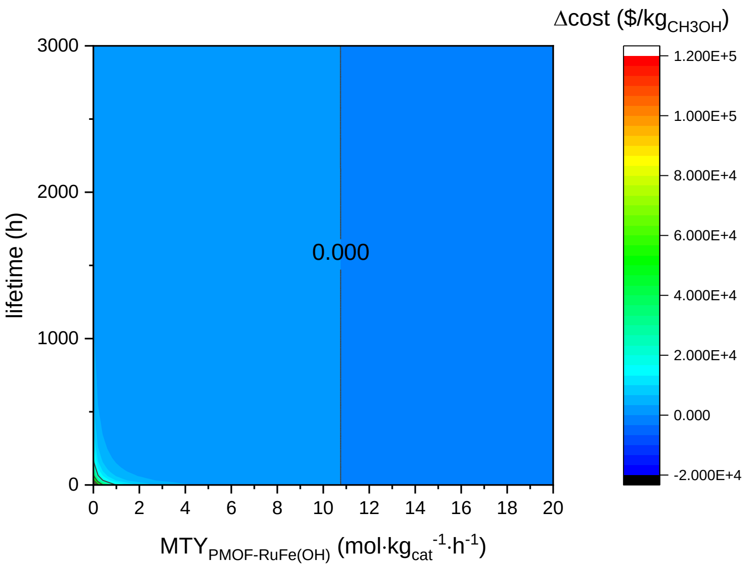


Figure S56. Costs of PMOF-RuFe(OH) compared to use of commercial $Cu/ZnO/Al_2O_3$ catalysts in producing 1 kg of CH_3OH. The PMOF-RuFe(OH) catalyst becomes competitive when MTY > 10.7 mol·kg_{cat}^{-1}·h^{-1}.

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